# Fundamental Theorem of Matrix Product States

A Formalization Blueprint

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# Chapter 1

# Introduction

This blueprint describes the fundamental theorem for periodic, translation-invariant matrix product vectors. The basic object is a family of matrices $\{A^i\}$, viewed as a tensor that generates vectors $|V^{(N)}(A)\rangle$ on chains of length $N$.

Chapters 2 and 4 describe the algebraic core. Chapter 2 introduces word evaluation, MPV coefficients, gauge equivalence, injectivity, blocking, overlap, and the block-injective canonical-form data. Chapter 4 proves the single-block fundamental theorem by a purely algebraic route through trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether.

The later chapters develop the spectral and canonical-form machinery needed for the full multi-block theorem. Chapter 5 studies positive maps and quantum channels. Chapters 7 and 8 supply Perron–Frobenius theory and overlap decay. Chapters 9 and 10 connect blocking, primitivity, irreducibility, and the normal/canonical-form reductions. Chapters 11 and 12 carry out block permutation and BNT theory; Chapter 13 proves the Fundamental Theorem.

# Chapter 2

# Matrix Product Vectors

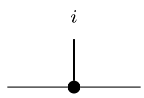
Fix a physical dimension $d$ and a bond dimension $D$. This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

## 2.1 Basic definitions

**Definition 2.1** (MPS tensor)**.** A (translation-invariant, PBC) *MPS tensor* with physical dimension $d$ and bond dimension $D$ is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \qquad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family. Diagrammatically,

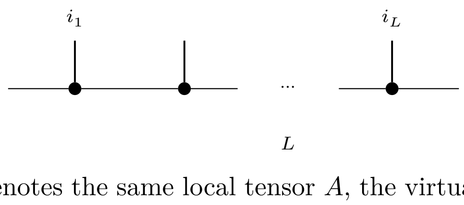



The black node denotes the tensor $A$, the horizontal legs are virtual, and the upper leg is the physical index $i$.

**Definition 2.2** (Word evaluation)**.** Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w \; := \; A^{i_1} A^{i_2} \cdots A^{i_L} \; \in \; M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^{\emptyset} = \mathbb{1}_D$. In tensor-network notation,




in which each black node denotes the same local tensor $A$, the virtual legs remain open, and the physical legs record the word $(i_1, \dots, i_L)$.

**Lemma 2.3** (Multiplicativity of word evaluation)**.** *For any words $w_1, w_2$, $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$.*

*Proof.* Immediate from the definition of $A^w$. □

**Definition 2.4** (Matrix product vector)**.** The *matrix product vector* (MPV) of a tensor $A$ at system size $N$ is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1,\dots,i_N=0}^{d-1} \operatorname{tr}(A^{i_1}A^{i_2}\cdots A^{i_N})|i_1,\dots,i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

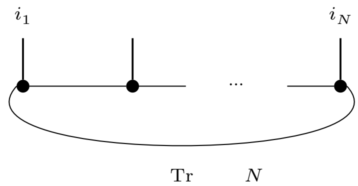
Equivalently, for a configuration $\sigma = (i_1,\dots,i_N) \in \{0,\dots,d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \operatorname{tr}(A^{i_1}A^{i_2}\cdots A^{i_N}).$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components; the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word $w$, we also write

$$c_w(A) := \operatorname{tr}(A^w),$$

The *MPV family* generated by $A$ is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N\ge 1}$. The coefficient $V^{(N)}(A)_\sigma$ is the periodic contraction

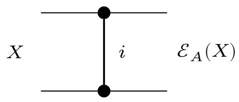



of $N$ copies of the local tensor $A$, with the outer virtual legs closed by the trace.

**Definition 2.5** (Transfer map)**.** The *transfer map* associated to a tensor $A$ is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) \;=\; \sum_{i=0}^{d-1} A^i\, X\, (A^i)^\dagger.$$

Diagrammatically, the transfer map is the double-layer contraction




in which the upper node denotes $A$, the lower node denotes $A^\dagger$, and the physical index is summed over between them.

*Remark* 2.6. The transfer map is automatically positive: if $X \ge 0$ then $\mathcal{E}_A(X) \ge 0$, because each summand $A^i X (A^i)^\dagger$ is positive semidefinite. The abstract positive-map framework appears in Chapter 5.

## 2.2 Periodic-chain aliases

**Definition 2.7** (Periodic-chain tensor alias)**.** For a fixed period $m$, a *periodic-chain tensor* is a site-dependent tensor family indexed by $\{0,\dots,m-1\}$, with local physical dimension $d$ and bond dimension $D$.

**Definition 2.8** (Periodic same-state and gauge-equivalence relations)**.** For periodic-chain tensors $A, B$ at fixed period $m$:

- equality of the induced periodic-chain states;
- cyclic gauge equivalence.

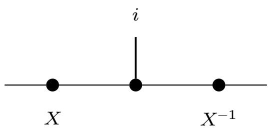
**Lemma 2.9** (Equivalence structures for periodic relations)**.** *Both periodic relations are equivalence relations (reflexive, symmetric, transitive).*

## 2.3 Gauge equivalence and same MPV

**Definition 2.10** (Gauge equivalence)**.** Two tensors $A$ and $B$ of the same bond dimension $D$ are *gauge equivalent* if there exists an invertible matrix $X \in \mathrm{GL}_D(\mathbb{C})$ such that

$$B^i \;=\; X\, A^i\, X^{-1} \qquad \text{for all } i \in \{0, \dots, d{-}1\}.$$

In tensor-network notation,




the black node denotes the tensor named in the surrounding formula, the upper leg is the physical index $i$, and the two side nodes indicate the gauge matrices acting on the virtual legs.

**Definition 2.11** (Same MPV)**.** Two tensors $A$ and $B$ (of the same bond dimension) *generate the same MPV family* if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size $N \geq 1$.

**Definition 2.12** (Same MPV — different bond dimensions)**.** For tensors $A$ and $B$ of possibly different bond dimensions $D_1$ and $D_2$, we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N \geq 1$.

*Remark* 2.13*.* Definition 2.11 is the same-bond-dimension specialization of Definition 2.12. We keep both because later results use both the same-dimension and different-dimension forms.

**Definition 2.14** (Proportional MPV)**.** Two tensors $A$ and $B$ (of possibly different bond dimensions) generate *proportional MPV families* if for every $N$ there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N\, |V^{(N)}(B)\rangle$.

**Lemma 2.15** (Scaling of word evaluation)**.** *Let $\zeta A$ denote the tensor with matrices $\{\zeta\, A^i\}_{i=0}^{d-1}$. Then for any word $w$,*

$$(\zeta A)^w \;=\; \zeta^{|w|}\, A^w,$$

*where $|w|$ is the length of $w$.*

*Proof.* Induction on $w$: the base case gives $\zeta^0 = 1$, and each step picks up one factor of $\zeta$. $\square$

**Definition 2.16** (Gauge-phase equivalence)**.** Two tensors $A$ and $B$ are *gauge-phase equivalent* if there exist $X \in \mathrm{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that

$$B^i \;=\; \zeta\, X\, A^i\, X^{-1} \qquad \text{for all } i.$$

*Remark* 2.17. After spectral-radius normalization one may restrict $\zeta$ to a unit-modulus phase $e^{i\phi}$. We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

**Lemma 2.18** (Gauge covariance of word evaluation). *If $B^i = X\,A^i\,X^{-1}$ for all $i$, then for every word $w$,*

$$B^w \;=\; X\,A^w\,X^{-1}.$$

*Proof.* Induction on the word length, using $XX^{-1} = \mathbb{1}$. □

**Theorem 2.19** (Gauge invariance of MPVs). *If $A$ and $B$ are gauge equivalent, then they generate the same MPV family.*

*Proof.* Let $B^i = XA^iX^{-1}$. By Lemma 2.18, $B^w = X\,A^w\,X^{-1}$ for every word $w$. Then $\mathrm{tr}(B^w) = \mathrm{tr}(X\,A^w\,X^{-1}) = \mathrm{tr}(A^w)$ by cyclicity of the trace. □

## 2.4 Injectivity and normality

**Definition 2.20** (Injectivity). A tensor $A$ is *injective* if its matrices span the full matrix algebra: $\mathrm{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d{-}1\} = M_D(\mathbb{C})$.

**Definition 2.21** ($L$-block injectivity). A tensor $A$ is *$L$-block injective* if $\mathrm{span}_{\mathbb{C}}\{A^{i_1} \cdots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d{-}1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

**Definition 2.22** (Normal tensor). A tensor $A$ is *normal* [CPGSV21] if there exists a finite $L_0$ such that $A$ is $L_0$-block injective.

*Remark* 2.23. The algebraic characterization of normality in Definition 2.22 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map $\mathcal{E}_A$ is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Theorem 9.49, Theorem 9.59, and Theorem 10.28. In particular, every subsequent use of "normal" here refers to this single notion; Definition 10.30 employs the spectral formulation, but it describes the same class of tensors.

## 2.5 Canonical form

**Definition 2.24** (Block-injective canonical form). Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i \;=\; \bigoplus_{k=1}^{r} \mu_k\, A_k^i.$$

This is the block-injective version of the canonical-form data; see [PGVWC07] and [CPGSV21].

*Remark* 2.25. Here we use the block-injective (biCF-style) canonical form: each block is injective in the sense of Definition 2.20. This is stronger than the NT-block canonical form of [CPGSV21], where the blocks are only normal in the paper's spectral sense. The paper-faithful NT/BNT/CFII route is developed later in Chapter 10; the present section records the stronger block-injective form used in the later block-injective canonical form construction.

**Definition 2.26** (Block-diagonal tensor from blocks)**.** Given $r$ blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i \;=\; \bigoplus_{k=1}^{r} \mu_k\, A_k^i.$$

**Theorem 2.27** (MPV decomposition)**.** *The MPV of a block-diagonal tensor decomposes as*

$$|V^{(N)}(A)\rangle \;=\; \sum_{k=1}^{r} \mu_k^N\, |V^{(N)}(A_k)\rangle.$$

*Equivalently, for each configuration $\sigma$ one has*

$$V^{(N)}(A)_\sigma \;=\; \sum_{k=1}^{r} \mu_k^N\, V^{(N)}(A_k)_\sigma.$$

*Proof.* For each physical index $i$, the block-diagonal tensor $A^i = \bigoplus_k \mu_k A_k^i$ is a block-diagonal matrix. By Lemma 2.15, the word evaluation of each block picks up a factor $\mu_k^{|w|}$, and the full word evaluation is block-diagonal with blocks $\mu_k^N A_k^w$. Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\mathrm{tr}(A^w) = \sum_k \mu_k^N\, \mathrm{tr}(A_k^w)$. □

## 2.6 Blocking

**Definition 2.28** (Blocked tensor)**.** Given a tensor $A$ and a blocking length $L$, the *$L$-blocked tensor* is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension $d^L$) and the same bond dimension $D$, whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1,\dots,i_L)} \;=\; A^{i_1} A^{i_2} \cdots A^{i_L}.$$

**Lemma 2.29** (Blocked word evaluation)**.** *If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each $J_k$ is an $L$-tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\widetilde{w}$ of length $mL$ in the original alphabet and one has*

$$(A^{[L]})^w = A^{\widetilde{w}}.$$

*Proof.* Induct on the blocked word $w$ and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations. □

**Lemma 2.30** (Blocking preserves MPV coefficients)**.** *For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each $\sigma_k$ is an $L$-tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that*

$$V^{(N)}(A^{[L]})_\sigma \;=\; V^{(NL)}(A)_{\tilde{\sigma}}.$$

*Proof.* Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.29). □

**Theorem 2.31** (Blocking preserves same MPV)**.** *If $A$ and $B$ generate the same MPV family, then for any blocking length $L$, the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.*

*Proof.* By Lemma 2.30, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of $A$ at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since $A$ and $B$ agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. □

## 2.7 MPV overlap

**Definition 2.32** (MPV as Hilbert-space vector)**.** We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \ldots, d-1\}^N$, with $\sigma$-component $V^{(N)}(A)_\sigma$ as in Definition 2.4. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

**Definition 2.33** (MPV inner product)**.** The *MPV inner product* between two tensors $A$ and $B$ at system size $N$ is

$$\langle V^{(N)}(A)|V^{(N)}(B)\rangle \ := \sum_{\sigma\in\{0,\ldots,d-1\}^N} \overline{V^{(N)}(A)_\sigma}\, V^{(N)}(B)_\sigma,$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

**Definition 2.34** (MPV overlap)**.** The *MPV overlap* between two tensors $A$ (of bond dimension $D_1$) and $B$ (of bond dimension $D_2$) at system size $N$ is

$$O_{AB}(N) \ := \sum_{\sigma\in\{0,\ldots,d-1\}^N} V^{(N)}(A)_\sigma\, \overline{V^{(N)}(B)_\sigma}.$$

This is the bilinear overlap (without complex conjugation) used later. By Lemma 2.35, it is the complex conjugate of the Hilbert-space inner product from Definition 2.33.

**Lemma 2.35** (Overlap equals conjugate inner product)**.** $O_{AB}(N) = \overline{\langle V^{(N)}(A)|V^{(N)}(B)\rangle}$.

*Proof.* The overlap sums $V_\sigma\, \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma}\, W_\sigma$. Complex conjugation swaps these. □

# Chapter 3

# Matrix Product Operators and Density Operators

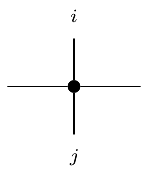
Fix a physical dimension $d$ and a bond dimension $D$. An MPO tensor carries two physical indices, one ket index and one bra index, together with the virtual left and right indices.

## 3.1 MPO tensors

**Definition 3.1** (MPO tensor)**.** An *MPO tensor* is a family of matrices

$$\{M^{ij}\}_{i,j=0}^{d-1}, \qquad M^{ij} \in M_D(\mathbb{C}),$$

indexed by a ket index $i$ and a bra index $j$. Diagrammatically,




The black node denotes the tensor $M$, the horizontal legs are virtual, the upper leg is the ket index $i$, and the lower leg is the bra index $j$.

**Definition 3.2** (Doubled-index MPS view)**.** By combining the pair $(i, j)$ into a single physical index in $\{0, \ldots, d^2 - 1\}$, one obtains an MPS tensor with physical dimension $d^2$ and bond dimension $D$.

**Definition 3.3** (Word evaluation)**.** For words $u = (i_1, \ldots, i_N)$ and $v = (j_1, \ldots, j_N)$ in $\{0, \ldots, d-1\}^N$, the *word evaluation* is

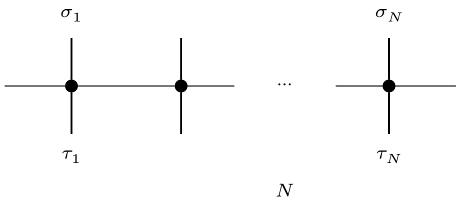
$$M^{u,v} := M^{i_1 j_1} M^{i_2 j_2} \cdots M^{i_N j_N} \in M_D(\mathbb{C}).$$

The empty pair of words evaluates to $\mathbb{1}_D$, and word pairs of different lengths evaluate to 0.

**Definition 3.4** (MPO operator family)**.** The operator generated by $M$ on $N$ sites is the matrix $\rho^{(N)}(M)$ indexed by configurations $\sigma, \tau \in \{0, \ldots, d-1\}^N$ and given by

$$\rho^{(N)}(M)_{\sigma,\tau} = \mathrm{tr}(M^{\sigma,\tau}).$$

Diagrammatically,




each black node denotes the same local tensor $M$, and the matrix entry $\rho^{(N)}(M)_{\sigma,\tau}$ is obtained by closing the outer virtual legs cyclically and taking the resulting trace.

**Definition 3.5** (Hermitian MPO tensor)**.** An MPO tensor is *Hermitian* if

$$M^{ij} = (M^{ji})^\dagger \qquad \text{for all } i, j \in \{0, \dots, d-1\}.$$

## 3.2 Transfer maps

**Definition 3.6** (MPO transfer map)**.** The *transfer map* associated to $M$ is the linear map $\mathcal{E}_M : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$\mathcal{E}_M(X) = \sum_{i,j=0}^{d-1} M^{ij} X (M^{ij})^\dagger.$$

**Lemma 3.7** (Identification with the doubled-index MPS transfer map)**.** *The transfer map of an MPO tensor agrees with the transfer map of its doubled-index MPS view.*

*Proof.* After identifying the single physical index of the doubled tensor with a pair $(i, j)$, the single sum defining the MPS transfer map is exactly the double sum defining $\mathcal{E}_M$. □

**Theorem 3.8** (Positivity of the MPO transfer map)**.** *If $X \geq 0$, then $\mathcal{E}_M(X) \geq 0$.*

*Proof.* By Lemma 3.7, $\mathcal{E}_M$ is the transfer map of the doubled-index MPS tensor. The corresponding MPS transfer map is positive, so $\mathcal{E}_M(X) \geq 0$ whenever $X \geq 0$. □

## 3.3 MPDOs and LPDOs

**Definition 3.9** (MPDO)**.** An MPO tensor is an *MPDO* if

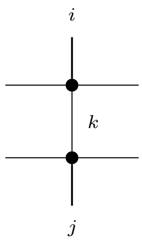
$$\rho^{(N)}(M) \geq 0 \qquad \text{for every } N \in \mathbb{N}.$$

This is the global positivity condition of [CPGSV17a, §4.1].

**Definition 3.10** (LPDO)**.** An MPO tensor is an *LPDO* if there exist an ancilla dimension $d_K$ and matrices $A^{(i,k)} \in M_D(\mathbb{C})$ such that

$$M^{ij} = \sum_{k=0}^{d_K-1} A^{(i,k)} (A^{(j,k)})^\dagger \qquad \text{for all } i, j.$$

This is the local purification form of [CPGSV17a, §4.3]. Diagrammatically, the local LPDO structure is




where the upper node denotes $A^{(i,k)}$, the lower node denotes $(A^{(j,k)})^\dagger$, the ancilla leg $k$ is contracted vertically, and the left and right outer legs are the MPO virtual legs.

**Theorem 3.11** (LPDOs are Hermitian)**.** *Every LPDO tensor is Hermitian.*

*Proof.* Write $M^{ij} = \sum_k A^{(i,k)}(A^{(j,k)})^\dagger$. Then

$$(M^{ji})^\dagger = \sum_k \left(A^{(j,k)}(A^{(i,k)})^\dagger\right)^\dagger = \sum_k A^{(i,k)}(A^{(j,k)})^\dagger = M^{ij}.$$

□

# Chapter 4

# The Single-Block Fundamental Theorem

This chapter proves that if an injective MPS tensor $A$ generates the same MPV family as another tensor $B$ of the same bond dimension, then $A$ and $B$ are gauge equivalent. The proof is purely algebraic and does not use the later overlap or normal-form reductions. The trace-pairing maps introduced below are auxiliary devices attached to the chosen generating families $\{A^i\}$ and $\{B^i\}$; they serve only to construct the linear map that is later shown to be a gauge transform.

## 4.1 Trace pairing

**Theorem 4.1** (Nondegeneracy of the trace pairing)**.** *For $M \in M_D(\mathbb{C})$, $\operatorname{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

*Proof.* Testing against the matrix units $E_{ji}$ (the matrix with 1 in position $(j, i)$ and 0 elsewhere) gives $\operatorname{tr}(E_{ji}M) = M_{ij}$. Hence if $\operatorname{tr}(MN) = 0$ for all $N$, then every entry of $M$ vanishes. □

**Lemma 4.2** (Same MPV implies trace agreement)**.** *If $A$ and $B$ generate the same MPV family, then $\operatorname{tr}(A^w) = \operatorname{tr}(B^w)$ for every word $w$.*

*Proof.* Viewing the word $w$ as a configuration of length $|w|$, Definition 2.4 gives the corresponding MPV coefficient as the trace coefficient $c_w(A) = \operatorname{tr}(A^w)$. The same-MPV hypothesis gives equality at every system size. □

**Definition 4.3** (Trace pairing map)**.** For a tensor $A$, attach to the chosen family $\{A^i\}$ the auxiliary linear map $\Phi_A : M_D(\mathbb{C}) \to \mathbb{C}^d$ defined by

$$\Phi_A(M)_i = \operatorname{tr}(M \cdot A^i).$$

**Theorem 4.4** (Injectivity of the trace pairing map)**.** *If $A$ is injective, then* $\ker \Phi_A = \{0\}$.

*Proof.* If $\Phi_A(M) = 0$, then $\operatorname{tr}(M \cdot A^i) = 0$ for all $i$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, this gives $\operatorname{tr}(MN) = 0$ for all $N$, so $M = 0$ by Theorem 4.1. □

**Theorem 4.5** (Nonvanishing trace on injective tensors)**.** *If $D \geq 1$, $A$ is injective, $A$ and $B$ generate the same MPV family, and $B^i = 0$ for all $i$, then we have a contradiction.*

*Proof.* If $B^i = 0$ for all $i$, then Lemma 4.2 gives $\mathrm{tr}(A^w) = 0$ for every word $w$, in particular for all length-1 words. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, the trace functional vanishes identically, contradicting $\mathrm{tr}(\mathbb{1}_D) = D \neq 0$. □

## 4.2 Linear extension

**Theorem 4.6** (Existence and uniqueness of the linear extension)**.** *If $A$ is injective and $A$, $B$ generate the same MPV family, then there exists a unique linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ satisfying $T(A^i) = B^i$ for all $i$.*

*Proof.* Write $\ell_A : \mathbb{C}^d \to M_D(\mathbb{C})$ for the map that sends a coefficient vector $c$ to $\sum_i c_i A^i$ (linear combination with the $A^i$). Applying Lemma 4.2 to all words of length 2 shows $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$. Since $A$ is injective, $\ell_A$ is surjective, giving $\mathrm{range}(\Phi_A) \subseteq \mathrm{range}(\Phi_B)$. By Theorem 4.4, $\ker \Phi_A = \{0\}$, so rank–nullity gives $\dim(\mathrm{range}(\Phi_A)) = \dim(M_D(\mathbb{C}))$. Hence $\dim(\mathrm{range}(\Phi_B)) \geq \dim(M_D(\mathbb{C}))$, and rank–nullity again gives $\ker \Phi_B = \{0\}$. A left inverse $g$ of $\Phi_B$ gives $T := g \circ \Phi_A$, and uniqueness follows because the $\{A^i\}$ span $M_D(\mathbb{C})$. □

**Theorem 4.7** (Multiplicativity of the linear extension)**.** *If $A$ is injective and $A$, $B$ generate the same MPV family, and $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is the unique linear map satisfying $T(A^i) = B^i$ for all $i$, then $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

*Proof.* **Step 1** ($\Phi_B \circ T = \Phi_A$). Applying Lemma 4.2 to words of length 2 gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ for all $i$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $\Phi_B \circ T = \Phi_A$.

**Step 2** ($T(A^iA^j) = B^iB^j$). Lemma 4.2 applied to the word $(i, j, k)$ gives $\mathrm{tr}(A^iA^jA^k) = \mathrm{tr}(B^iB^jB^k)$ for all $i, j, k$. So $\Phi_B(T(A^iA^j))_k = \Phi_A(A^iA^j)_k = \mathrm{tr}(A^iA^jA^k) = \mathrm{tr}(B^iB^jB^k) = \Phi_B(B^iB^j)_k$. The same rank–nullity argument used in the proof of Theorem 4.6 gives $\ker \Phi_B = \{0\}$. Hence $T(A^iA^j) = B^iB^j$.

**Step 3** (bilinear extension). For fixed $j$, the maps $M \mapsto T(M \cdot A^j)$ and $M \mapsto T(M) \cdot B^j$ agree on the spanning set $\{A^i\}$, hence on all of $M_D(\mathbb{C})$. Applying this once more with $j$ replaced by a general matrix gives full multiplicativity. □

## 4.3 Simplicity and Skolem–Noether

**Theorem 4.8** (Simplicity of $M_D(\mathbb{C})$)**.** *A nonzero multiplicative $\mathbb{C}$-linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

*Proof.* The kernel of a multiplicative linear map is a two-sided ideal. Since $M_D(\mathbb{C})$ is a simple ring (it has no nontrivial two-sided ideals), the kernel is either $\{0\}$ or $M_D(\mathbb{C})$. The latter would force the map to be zero, contradicting the hypothesis. Hence the map is injective, and surjectivity follows from finite-dimensionality. □

**Theorem 4.9** (Skolem–Noether)**.** *Every $\mathbb{C}$-algebra automorphism of $M_D(\mathbb{C})$ is inner: for any $f \in \mathrm{Aut}_{\mathbb{C}\text{-}alg}(M_D(\mathbb{C}))$, there exists $X \in \mathrm{GL}_D(\mathbb{C})$ such that $f(M) = XMX^{-1}$ for all $M$.*

*Proof.* Via the canonical identification $M_D(\mathbb{C}) \cong \mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$, the automorphism $f$ induces an automorphism of $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. Every algebra automorphism of $\mathrm{End}(V)$ is conjugation by an invertible linear map (this is a standard result in the theory of central simple algebras). Translating back to matrices gives the invertible $X$. □

**Lemma 4.10** (Promotion to algebra homomorphism)**.** *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a multiplicative surjective $\mathbb{C}$-linear map, then $T(\mathbb{1}) = \mathbb{1}$. Equivalently, the same underlying function defines a $\mathbb{C}$-algebra homomorphism.*

*Proof.* By surjectivity, there exists $X$ with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X) \cdot T(\mathbb{1}) = T(X \cdot \mathbb{1}) = T(X) = \mathbb{1}$, using multiplicativity and $T(X) = \mathbb{1}$. With $T(\mathbb{1}) = \mathbb{1}$ established, $T$ preserves the unit and hence is a $\mathbb{C}$-algebra homomorphism. □

## 4.4 Proof of the single-block theorem

**Theorem 4.11** (Single-block Fundamental Theorem)**.** *Let $A$ be an injective MPV tensor and let $B$ be an MPV tensor of the same bond dimension. If $A$ and $B$ generate the same MPV family, then they are gauge equivalent: there exists $X \in \mathrm{GL}_D(\mathbb{C})$ such that $B^i = XA^iX^{-1}$ for all $i$.*

*Proof.* By Theorem 4.6, there is a unique linear $T$ with $T(A^i) = B^i$. By Theorem 4.7, $T$ is multiplicative. $T$ is nonzero: if $T = 0$ then $B^i = 0$ for all $i$, contradicting Theorem 4.5. By Theorem 4.8, $T$ is bijective. By Lemma 4.10, $T$ is a $\mathbb{C}$-algebra automorphism. Theorem 4.9 gives an invertible $X$ with $T(M) = XMX^{-1}$, and evaluating at $M = A^i$ yields $B^i = XA^iX^{-1}$. □

## 4.5 Alternative fixed-length non-TI setup

The proof above treats a single translation-invariant tensor and equality of the full MPV family for all system sizes. The alternative algebraic route of [MGRPG+18] instead keeps a fixed periodic chain length and allows the local tensors to vary from site to site. The first step is therefore a separate bookkeeping layer for finite non-translation-invariant chains.

**Definition 4.12** (Periodic non-TI chain data)**.** Fix a chain length $n$. A *non-translation-invariant periodic MPS chain* consists of sitewise tensors

$$A_k = \{A_k^i\}_{i=0}^{d-1}, \qquad k = 0, \dots, n-1,$$

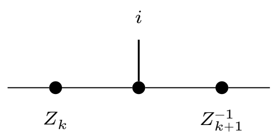
where each $A_k^i \in M_D(\mathbb{C})$. For a basis configuration $\sigma = (\sigma_0, \dots, \sigma_{n-1})$, its coefficient is

$$c_{A,\sigma} := \mathrm{tr}\big(A_0^{\sigma_0} A_1^{\sigma_1} \cdots A_{n-1}^{\sigma_{n-1}}\big).$$

Two such chains generate the *same chain state* if these coefficients agree for every $\sigma$. The chain is *injective* if each local tensor $A_k$ is injective in the usual single-site sense. Finally, two chains are *cyclically gauge equivalent* if there are invertible bond matrices $Z_0, \dots, Z_{n-1}$ such that

$$B_k^i = Z_k A_k^i Z_{k+1}^{-1}$$

for every site $k$ and physical index $i$, with indices understood modulo $n$ so that $Z_n = Z_0$. Locally, this gauge relation is represented by




where the black node denotes the site tensor named in the surrounding formula and the two side nodes mark the bond gauges on the neighboring virtual legs.

**Lemma 4.13** (Cyclic gauge equivalence is an equivalence relation)**.** *On fixed-length periodic chains, cyclic gauge equivalence is reflexive, symmetric, and transitive.*

*Proof.* Reflexivity uses the identity gauge on every bond. Symmetry inverts each bond gauge, and transitivity multiplies the gauges pointwise along the chain. □

# Chapter 5

# Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

## 5.1 Positive maps and channels

**Definition 5.1** (Positive map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that $X$ is positive semidefinite.

*Remark* 5.2. We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

**Definition 5.3** (Trace-preserving map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *trace-preserving* if $\operatorname{tr}(E(X)) = \operatorname{tr}(X)$ for all $X$.

**Definition 5.4** (Completely positive map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger \quad \text{for all } X \in M_D(\mathbb{C}).$$

We use the Kraus characterisation because it matches the later Kadison–Schwarz arguments. By Choi's theorem, this is equivalent to $E \otimes \mathrm{id}_n$ being positive for all $n$.

**Theorem 5.5** (CP maps are positive)**.** *Every completely positive map is positive.*

*Proof.* If $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ (conjugation preserves positive semidefiniteness), so the sum is PSD. □

**Definition 5.6** (Quantum channel)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

**Definition 5.7** (Density matrices)**.** The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \operatorname{tr}(\rho) = 1\}.$$

**Theorem 5.8** (Density matrices are compact)**.** *$\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).*

*Proof.* The PSD cone is closed (preimage of the closed nonnegative cone under continuous quadratic forms), and if $\rho \geq 0$ with $\mathrm{tr}(\rho) = 1$ then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are nonnegative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. □

**Theorem 5.9** (Density matrices are convex)**.** *$\mathcal{D}_D$ is convex.*

*Proof.* Convex combination of PSD matrices is PSD, and trace is linear. □

**Theorem 5.10** (Density matrices are nonempty)**.** *For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.*

*Proof.* $\frac{1}{D}\mathbb{1}_D$ is a density matrix. □

**Theorem 5.11** (Channels preserve density matrices)**.** *If $E$ is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.*

*Proof.* Complete positivity implies positivity (Theorem 5.5), so $E(\rho) \geq 0$; trace preservation gives $\mathrm{tr}(E(\rho)) = \mathrm{tr}(\rho) = 1$. □

## 5.2 Irreducibility

**Definition 5.12** (Irreducible map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *irreducible* if whenever $P$ is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Thm. 6.2, condition (1)]. The definition applies to any linear map; complete positivity is not required.

**Theorem 5.13** (Injectivity implies irreducibility)**.** *If $A$ is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.*

*Proof.* If $P$ is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^iP = 0$ for all $i$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all $M$, so $P = 0$ or $P = \mathbb{1}$. □

*Remark* 5.14 (Transfer map is CP). The transfer map $\mathcal{E}_A(X) = \sum_i A^iX(A^i)^\dagger$ is completely positive: it is already written in Kraus form, with Kraus operators $\{A^i\}$.

**Theorem 5.15** (Transfer map is a channel)**.** *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^iX(A^i)^\dagger$ is a quantum channel (CPTP).*

*Proof.* Complete positivity is the observation of Remark 5.14. Trace preservation: $\mathrm{tr}(\mathcal{E}_A(X)) = \sum_i \mathrm{tr}(A^iX(A^i)^\dagger) = \mathrm{tr}((\sum_i (A^i)^\dagger A^i)X) = \mathrm{tr}(X)$ by cyclicity and the normalization hypothesis. □

## 5.3 Cesàro fixed-point theorem

**Definition 5.16** (Cesàro mean)**.** The *Cesàro mean* of a linear map $E$ at order $N$, applied to a starting matrix $\rho_0$, is

$$\sigma_N = \frac{1}{N}\sum_{n=0}^{N-1} E^n(\rho_0).$$

**Theorem 5.17** (Cesàro telescope)**.** $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$.

*Proof.* Telescoping: $E(\sigma_N) = \frac{1}{N}\sum_{n=1}^{N} E^n(\rho_0)$ and subtracting $\sigma_N$ cancels all but the first and last terms. □

**Theorem 5.18** (Hermitian fixed-point decomposition)**.** *Let $E$ be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_\pm \geq 0$, $\mathrm{tr}(P_+P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Prop. 6.8].*

*Proof.* Let $Q$ be the support projection of $P_+$. Trace preservation and positivity give $\mathrm{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_\pm) = P_\pm$. □

**Theorem 5.19** (Cesàro fixed-point theorem)**.** *Every quantum channel $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.*

*Proof.* Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 5.10). Each Cesàro mean $\sigma_N$ lies in $\mathcal{D}_D$ (by convexity and channel preservation). By compactness (Theorem 5.8), extract a convergent subsequence $\sigma_{N_k} \to \rho$. The telescope identity (Theorem 5.17) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N}\|E^N(\rho_0) - \rho_0\| \to 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Thm. 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. □

## 5.4 Fixed-point projection

**Definition 5.20** (Fixed-point projection)**.** Given a linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a fixed point $\rho$ with $E(\rho) = \rho$ and $\mathrm{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho.$$

This projects onto the span of $\rho$ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

**Theorem 5.21** (Power decomposition)**.** *If $E$ is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,*

$$E^n = P + (E - P)^n,$$

*where $P$ is the fixed-point projection.*

*Proof.* $P$ is idempotent ($P^2 = P$), the fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, leaving $E^n = P^n + (E - P)^n = P + (E - P)^n$. □

## 5.5 Peripheral spectrum and primitivity

**Definition 5.22** (Peripheral spectrum)**.** The *peripheral spectrum* of a bounded linear operator $T$ is the set of spectral values $\mu$ with $|\mu|$ equal to the spectral radius of $T$.

**Definition 5.23** (Peripheral eigenvalues)**.** The *peripheral eigenvalues* of a linear map $E$ on a finite-dimensional space are the eigenvalues $\lambda$ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Prop. 6.1], so these coincide with the eigenvalues of maximal modulus.

**Note:** The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1. For a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

**Theorem 5.24** (1 is a peripheral eigenvalue)**.** *If $E$ has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then* 1 *is a peripheral eigenvalue of $E$.*

*Proof.* $\rho$ is an eigenvector with eigenvalue 1, and $|1| = 1$. ☐

*Remark* 5.25 (Peripheral eigenvalues lie on the unit circle)*.* Every peripheral eigenvalue $\lambda$ satisfies $|\lambda| = 1$. This is simply the defining condition in Definition 5.23.

**Theorem 5.26** (Finiteness of peripheral eigenvalues)**.** *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

*Proof.* Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. ☐

**Theorem 5.27** (Power-stable peripheral eigenvalues are roots of unity)**.** *Let $E$ be a linear endomorphism on a finite-dimensional space. If $\mu$ is a peripheral eigenvalue of $E$ and $\mu^n$ is an eigenvalue of $E$ for every $n \geq 1$, then $\mu$ is a root of unity.*

*Proof.* Since $\mu^n$ is an eigenvalue for every $n$, the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. ☐

**Theorem 5.28** (Peripheral powers imply root of unity)**.** *If the set of peripheral eigenvalues of a linear endomorphism $E$ on a finite-dimensional space is* closed under powers *($\mu \in \mathrm{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \mathrm{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

*Proof.* The closure hypothesis provides $\mu^n \in \mathrm{peripheral}(E)$ for all $n$, which in particular means $\mu^n$ is an eigenvalue. Theorem 5.27 then gives $\mu^p = 1$ for some $p > 0$. ☐

### 5.5.1 Periodicity removal by powering

*Remark* 5.29*.* The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its $p$-blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

**Lemma 5.30** (Common exponent for a finite set of roots of unity)**.** *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

*Proof.* Take $p$ to be the least common multiple of the exponents $p_\mu$. ☐

**Lemma 5.31** (Powering collapses peripheral eigenvalues)**.** *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a linear map, and assume that $E$ has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue $\mu$ of $E$ satisfies $\mu^p = 1$, then* $\mathrm{peripheral}(E^p) = \{1\}$.

*Proof.* The fixed point $\rho$ gives $1 \in \mathrm{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \mathrm{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \sigma(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \mathrm{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. ☐

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

*Remark* 5.32 (Bi-canonical special case). When a Kraus map happens to be both unital and trace-preserving (the "bi-canonical" case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 6.13) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 5.28. This is the simplest route, but it requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

### 5.5.2 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV17a, Appendix A].

**Lemma 5.33** (Peripheral closure via adjoint fixed point). *Let* $E(X) = \sum_i K_i X K_i^\dagger$ *be a Kraus map that is unital (*$\sum_i K_i K_i^\dagger = \mathbb{1}$*), and assume:*

1. *the adjoint Kraus map* $E^*(X) = \sum_i K_i^\dagger X K_i$ *has a positive definite fixed point* $\rho > 0$*, and*
2. $E$ *is irreducible.*

*Then* $\mathrm{peripheral}(E)$ *is closed under powers: if* $\mu \in \mathrm{peripheral}(E)$ *then* $\mu^n \in \mathrm{peripheral}(E)$ *for every* $n \in \mathbb{N}$*.*

*Proof.* Let $E(X) = \mu X$ with $|\mu| = 1$, and set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By Kadison–Schwarz, $G \geq 0$. Since $E^*(\rho) = \rho$,

$$\mathrm{tr}(\rho G) = \mathrm{tr}(\rho E(X^\dagger X)) - \mathrm{tr}(\rho E(X)^\dagger E(X)) = \mathrm{tr}(E^*(\rho) X^\dagger X) - \mathrm{tr}(\rho E(X)^\dagger E(X)) = \mathrm{tr}(\rho X^\dagger X) - |\mu|^2\,\mathrm{tr}(\rho X^\dagger X) = 0.$$

Thus the trace of the Kadison–Schwarz gap against the adjoint fixed point vanishes. Because $\rho > 0$ and $G \geq 0$, this forces $G = 0$, so $E(X^\dagger X) = E(X)^\dagger E(X)$. Hence $X^\dagger X$ is a PSD fixed point, which is positive definite by irreducibility (Theorem 7.3). So $X$ is invertible. By Theorem 6.12 (KS equality implies Kraus commutation), $X$ commutes with each Kraus operator; iterating via the left multiplicative identity (Theorem 6.14) gives $E(X^n) = \mu^n X^n$, so $\mu^n$ is peripheral. ☐

**Theorem 5.34** (Roots of unity via adjoint fixed point). *Let* $E(X) = \sum_i K_i X K_i^\dagger$ *be a Kraus map that is unital, and assume that the adjoint Kraus map* $E^*(X) = \sum_i K_i^\dagger X K_i$ *has a positive definite fixed point and that* $E$ *is irreducible. Then every peripheral eigenvalue of* $E$ *is a root of unity.*

*Proof.* Combine the closure-under-powers result (Lemma 5.33) with Theorem 5.28. ☐

*Remark* 5.35. Lemma 5.33 and Theorem 5.34 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV17a, Appendix A], which uses a faithful invariant state rather than bi-canonicality.

**Definition 5.36** (Channel period). The *period* of a channel $E$ is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\mathrm{peripheral}(E)|$.

**Definition 5.37** (Primitive map)**.** A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\text{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Thm. 6.7]. This definition applies to any linear endomorphism, not only to channels.

**Theorem 5.38** (Primitivity equals period one)**.** *A linear map with a nonzero fixed point is primitive if and only if its period is* 1*.*

*Proof.* If $\text{peripheral}(E) = \{1\}$, then $|\text{peripheral}(E)| = 1$. Conversely, if $|\text{peripheral}(E)| = 1$, then since $1 \in \text{peripheral}(E)$ (Theorem 5.24), the only element is 1. □

**Theorem 5.39** (Primitive maps have a spectral gap on the complement)**.** *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be trace-preserving, and let $\rho \neq 0$ satisfy $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$. Let*

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)}\,\rho$$

*be the fixed-point projection associated to $\rho$. Assume that:*

1. *$E$ is primitive;*
2. *every eigenvalue $\mu$ of $E$ satisfies $|\mu| \leq 1$;*
3. *whenever $E(X) = X$ and $\text{tr}(X) = 0$, one has $X = 0$.*

*Then every eigenvalue $\nu$ of $E - P$ satisfies $|\nu| < 1$.*

*Proof.* Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then

$$E(X) = \nu X + P(X).$$

If $\text{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence $\nu$ is an eigenvalue of $E$ with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so $X$ is a trace-zero fixed point of $E$; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\text{tr}(X) \neq 0$, trace preservation and $\text{tr}(P(X)) = \text{tr}(X)$ give

$$\text{tr}(X) = \text{tr}(E(X)) = \nu\,\text{tr}(X) + \text{tr}(P(X)) = \nu\,\text{tr}(X) + \text{tr}(X),$$

so $\nu\,\text{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. □

## 5.6 Representations: infrastructure

The following infrastructure supports the Choi–Jamiolkowski isomorphism (Theorem 5.44) and the representation theorems of [Wol12, Ch. 2].

**Definition 5.40** (Partial traces)**.** Let $X \in M_{d \cdot d'}(\mathbb{C})$ be a bipartite matrix indexed by $(\text{Fin}\, d \times \text{Fin}\, d')^2$. The *left partial trace* $\text{tr}_A(X)$ and *right partial trace* $\text{tr}_B(X)$ are the $d' \times d'$ and $d \times d$ matrices defined by

$$(\text{tr}_A(X))_{ij} = \sum_k X_{(k,i),(k,j)}, \qquad (\text{tr}_B(X))_{ij} = \sum_k X_{(i,k),(j,k)}.$$

**Definition 5.41** (Maximally entangled state)**.** The *maximally entangled state* on $\mathbb{C}^d \otimes \mathbb{C}^d$ is the rank-one projector

$$|\Omega\rangle\langle\Omega|, \quad |\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j,j\rangle.$$

Concretely, $(|\Omega\rangle\langle\Omega|)_{(i_1,i_2),(j_1,j_2)} = \frac{1}{d}\,\delta_{i_1 j_1}\delta_{i_2 j_2}$ and $\text{tr}(|\Omega\rangle\langle\Omega|) = 1$.

**Definition 5.42** (Tensor product of a map with identity)**.** Given a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the *tensor extension* $T \otimes \mathrm{id}$ acts on bipartite matrices $X \in M_{D\times D}(\mathbb{C})$ by applying $T$ to each "slice":

$$(T \otimes \mathrm{id})(X)_{(i_1,i_2),(j_1,j_2)} = (T(X^{(i_2,j_2)}))_{i_1 j_1},$$

where $X^{(i_2,j_2)}_{ab} = X_{(a,i_2),(b,j_2)}$ is the bipartite slice.

## 5.7 Choi–Jamiołkowski isomorphism (Wolf Prop. 2.1)

**Definition 5.43** (Choi matrix)**.** The *Choi matrix* of a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is

$$\tau = (T \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \in M_{D\times D}(\mathbb{C}).$$

Concretely, $\tau_{(i_1,i_2),(j_1,j_2)} = \frac{1}{D}(T(E_{i_2 j_2}))_{i_1 j_1}$ where $E_{i_2 j_2}$ is the matrix unit. This is the Choi–Jamiolkowski convention of [Wol12, Prop. 2.1].

**Theorem 5.44** (Prop. 2.1: CP iff Choi matrix is PSD)**.** *A linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is completely positive (in the Kraus sense) if and only if its Choi matrix $\tau \geq 0$.*

*Proof.* ($\Rightarrow$) If $T(X) = \sum_i K_i X K_i^\dagger$, then $\tau = \sum_i |v_i\rangle\langle v_i|$ where $(v_i)_{(a,b)} = c\, K_i(a,b)$ and $c = 1/\sqrt{D}$. This is a sum of rank-one PSD matrices.

($\Leftarrow$) If $\tau \geq 0$, write $\tau = \sum_m |v_m\rangle\langle v_m|$ by spectral decomposition, define $K_m(a,b) = v_m(a,b)/c$, and recover $T(X) = \sum_m K_m X K_m^\dagger$ by linearity on matrix units. □

**Theorem 5.45** (Prop. 2.1: TP implies partial trace condition)**.** *If $T$ is trace-preserving, then* $\mathrm{tr}_A(\tau) = \frac{1}{D}\mathbb{1}_D$.

*Proof.* $(\mathrm{tr}_A(\tau))_{ij} = \mathrm{tr}(T(\Omega^{(ij)})) = \mathrm{tr}(\Omega^{(ij)})$ by trace preservation, where $\Omega^{(ij)} = \frac{1}{D}E_{ij}$ is the $(i,j)$-slice of $|\Omega\rangle\langle\Omega|$. Taking the trace gives $\frac{1}{D}\delta_{ij}$. □

**Theorem 5.46** (Prop. 2.1: Hermiticity correspondence)**.** *The Choi matrix $\tau$ is Hermitian if and only if $T(B^\dagger) = T(B)^\dagger$ for all $B \in M_D(\mathbb{C})$.*

*Proof.* The Choi matrix entries satisfy $\tau_{(a,i),(b,j)} = \frac{1}{D}(T(E_{ij}))_{ab}$. Hermiticity of $\tau$ says $\overline{\tau_{(b,j),(a,i)}} = \tau_{(a,i),(b,j)}$, which unravels to $T(E_{ji})_{ba} = \overline{T(E_{ij})_{ab}}$, i.e., $T(E_{ij}^\dagger) = T(E_{ij})^\dagger$. By linearity this extends to all $B$. □

**Theorem 5.47** (Prop. 2.1: Trace of Choi matrix for TP map)**.** *If $T$ is trace-preserving, then* $\mathrm{tr}(\tau) = 1$.

*Proof.* $\mathrm{tr}(\tau) = \mathrm{tr}(\mathrm{tr}_A(\tau)) = \mathrm{tr}(\frac{1}{D}\mathbb{1}_D) = 1$. □

## 5.8 Kraus representation theorem (Wolf Thm. 2.1)

**Theorem 5.48** (Thm. 2.1: TP implies Kraus normalization)**.** *If $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving, then $\sum_i K_i^\dagger K_i = \mathbb{1}$.*

*Proof.* For any $N$, $\mathrm{tr}((\sum_i K_i^\dagger K_i)\, N) = \sum_i \mathrm{tr}(K_i^\dagger K_i N) = \sum_i \mathrm{tr}(K_i N K_i^\dagger) = \mathrm{tr}(T(N)) = \mathrm{tr}(N)$. Non-degeneracy of the trace pairing forces $\sum_i K_i^\dagger K_i = \mathbb{1}$. □

**Theorem 5.49** (Thm. 2.1: Kraus normalization implies TP). *If $\sum_i K_i^\dagger K_i = \mathbb{1}$, then $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving.*

*Proof.* $\operatorname{tr}(T(X)) = \sum_i \operatorname{tr}(K_i X K_i^\dagger) = \sum_i \operatorname{tr}(K_i^\dagger K_i X) = \operatorname{tr}((\sum_i K_i^\dagger K_i)\, X) = \operatorname{tr}(X)$. □

**Theorem 5.50** (Thm. 2.1: Unitary freedom in Kraus operators). *Let $U$ be a unitary $r \times r$ matrix ($U^\dagger U = \mathbb{1}$) and suppose $K_j = \sum_\ell U_{j\ell}\, \tilde{K}_\ell$. Then $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same Kraus map:*

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger \quad \forall\, X \in M_D(\mathbb{C}).$$

*Proof.* Substitute the combination formula, expand the triple sum over $j, \ell, \ell'$, swap summation order, and use the entry-wise form of $U^\dagger U = \mathbb{1}$ to collapse to $\ell = \ell'$ diagonal terms. □

**Theorem 5.51** (Thm. 2.1: Orthonormal Kraus transition is unitary). *Let $\{K_j\}_{j=0}^{r-1}$ and $\{K'_j\}_{j=0}^{r-1}$ be two Hilbert–Schmidt orthonormal Kraus families, and suppose*

$$K_j = \sum_{\ell=0}^{r-1} U_{j\ell}\, K'_\ell.$$

*Then the transition matrix $U$ is unitary: $U^\dagger U = \mathbb{1}_r$.*

*Proof.* Expand $\operatorname{tr}(K_j^\dagger K_i)$ using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity $UU^\dagger = \mathbb{1}_r$, hence also $U^\dagger U = \mathbb{1}_r$. □

**Theorem 5.52** (Thm. 2.1: Dual map equality from primal map equality). *If two Kraus families satisfy $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X$, then $\sum_\alpha B_\alpha^\dagger Y B_\alpha = \sum_j A_j^\dagger Y A_j$ for all $Y$.*

*Proof.* Use the trace pairing: for all $X$, $\operatorname{tr}(X^\dagger(\sum B_\alpha^\dagger Y B_\alpha)) = \operatorname{tr}((\sum B_\alpha X^\dagger B_\alpha^\dagger)Y) = \operatorname{tr}((\sum A_j X^\dagger A_j^\dagger)Y) = \operatorname{tr}(X^\dagger(\sum A_j^\dagger Y A_j))$. Nondegeneracy of the trace pairing gives the result. □

**Theorem 5.53** (Thm. 2.1: Equal Stinespring Gramians). *If two Kraus families define the same CPM, then $\sum_\alpha B_\alpha^\dagger B_\alpha = \sum_j A_j^\dagger A_j$.*

*Proof.* Specialise `kraus_dual_eq_of_map_eq` at $Y = \mathbb{1}$. □

**Theorem 5.54** (Thm. 2.1 item 4: Rectangular Kraus freedom (necessary direction)). *If $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X$ and $r_2 \le r_1$, then there exists a rectangular isometry $V$ ($r_1 \times r_2$, $V^\dagger V = \mathbb{1}$) such that $B_\alpha = \sum_j V_{\alpha j}\, A_j$.*

*Proof.* Pad the smaller family to size $r_1$, establish Gram matrix equality $M_B^\dagger M_B = M_{A'}^\dagger M_{A'}$ from dual map equality, construct a partial isometry via inner product preservation on the range, and extend to a full isometry $V$ with $V^\dagger V = \mathbb{1}$. □

**Theorem 5.55** (Thm. 2.1 item 4: Rectangular Kraus freedom (general index variant)). *Variant of Theorem 5.54 with general finite index types $\iota_1, \iota_2$ in place of $\operatorname{Fin} r_1, \operatorname{Fin} r_2$.*

*Proof.* Reindex via equivalences to Fin and apply Theorem 5.54. □

## 5.9 Stinespring dilation (Wolf Thm. 2.2)

**Definition 5.56** (Stinespring isometry)**.** Given Kraus operators $\{K_j\}_{j=0}^{r-1}$ with $K_j \in M_D(\mathbb{C})$, the *Stinespring isometry* $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^r$ is the $(D \cdot r) \times D$ matrix

$$V_{(i,j),k} = (K_j)_{ik}.$$

This implements $V = \sum_j K_j \otimes |j\rangle$.

**Lemma 5.57** (Entry formula for the Stinespring isometry)**.** *For indices $i, k \in \{0, \ldots, D-1\}$ and $j \in \{0, \ldots, r-1\}$,*

$$V_{(i,j),k} = (K_j)_{ik}.$$

**Theorem 5.58** (Stinespring Gram identity)**.** *The Stinespring isometry satisfies*

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j.$$

**Theorem 5.59** (Thm. 2.2: Stinespring dual representation)**.** *For Kraus operators $\{K_j\}$ and $A \in M_D(\mathbb{C})$,*

$$V^\dagger (A \otimes \mathbb{1}_r) V = \sum_j K_j^\dagger A K_j = T^*(A).$$

*This is the Stinespring representation of the* dual *(Heisenberg picture) map $T^*$.*

*Proof.* Compute $(V^\dagger(A \otimes \mathbb{1}_r)V)_{ab}$ by summing over the product index $(i,j)$; the $\delta_{jl}$ from $\mathbb{1}_r$ collapses the sum over $l$, giving $\sum_j K_j^\dagger A K_j$. □

**Theorem 5.60** (Thm. 2.2: Isometry condition iff TP)**.** $V^\dagger V = \mathbb{1}_D$ *if and only if* $\sum_j K_j^\dagger K_j = \mathbb{1}_D$. *In particular, $V$ is an isometry precisely when the Kraus map is trace-preserving.*

*Proof.* $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}}\, V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}}\, (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}$. □

**Theorem 5.61** (Thm. 2.2: Stinespring Schrödinger representation)**.** *The Kraus map $T(\rho) = \sum_j K_j \rho K_j^\dagger$ equals the partial trace over the dilation space:*

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V\rho V^\dagger)_{(i,k),(j,k)} = \mathrm{tr}_r(V\rho V^\dagger).$$

*This is the Schrödinger-picture Stinespring representation.*

*Proof.* Expand $(V\rho V^\dagger)_{(i,k),(j,k)}$ and sum over $k$ to recover $\sum_k K_k \rho K_k^\dagger$. □

### 5.9.1 Trace-pairing expansion in transfer-matrix form

**Definition 5.62** (Trace-self-dual basis)**.** A basis $\{\sigma_i\}_i$ of $M_D(\mathbb{C})$ is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \mathrm{tr}(\sigma_i X)\, \sigma_i.$$

**Definition 5.63** (Trace-pairing coefficients)**.** For a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a trace-self-dual basis $\{\sigma_i\}$, define coefficients

$$t_{ij} := \operatorname{tr}(\sigma_i\, T(\sigma_j)).$$

**Theorem 5.64** (Trace-pairing expansion of a linear map)**.** *If $\{\sigma_i\}$ is trace-self-dual, then every linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits the expansion*

$$T(\rho) = \sum_i \sum_j t_{ij} \operatorname{tr}(\sigma_j \rho)\, \sigma_i, \qquad t_{ij} = \operatorname{tr}(\sigma_i T(\sigma_j)).$$

*Proof.* Expand $\rho$ and each $T(\sigma_j)$ in the chosen basis and substitute. The trace-self-dual identity replaces coordinates by traces, giving the stated double-sum formula. □

## 5.10 Determinant of a quantum channel (Wolf §6.1.1)

**Definition 5.65** (Channel determinant)**.** For a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the *channel determinant* $\det T$ is the determinant of the matrix of $T$ with respect to the standard matrix-unit basis of $M_D(\mathbb{C})$. This is the quantity studied in [Wol12, §6.1.1].

**Definition 5.66** (Unitary channel)**.** For a unitary matrix $U \in \mathcal{U}(D)$, the *unitary channel* is $T(\rho) = U\rho U^\dagger$.

**Theorem 5.67** (Thm. 6.1(1): Determinant bound for positive TP maps)**.** *For any positive trace-preserving map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, $|\det T| \le 1$. This is [Wol12, Thm. 6.1(1)].*

*Proof.* Every eigenvalue $\mu$ of $T$ satisfies $|\mu| \le 1$ (positivity and trace preservation force spectral radius $\le 1$). Since $\det T$ is the product of the eigenvalues (counted with algebraic multiplicity), the bound $|\det T| = \prod_i |\mu_i| \le 1$ follows by induction. □

**Theorem 5.68** (Thm. 6.1(2): Determinant one iff unitary channel)**.** *For a CPTP map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$,*

$$|\det T| = 1 \iff \exists\, U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger.$$

*This is [Wol12, Thm. 6.1(2)].*

*Proof.* Reverse direction: $|\det(U \cdot U^\dagger)| = |\det U|^{2d^2} = 1$. Forward direction: $|\det T| = 1$ and Kadison–Schwarz forces each Kraus operator $K_i$ to be a scalar multiple of a unitary; the trace-preservation constraint then pins $T = U \cdot U^\dagger$. □

## 5.11 Fixed-point algebra (Wolf Thms. 6.12–6.13)

**Definition 5.69** (Kraus fixed points)**.** The *fixed-point set* of a Kraus map $E(X) = \sum_i K_i X K_i^\dagger$ is $\operatorname{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$.

**Theorem 5.70** (Thm. 6.12: Fixed points form a $*$-subalgebra)**.** *Let $E$ be a unital Kraus map whose adjoint map $E^*$ has a positive definite fixed point $\rho > 0$. Then $\operatorname{Fix}(E)$ is a $*$-subalgebra of $M_D(\mathbb{C})$. This is [Wol12, Thm. 6.12].*

*Proof.* Closure under addition and scalar multiplication is immediate from linearity. Closure under †: if $E(X) = X$ then $E(X^\dagger) = X^\dagger$ because $E$ preserves Hermitian and skew-Hermitian parts. Closure under multiplication: if $E(X) = X$ and $E(Y) = Y$, the Kadison–Schwarz gap $E(X^\dagger X) - X^\dagger X$ is PSD and has zero weighted trace against $\rho > 0$, so it vanishes. This places $X$ in the multiplicative domain, and then $E(XY) = E(X)E(Y) = XY$. □

**Definition 5.71** (Adjoint fixed points)**.** The *adjoint fixed-point set* $\mathrm{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$, where $E^*(X) = \sum_i K_i^\dagger X K_i$.

**Theorem 5.72** (Thm. 6.12 in the Heisenberg picture)**.** *If the Kraus map is trace-preserving and has a positive definite fixed point $\rho > 0$ with $E(\rho) = \rho$, then $\mathrm{Fix}(E^*)$ is a $*$-subalgebra of $M_D(\mathbb{C})$.*

*Proof.* Apply Theorem 5.70 to the adjoint family $\{K_i^\dagger\}$. □

**Definition 5.73** (Kraus commutant)**.** The *Kraus commutant* is $\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0\ \forall i\}$. It is a $*$-subalgebra of $M_D(\mathbb{C})$.

**Theorem 5.74** (Thm. 6.13: Adjoint fixed points = Kraus commutant)**.** *Under the hypotheses of Theorem 5.72, the adjoint fixed-point $*$-subalgebra equals the Kraus commutant:*

$$\mathrm{Fix}(E^*) = \{K_i, K_i^\dagger\}'.$$

*This is [Wol12, Thm. 6.13].*

*Proof.* ($\supseteq$): if $X$ commutes with all $K_i$ and $K_i^\dagger$, then $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$ by trace preservation. ($\subseteq$): any $*$-subalgebra inside $\mathrm{Fix}(E^*)$ lies in the Kraus commutant, because for $X \in \mathrm{Fix}(E^*)$ with $X^\dagger X \in \mathrm{Fix}(E^*)$ the Kadison–Schwarz gap vanishes, which forces $XK_i = K_i X$ and $XK_i^\dagger = K_i^\dagger X$. □

## 5.12 Quantum dynamical semigroups

The one-parameter semigroup theory of Wolf Chapter 7 is treated in Chapter 14. That later chapter records the exponential representation of norm-continuous semigroups, perturbation bounds, and the GKSL/Lindblad description of generators. We do not duplicate that material here.

# Chapter 6

# Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory from Wolf's Chapter 5 [Wol12]. We begin with the Kadison–Schwarz inequality for Kraus maps, then describe the multiplicative domain and its algebraic structure. We also establish the extension to normal operators for positive subunital maps, the resulting order and spectral contractivity statements, and Wolf's standard example of a Schwarz map that is not completely positive.

## 6.1 Kadison–Schwarz inequality

A Kraus map is the concrete realisation of a completely positive map (Definition 5.4). The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map is a Kraus map by construction (Remark 5.14), so these results apply directly to transfer maps.

Definitions 6.1–6.4 below restate the Kraus-map notions from Chapter 5 for the reader's convenience.

**Definition 6.1** (Kraus map)**.** Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$. A linear map is completely positive (Definition 5.4) if and only if it can be written in this form.

**Definition 6.2** (Adjoint Kraus map)**.** The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor $A$, this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

**Definition 6.3** (Unital Kraus map)**.** A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$. In the later gauge language this is the right-canonical condition.

**Definition 6.4** (Trace-preserving Kraus map)**.** A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

**Theorem 6.5** (Kadison–Schwarz inequality). *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a* unital *Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{1}$). Then for all $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

*This is [Wol12, Eq. (5.2)].*

*Proof.* The $2 \times 2$ block matrix $\left[\begin{smallmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{smallmatrix}\right]$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{1}]^T$). Applying the unital CP map blockwise preserves positive semidefiniteness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2,2)$-block maps to $\mathbb{1}$). The Schur complement of the $(2,2)$-block $\mathbb{1}$ gives the result. ☐

**Theorem 6.6** (Kadison–Schwarz for the adjoint channel). *If $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.*

*Proof.* The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 6.5 to $\{K_i^\dagger\}$. ☐

**Theorem 6.7** (Hilbert–Schmidt contraction). *If a Kraus map is both unital and trace-preserving, then $\operatorname{tr}(E(X)^\dagger E(X)) \leq \operatorname{tr}(X^\dagger X)$ for all $X \in M_D(\mathbb{C})$.*

*Proof.* By Theorem 6.5, $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\operatorname{tr}(E(X^\dagger X)) \geq \operatorname{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\operatorname{tr}(E(X^\dagger X)) = \operatorname{tr}(X^\dagger X)$. ☐

**Lemma 6.8** (Trace pairing for Kraus map and adjoint map). *For all $X, Y \in M_D(\mathbb{C})$,*

$$\operatorname{tr}\big(X\, E(Y)\big) = \operatorname{tr}\big(E^*(X)\, Y\big).$$

**Lemma 6.9** (Positive-definite trace test). *If $A > 0$, $X \geq 0$, and $\operatorname{tr}(AX) = 0$, then $X = 0$.*

**Theorem 6.10** (Fixed-point peripheral eigenvectors satisfy KS equality). *Let $E$ be unital and let $E^*$ be trace-preserving with a positive definite fixed point $\rho > 0$. If $E(X) = \mu X$ with $|\mu| = 1$, then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

## 6.2 Multiplicative domain

**Theorem 6.11** (KS gap decomposition). *For a unital Kraus map $E$, the Kadison–Schwarz gap decomposes as*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

*where $R_i := XK_i^\dagger - K_i^\dagger E(X)$.*

*Proof.* Expand the right-hand side $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to identify the result with the KS gap. ☐

**Theorem 6.12** (KS equality implies Kraus commutation). *Let $E$ be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$ for some $X$, then $XK_i^\dagger = K_i^\dagger E(X)$ for all $i$.*

*Proof.* By Theorem 6.11, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. □

**Theorem 6.13** (KS equality for peripheral eigenvectors)**.** *Let $E$ be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then the KS gap vanishes: $E(X^\dagger X) = E(X)^\dagger E(X)$.*

*Proof.* The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 6.5. Its trace satisfies $\operatorname{tr}(G) = \operatorname{tr}(E(X^\dagger X)) - \operatorname{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\operatorname{tr}(E(X^\dagger X)) = \operatorname{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\operatorname{tr}(E(X)^\dagger E(X)) = |\mu|^2 \operatorname{tr}(X^\dagger X) = \operatorname{tr}(X^\dagger X)$. So $\operatorname{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. □

**Theorem 6.14** (Left multiplicative identity from KS equality)**.** *Let $E$ be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.*

*Proof.* From Theorem 6.12, $XK_i^\dagger = K_i^\dagger E(X)$ for all $i$. Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \end{aligned}$$

□

### 6.2.1 Full multiplicative-domain structure

**Definition 6.15** (Right and left multiplicative domains)**.** For a Kraus map $E$, define

$$\mathcal{A}_R(E) := \{X \in M_D(\mathbb{C}) : E(XY) = E(X)E(Y) \text{ for all } Y \in M_D(\mathbb{C})\}$$

and

$$\mathcal{A}_L(E) := \{X \in M_D(\mathbb{C}) : E(YX) = E(Y)E(X) \text{ for all } Y \in M_D(\mathbb{C})\}.$$

We follow the convention that $\mathcal{A}_R(E)$ controls right multiplication and $\mathcal{A}_L(E)$ controls left multiplication. These are the two one-sided multiplicative domains from [Wol12, Eqs. (5.11)–(5.12)].

**Definition 6.16** (Full multiplicative domain)**.** The *multiplicative domain* of $E$ is

$$\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E).$$

**Theorem 6.17** (Multiplicative-domain characterization)**.** *Let $E$ be a unital Kraus map. Then*

$$X \in \mathcal{A}_R(E) \iff E(XX^\dagger) = E(X)E(X)^\dagger$$

*and*

$$X \in \mathcal{A}_L(E) \iff E(X^\dagger X) = E(X)^\dagger E(X).$$

*Together these are the two one-sided characterizations from [Wol12, Thm. 5.7].*

*Proof.* If $X$ lies in a one-sided multiplicative domain, evaluate the defining identity at $Y = X^\dagger$. Conversely, the left-domain implication is exactly Theorem 6.14, and the right-domain implication follows by applying Theorem 6.14 to $X^\dagger$. □

**Theorem 6.18** (One-sided multiplicative domains are subalgebras)**.** *For a unital Kraus map $E$, both $\mathcal{A}_R(E)$ and $\mathcal{A}_L(E)$ are unital subalgebras of $M_D(\mathbb{C})$.*

*Proof.* Closure under addition follows from linearity of $E$. Closure under multiplication: if $X, Y \in \mathcal{A}_R(E)$, then $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$ for all $Z$, using Theorem 6.17 iteratively, and the $\mathcal{A}_L(E)$ case is analogous. Containment of $\mathbb{1}$ follows from unitality of $E$. □

**Theorem 6.19** (Multiplicative domain is a $*$-subalgebra)**.** *For a unital Kraus map $E$, the full multiplicative domain $\mathcal{A}(E)$ is a $*$-subalgebra of $M_D(\mathbb{C})$. This is the algebraic content of [Wol12, Thm. 5.7].*

*Proof.* If $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$, then $E(X^\dagger X) = E(X)^\dagger E(X)$ and $E(XX^\dagger) = E(X)E(X)^\dagger$. The characterization of Theorem 6.17 shows $X^\dagger \in \mathcal{A}(E)$. Combined with Theorem 6.18, this gives a $*$-subalgebra. □

**Theorem 6.20** (Restriction to multiplicative domain is a $*$-homomorphism)**.** *If $E$ is unital, then the restricted map $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \to M_D(\mathbb{C})$ is a $*$-algebra homomorphism. This is the homomorphism conclusion of [Wol12, Thm. 5.7].*

*Proof.* Multiplicativity on $\mathcal{A}(E)$ is immediate from the definition. For $X \in \mathcal{A}(E)$, the Kraus commutation relation from Theorem 6.12 gives $E(X^\dagger) = E(X)^\dagger$, so the restriction preserves adjoints. □

### 6.2.2 Schwarz inequality for normal operators

**Theorem 6.21** (CP Schwarz inequality for normal operators)**.** *Let $E^*(X) = \sum_i K_i^\dagger X K_i$ be the adjoint Kraus map of a trace-preserving Kraus family, and let $A \in M_D(\mathbb{C})$ be normal. Then the Schwarz gap*

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A)$$

*is positive semidefinite. Equivalently,*

$$E^*(A^\dagger)E^*(A) \leq E^*(A^\dagger A).$$

*This is the completely positive case of [Wol12, Prop. 5.1].*

*Proof.* The positivity statement is exactly Theorem 6.6. The second formulation is just the corresponding Loewner-order reformulation. □

**Theorem 6.22** (Prop. 5.1: Schwarz inequality for normal operators)**.** *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is normal, then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

*This is [Wol12, Prop. 5.1].*

*Proof.* Since $A$ is normal, diagonalize $A = UDU^\dagger$ with $D$ diagonal and $U$ unitary. Express $A^\dagger A = U|D|^2U^\dagger$ as a sum $\sum_{i,j} \bar{d}_i d_j P_i P_j$ where $P_i = Ue_ie_i^\dagger U^\dagger$. Each $P_i$ is rank-one PSD, and the images $T(P_i)$ pairwise commute because they are images of mutually orthogonal projectors under a positive map. The Schwarz gap then reduces to a block quadratic form in the images, which is shown nonneg by a simultaneous-diagonalization argument: positivity of $T$ ensures the scalar PSD matrix $(\langle v_k, T(P_iP_j)v_k\rangle)_{i,j}$ dominates the product $(\langle v_k, T(P_i)v_k\rangle\langle v_k, T(P_j)v_k\rangle)_{i,j}$, and combining over a common eigenbasis of the $T(P_i)$ yields the result. □

*Remark* 6.23. The formalized proof follows the positivity-on-abelian route of [Wol12, Prop. 1.6]: a positive map restricted to the commutative $*$-subalgebra generated by the spectral projectors of $A$ satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

### 6.2.3 Schwarz inequality for subnormal and commuting-dominant operators

**Theorem 6.24** (Thm. 5.5: Schwarz inequality for subnormal operators)**.** *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \le \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is subnormal (i.e. there exists a normal operator on a larger space whose compression to $\mathbb{C}^D$ is $A$), then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \ge 0.$$

*This is [Wol12, Thm. 5.5].*

*Proof.* Embed $A$ as the northwest corner of a normal operator $N$ on $\mathbb{C}^D \oplus \mathbb{C}^E$. Extend $T$ to a positive subunital map $\tilde{T}$ on the larger space (acting as $T$ on the northwest block, zero elsewhere). Apply Theorem 6.22 to $\tilde{T}$ and $N$, then compress the resulting inequality back to $\mathbb{C}^D$. □

**Theorem 6.25** (Thm. 5.6: Schwarz inequality for commuting-dominant operators)**.** *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \le \mathbb{1}$. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite* dominant *$B \ge 0$ with $A^\dagger A \le B$ and $B$ commuting with $A$ (in the sense $BA = AB$, $BA^\dagger = A^\dagger B$), then*

$$T(A^\dagger)T(A) \le T(B).$$

*This is [Wol12, Thm. 5.6].*

*Proof.* Let $B^{1/2}$ be the positive square root of $B$. Since $B$ commutes with $A$ and $A^\dagger$, the contraction $C = B^{-1/2}A$ (defined on the range of $B^{1/2}$) is subnormal. An $\varepsilon$-regularisation $B_\varepsilon = B + \varepsilon\mathbb{1}$ makes $B_\varepsilon > 0$ and the contraction $C_\varepsilon$ globally defined. Apply Theorem 6.24 to $C_\varepsilon$, multiply both sides by $T(B_\varepsilon)$, use positivity and monotonicity, and take $\varepsilon \to 0$. □

**Theorem 6.26** (CP variant: Kadison–Schwarz for commuting-dominant inputs)**.** *Let $E^*$ be the adjoint of a trace-preserving Kraus map. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant $B$ commuting with $A$, $A^\dagger$, and satisfying $A^\dagger A \le B$, then $E^*(A^\dagger)E^*(A) \le E^*(B)$.*

*Proof.* Apply Theorem 6.25 to the adjoint Kraus map (which is unital under the trace-preservation hypothesis). □

### 6.2.4 Positive maps preserve order and spectral intervals

**Theorem 6.27** (Positive maps are monotone)**.** *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is positive and $A \le B$, then $T(A) \le T(B)$.*

*Proof.* Since $B - A \ge 0$, positivity gives $T(B - A) \ge 0$. By linearity, this is $T(B) - T(A) \ge 0$. □

**Theorem 6.28** (Positive maps preserve adjoints)**.** *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is positive, then $T(A^\dagger) = T(A)^\dagger$ for all $A \in M_D(\mathbb{C})$.*

*Proof.* Write $A = B + iC$ with $B$ and $C$ Hermitian. Positive maps send Hermitian matrices to Hermitian matrices, so they preserve this decomposition and commute with taking adjoints. □

**Theorem 6.29** (Order intervals are preserved)**.** *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \le \mathbb{1}$. If $a \le 0 \le b$ and $a\mathbb{1} \le A \le b\mathbb{1}$, then*

$$a\mathbb{1} \le T(A) \le b\mathbb{1}.$$

*This is the matrix form of [Wol12, Eq. (5.21)].*

*Proof.* By monotonicity, $T(a\mathbb{1}) \le T(A) \le T(b\mathbb{1})$. For the lower bound, $T(a\mathbb{1}) = a\,T(\mathbb{1})$. Since $a \le 0$ and $T(\mathbb{1}) \le \mathbb{1}$, we have $a\,T(\mathbb{1}) \ge a\mathbb{1}$. For the upper bound, $T(b\mathbb{1}) = b\,T(\mathbb{1}) \le b\mathbb{1}$ since $b \ge 0$. Therefore $a\mathbb{1} \le T(A) \le b\mathbb{1}$. ☐

**Theorem 6.30** (Spectral interval contractivity). *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \le \mathbb{1}$, and let $A \in M_D(\mathbb{C})$ be Hermitian. If* $\operatorname{spec}(A) \subseteq [a, b]$ *with $a \le 0 \le b$, then*

$$\operatorname{spec}(T(A)) \subseteq [a, b].$$

*This is [Wol12, Eq. (5.21)].*

*Proof.* For Hermitian matrices, the inclusion $\operatorname{spec}(A) \subseteq [a, b]$ is equivalent to the order bounds $a\mathbb{1} \le A \le b\mathbb{1}$. Apply Theorem 6.29 and translate the resulting inequalities back into spectral bounds. ☐

### 6.2.5 Wolf Example 5.3

*Example* 6.31 (Wolf Example 5.3 map). Consider the linear map $T : M_2(\mathbb{C}) \to M_2(\mathbb{C})$ defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\operatorname{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Ex. 5.3].

**Theorem 6.32** (Wolf Example 5.3 is positive). *The map from Example 6.31 is positive.*

*Proof.* It is a nonnegative linear combination of the transpose map and the map $A \mapsto \operatorname{tr}(A)\mathbb{1}$, and both of these maps send positive semidefinite matrices to positive semidefinite matrices. ☐

**Theorem 6.33** (Wolf Example 5.3 satisfies the Schwarz inequality). *For every $A \in M_2(\mathbb{C})$,*

$$T(A^\dagger A) - T(A^\dagger)T(A) \ge 0,$$

*where $T$ is the map from Example 6.31.*

*Proof.* Write the Schwarz gap as $F(A)$. One checks that $F(A + \lambda\mathbb{1}) = F(A)$ for every scalar $\lambda$, so it is enough to treat the case $\operatorname{tr}(A) = 0$. In that case a direct $2 \times 2$ computation shows that $F(A)$ dominates $\frac{1}{2}(A^\dagger A)^T$, hence is positive semidefinite. ☐

**Theorem 6.34** (Wolf Example 5.3 is not completely positive). *The map from Example 6.31 is not completely positive.*

*Proof.* Compute the Choi matrix of $T$ and test it on the antisymmetric vector $|01\rangle - |10\rangle$. The resulting expectation value is negative, so the Choi matrix is not positive semidefinite. ☐

# Chapter 7

# Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory used throughout the blueprint: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, ergodicity of irreducible channels, the spectral-radius identification of Perron eigenvalues, and Wolf's spectral characterization of irreducibility. The conceptual source is Wolf's treatment of irreducible positive maps [Wol12, Ch. 6, especially Thms. 6.2–6.5 and Cor. 6.3] and [EHK78].

## 7.1 Positive definiteness

*Remark* 7.1. Wolf's general theory treats irreducible positive maps first ([Wol12, Thm. 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Thm. 6.3]). The injective-tensor version (Theorem 7.2) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 7.3) follows via the support-projection route of [Wol12, Thm. 6.2].

**Theorem 7.2** (PSD fixed point is positive definite)**.** *Let $A$ be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then $\rho$ is positive definite.*

*Proof.* Suppose $v^\dagger \rho v = 0$ for some $v \neq 0$. Since $\rho \geq 0$, this implies $\rho v = 0$. The fixed-point equation gives $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$. Each term is nonneg; if the sum is zero, then $\rho (A^i)^\dagger v = 0$ for all $i$: the kernel of $\rho$ is invariant under all $(A^i)^\dagger$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, we have $\rho M v = 0$ for every matrix $M$. Given any vector $w$, choose $M$ with $Mv = w$. Then $\rho w = 0$ for every $w$, so $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Thm. 6.3, item 2], the same conclusion is reached via the growth condition $(\mathbb{1} + T)^{d-1}(X) > 0$ of [Wol12, Thm. 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity. □

**Theorem 7.3** (PSD fixed point is PD — irreducible version)**.** *If $\mathcal{E}_A$ is irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point, then $\rho$ is positive definite.*

*Proof.* Suppose $\rho$ is not positive definite. Then $\rho$ has a nontrivial kernel; let $Q$ be the orthogonal projection onto the support of $\rho$ (the complement of the kernel). A direct computation using the fixed-point equation $\mathcal{E}_A(\rho) = \rho$ shows that $Q$ is an invariant projection for $\mathcal{E}_A$: it satisfies $Q\,\mathcal{E}_A(QXQ)\,Q = \mathcal{E}_A(QXQ)$ for all $X$. By irreducibility of $\mathcal{E}_A$ (Definition 5.12), every invariant projection is 0 or $\mathbb{1}$. If $Q = 0$ then $\rho = 0$, contradicting $\rho \neq 0$. If $Q = \mathbb{1}$ then $\rho$ is positive definite, contradicting our assumption. Hence $\rho$ must be positive definite. This is the support-projection direction of [Wol12, Thm. 6.2, condition (1)]. □

**Theorem 7.4** (Orthogonal trace condition)**.** *Let $E$ be an irreducible completely positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with $\mathrm{tr}(BA) = 0$. Then there exists $t$ with $1 \leq t \leq D-1$ such that*

$$\mathrm{tr}(B \cdot E^t(A)) > 0.$$

*Proof.* The growth theorem for irreducible CP maps ([Wol12, Thm. 6.2, item 2]) gives $(\mathbb{1} + E)^{D-1}(A) > 0$ (positive definite). Since $B \geq 0$ is nonzero, $\mathrm{tr}(B \cdot (\mathbb{1}+E)^{D-1}(A)) > 0$. Expanding by the binomial theorem:

$$\mathrm{tr}\Big(B \cdot \sum_{k=0}^{D-1} \tbinom{D-1}{k} E^k(A)\Big) > 0.$$

Each term $\mathrm{tr}(B \cdot E^k(A)) \geq 0$ since both factors are PSD. The $k = 0$ term vanishes by hypothesis ($\mathrm{tr}(BA) = 0$). Hence at least one $k \in \{1, \dots, D-1\}$ contributes a strictly positive term. This is [Wol12, Thm. 6.2, item 4]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument. □

## 7.2 Uniqueness

**Theorem 7.5** (Uniqueness of PSD fixed point)**.** *Let $A$ be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

*Proof.* Both $\rho$ and $\sigma$ are positive definite by Theorem 7.2. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1}\sigma S^{-1}$, which is Hermitian and positive definite. Let $c_0$ be the minimal eigenvalue of $H$. Then $\tau := \sigma - c_0\rho$ is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of $c_0$). By Theorem 7.2, $\tau$ must be zero, giving $\sigma = c_0\rho$. **Relation to Wolf:** In [Wol12, Thm. 6.3, item 2], uniqueness is proved via the growth condition of [Wol12, Thm. 6.2]. The same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 7.2) in place of the growth condition. □

*Remark* 7.6. The proof produces a positive real scalar $c_0 > 0$: it is the minimal eigenvalue of the positive definite Hermitian matrix $H = (S^\dagger)^{-1}\sigma S^{-1}$.

**Theorem 7.7** (Uniqueness of positive eigenvalue for irreducible CP maps)**.** *Let $D \geq 1$ and let $E$ be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

*Proof.* Extract Kraus operators $K$ for $E$. Irreducibility of $E$ passes to the adjoint transfer map $E^\dagger(\,\cdot\,) = \sum_i K_i^\dagger\,(\,\cdot\,)\,K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 7.15) together with the irreducible PSD-to-PosDef upgrade (Theorem 7.3) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\,\tau$. For any nonzero PSD $X$ with $E(X) = sX$, the trace pairing identity yields

$$s \cdot \mathrm{tr}(\tau X) = \mathrm{tr}(\tau \cdot E(X)) = \mathrm{tr}(E^\dagger(\tau) \cdot X) = t \cdot \mathrm{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\mathrm{tr}(\tau X) > 0$, so $s = t$. Applying this to both $(\rho, r_1)$ and $(\sigma, r_2)$ gives $r_1 = t = r_2$. **Relation to Wolf:** This is [Wol12, Thm. 6.3, item 3], establishing non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps. The proof follows Wolf's dual-map trace argument. □

## 7.3 Existence and the Perron–Frobenius theorem

**Theorem 7.8** (Existence of PSD fixed point)**.** *Assume $D \geq 1$. Let $A$ be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.*

*Proof.* Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 5.15). The Cesàro fixed-point theorem (Theorem 5.19) then provides the fixed point; injectivity is not needed for this step. □

**Theorem 7.9** (Quantum Perron–Frobenius)**.** *Assume $D \geq 1$. Let $A$ be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point $\rho$ (up to scaling), and $\rho$ is positive definite.*

*Proof.* Combine Theorems 7.8, 7.2, and 7.5. □

## 7.4 Right- and left-canonical gauges

**Theorem 7.10** (Right-canonical (unital) gauge)**.** *Let $\mathcal{E}_A(\rho) = \rho$ and let $S$ be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1}A^iS$. Then $\sum_i A'^i(A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization used later in the blueprint.*

*Proof.* Expand each summand: $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i \underbrace{SS^\dagger}_{\rho}(A^i)^\dagger(S^\dagger)^{-1}$. Summing over $i$ and using $\sum_i A^i\rho(A^i)^\dagger = \rho$ gives $S^{-1}\rho(S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$. □

**Theorem 7.11** (Left-canonical (trace-preserving) gauge)**.** *Let $\sigma$ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger\sigma A^i = \sigma$, and let $S$ be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization used later in the blueprint.*

*Proof.* Expand each summand: $(SA^iS^{-1})^\dagger(SA^iS^{-1}) = (S^\dagger)^{-1}(A^i)^\dagger \underbrace{S^\dagger S}_{\sigma} A^iS^{-1}$. Summing over $i$ and using $\sum_i (A^i)^\dagger\sigma A^i = \sigma$ gives $(S^\dagger)^{-1}\sigma S^{-1} = (S^\dagger)^{-1}S^\dagger SS^{-1} = \mathbb{1}$. □

*Remark* 7.12. The right-canonical gauge of Theorem 7.10 and the left-canonical gauge of Theorem 7.11 are generally different similarity transforms: one is built from a fixed point of $\mathcal{E}_A$, the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

## 7.5 Similarity preserves irreducibility

**Lemma 7.13** (Similarity transform preserves irreducibility)**.** *Let $E$ be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible ($\det C \neq 0$), and let $c > 0$. Define the similarity-transformed map*

$$E'(X) \;=\; c \cdot C^{-1}\, E(C\,X\,C^\dagger)\,(C^\dagger)^{-1}.$$

*Then $E'$ is also irreducible.*

*Proof.* Suppose $Q$ is an invariant projection for $E'$: $Q \neq 0$, $Q \neq \mathbb{1}$, and $(\mathbb{1} - Q)E'(QXQ)Q = 0$ for all $X$. Setting $R = CQC^\dagger$, the support projection $P$ of $R$ is an invariant projection for $E$. Irreducibility of $E$ forces $P = 0$ or $P = \mathbb{1}$. The invertibility of $C$ then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the full similarity version of [Wol12, Prop. 6.6]; the scalar case ($C = \mathbb{1}$) gives the scaling result stated next. □

**Theorem 7.14** (Scaling preserves irreducibility)**.** *If $E$ is an irreducible map and $c \neq 0$, then $cE$ is also irreducible. This is the scalar special case ($C = \mathbb{1}$) of [Wol12, Prop. 6.6].*

*Proof.* An invariant projection for $cE$ is an invariant projection for $E$ (scale out the nonzero constant from the commutation relation). □

## 7.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Thm. 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Thm. 6.3]. The TP-gauge application follows [CPGSV17a, Appendix A].

**Theorem 7.15** (PSD eigenvector for positive maps)**.** *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero PSD matrix $\sigma$. Then there exist a nonzero PSD matrix $\rho$ and a positive real $r > 0$ such that $E(\rho) = r\rho$.*

*This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Thm. 6.5]; see also [EHK78]). The proof applies Brouwer's fixed-point theorem to the normalization map $\rho \mapsto E(\rho)/\operatorname{tr}(E(\rho))$ on the compact convex set of density matrices.*

**Theorem 7.16** (Adjoint transfer map is nonzero)**.** *If some Kraus operator $A^i \neq 0$, then the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is nonzero.*

*Proof.* If $\mathcal{E}_A^\dagger = 0$ then $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$, so $\sum_i (A^i)^\dagger A^i = 0$. Since each summand $(A^i)^\dagger A^i$ is PSD, each $(A^i)^\dagger = 0$ and hence each $A^i = 0$. □

**Theorem 7.17** (PosDef adjoint eigenvector for irreducible tensors)**.** *Let $A$ be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix $\sigma$ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.*

*This combines the general PF existence ([Wol12, Thm. 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Thm. 6.3, item 2]); the application to the adjoint transfer map follows [CPGSV17a, Appendix A].*

*Proof.* By Theorem 7.16, $\mathcal{E}_A^\dagger \neq 0$. Since $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is a Kraus map, it is positive. Irreducibility of $A$ transfers to the adjoint transfer map (if $\mathcal{E}_A^\dagger$ had a nontrivial invariant projection $P$, then taking adjoints would show that $\mathbb{1} - P$ is invariant for $\mathcal{E}_A$, contradicting irreducibility of $\mathcal{E}_A$), and together with $\mathcal{E}_A^\dagger \neq 0$ this implies $\mathcal{E}_A^\dagger(\tau) \neq 0$ for every nonzero PSD matrix $\tau$. Hence Theorem 7.15 applies to $\mathcal{E}_A^\dagger$, giving $\sigma_0 \geq 0$ (PSD, nonzero) and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$. Rescale: set $T^i = r^{-1/2}(A^i)^\dagger$. Then $\mathcal{E}_T(\sigma_0) = \sigma_0$, and $\mathcal{E}_T$ is irreducible (by Theorem 7.14, since irreducibility of $\mathcal{E}_A$ transfers to the adjoint and is preserved under scaling). Since $\mathcal{E}_T$ is irreducible with a PSD fixed point, the PSD-to-PosDef upgrade (Theorem 7.3) gives $\sigma_0$ positive definite. □

**Theorem 7.18** (TP-gauge existence for irreducible tensors)**.** *Let $A$ be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real $r$, a positive definite matrix $\sigma$, and a gauge-equivalent tensor $B$ such that*

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

*is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.*

*This is the "spectral rescaling + TP gauge" step of [CPGSV17a, Appendix A]. The similarity is generally non-unitary.*

*Proof.* By Theorem 7.17, obtain $\sigma$ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A'^i = cA^i$. Then $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$. By the TP gauge construction (Theorem 7.11), $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$ satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. □

## 7.7 Exponential positivity for irreducible CP maps

**Theorem 7.19** (Exponential truncation is positive definite)**.** *Let $E$ be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then the finite exponential truncation*

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A)$$

*is positive definite. This is the finite-sum core of [Wol12, Thm. 6.2, item (3)].*

*Proof.* If a nonzero vector $v$ annihilated this quadratic form, then every term $E^k(A)$ with $0 \leq k \leq D-1$ would annihilate $v$. The growth theorem for irreducible CP maps (that is, $(\mathbb{1}+E)^{D-1}(A) > 0$ for irreducible $E$ and nonzero PSD $A$, as established in the proof of Theorem 7.4) then forces $(\mathbb{1} + E)^{D-1}(A)$ to annihilate $v$, contradicting its positive definiteness. □

**Theorem 7.20** (Exponential positivity for irreducible CP maps)**.** *Let $E$ be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then*

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A)$$

*is positive definite. This is [Wol12, Thm. 6.2, item (3)].*

*Proof.* By Theorem 7.19, the first $D$ terms already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail. □

## 7.8 Ergodicity of irreducible channels

**Theorem 7.21** (Unique density-matrix fixed point for an irreducible channel)**.** *Let $E$ be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$. Then there exists a unique density matrix $\sigma$ such that $E(\sigma) = \sigma$. Moreover $\sigma$ is positive definite. This is the qualitative form of [Wol12, Cor. 6.3].*

*Proof.* Every Cesàro mean of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges. Its limit is a fixed point by the telescoping argument, positive definiteness follows from irreducibility, and uniqueness comes from the uniqueness of positive semidefinite Perron eigenvectors at eigenvalue 1. □

**Theorem 7.22** (Cesàro convergence for irreducible channels)**.** *Let $E$ be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$, and let $\rho$ be any density matrix. Then the Cesàro means*

$$\frac{1}{N}\sum_{n=0}^{N-1} E^n(\rho)$$

*converge to the unique positive definite density-matrix fixed point of $E$. This is [Wol12, Cor. 6.3].*

*Proof.* Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 7.21, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit. □

## 7.9 Spectral radius at the Perron eigenvalue

**Theorem 7.23** (Spectral radius is the Perron eigenvalue)**.** *Let $E$ be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho > 0$ and $E(\rho) = r\rho$ with $r > 0$. Then the spectral radius of $E$ is $r$. This is [Wol12, Thm. 6.3, item (4)].*

*Proof.* Choose a positive definite eigenvector for the adjoint map with the same eigenvalue, gauge by its square root, and rescale by $r^{-1}$. The resulting map is trace-preserving and has a positive definite fixed point, so 1 is an eigenvalue. The spectral radius of a trace-preserving map $F$ satisfies $\rho(F) \leq 1$, since $\mathrm{tr}(F^n(X)) = \mathrm{tr}(X)$ for all $n$ bounds the growth of iterates. Hence its spectral radius is 1. Undoing the similarity and scalar rescaling recovers the spectral radius of $E$. □

**Theorem 7.24** (Real spectral-radius identity)**.** *Under the hypotheses of Theorem 7.23, the real-valued spectral radius of $E$ is also equal to $r$.*

*Proof.* This is the real-valued reformulation of Theorem 7.23. □

## 7.10 Spectral characterization of irreducibility

**Definition 7.25** (Wolf spectral properties)**.** A completely positive map $E$ on $M_D(\mathbb{C})$ has the *spectral properties* of [Wol12, Thm. 6.4] if there exist a positive real number $r$, a positive definite right eigenvector $\rho$, and a positive definite left eigenvector $\sigma$ for the adjoint map such that $E(\rho) = r\rho$, $E^\dagger(\sigma) = r\sigma$, every positive semidefinite right eigenvector for eigenvalue $r$ is a scalar multiple of $\rho$, and the spectral radius of $E$ is equal to $r$.

**Theorem 7.26** (Irreducibility gives the spectral properties)**.** *Every nonzero irreducible completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 7.25.*

*Proof.* Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 7.23. □

**Theorem 7.27** (Spectral properties imply irreducibility)**.** *If a completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 7.25, then it is irreducible.*

*Proof.* Gauge by the positive definite left eigenvector and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 7.25. □

**Theorem 7.28** (Irreducibility iff Wolf's spectral properties)**.** *Let $E$ be a nonzero completely positive map on $M_D(\mathbb{C})$. Then $E$ is irreducible if and only if it has the spectral properties of Definition 7.25. This is [Wol12, Thm. 6.4].*

*Proof.* Combine Theorems 7.26 and 7.27. $\square$

# Chapter 8

# Spectral Gap and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary spectral-gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

## 8.1 Mixed transfer operator

**Definition 8.1** (Mixed transfer operator)**.** The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors $A$ and $B$ of the same bond dimension $D$ is the linear map $F_{AB} : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

**Definition 8.2** (Rectangular mixed transfer operator)**.** For MPS tensors $A$ of bond dimension $D_1$ and $B$ of bond dimension $D_2$ (possibly $D_1 \neq D_2$), the *rectangular mixed transfer operator* is $F_{AB}^{\text{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \to M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\text{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

*Remark* 8.3 (Self-transfer). Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.

**Theorem 8.4** (Iterated mixed transfer)**.** *For all $X \in M_D(\mathbb{C})$ and $N \geq 0$,*

$$F_{AB}^N(X) = \sum_{\sigma \in \{0,\ldots,d-1\}^N} A^\sigma X (B^\sigma)^\dagger,$$

*where $A^\sigma = A^{\sigma_1} \cdots A^{\sigma_N}$ denotes the word evaluation (Definition 2.2).*

*Proof.* Induction on $N$: the base case is $F^0(X) = X$; the inductive step applies $F_{AB}$ to each summand and re-indexes via $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$. $\square$

**Theorem 8.5** (Matrix trace of iterated mixed transfer at the identity)**.** *For $N \geq 0$,*

$$\mathrm{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \mathrm{tr}(A^\sigma (B^\sigma)^\dagger).$$

*This is the matrix trace of $F_{AB}^N$ evaluated at the identity matrix. It should* not *be confused with the operator trace $\mathrm{Tr}(F_{AB}^N)$, which equals the MPV overlap (Theorem 8.7).*

*Proof.* Apply the matrix trace to each summand in Theorem 8.4 with $X = \mathbb{1}$: $\mathrm{tr}(F_{AB}^N(\mathbb{1})) = \sum_\sigma \mathrm{tr}(A^\sigma \cdot \mathbb{1} \cdot (B^\sigma)^\dagger) = \sum_\sigma \mathrm{tr}(A^\sigma (B^\sigma)^\dagger)$. $\square$

## 8.2 MPV overlap as transfer trace

Recall from Definition 2.34 that

$$O_{AB}(N) = \sum_{\sigma \in \{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

**Lemma 8.6** (Trace expansion over matrix units)**.** *For any linear endomorphism $T$ on $M_D(\mathbb{C})$, the operator trace can be expanded as*

$$\mathrm{Tr}(T) = \sum_{p,q=0}^{D-1} (T(E_{pq}))_{pq},$$

*where $E_{pq}$ is the matrix with $1$ in position $(p, q)$ and $0$ elsewhere.*

*Proof.* The operator trace $\mathrm{Tr}(T)$ equals the sum of $\mathrm{tr}(T(E_{pq}) \cdot E_{qp})$ over all $(p, q)$, and $\mathrm{tr}(M \cdot E_{qp}) = M_{pq}$. $\square$

**Theorem 8.7** (Overlap equals transfer trace)**.** *For tensors $A, B$ of bond dimension $D \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}(F_{AB}^N).$$

*Proof.* Expand the operator trace using Lemma 8.6 as a sum over matrix units $E_{pq}$. Apply Theorem 8.4 to evaluate $F_{AB}^N(E_{pq})$, extract the $(p, q)$-entry, and reassemble. The resulting double sum over $(p, q)$ and $\sigma$ equals $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N)$. $\square$

**Theorem 8.8** (Rectangular overlap as transfer trace)**.** *For tensors $A$ of bond dimension $D_1 \geq 1$ and $B$ of bond dimension $D_2 \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}((F_{AB}^{\mathrm{rect}})^N).$$

*Proof.* Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Theorem 8.7. $\square$

**Lemma 8.9** (Word trace as entrywise inner product)**.** *For a word $\sigma \in \{0, \dots, d-1\}^N$,*

$$\mathrm{tr}(A^\sigma (B^\sigma)^\dagger) = \sum_{j,k=0}^{D-1} (A^\sigma)_{jk} \overline{(B^\sigma)_{jk}}.$$

*Proof.* Expand the matrix trace and the $(j, j)$-entry of $A^\sigma (B^\sigma)^\dagger$. $\square$

## 8.3 Frobenius norm infrastructure

The spectral-gap proofs use the Frobenius (Hilbert–Schmidt) norm as the natural norm for the finite-dimensional matrix algebra. We write $\|\cdot\|_F$ for this norm; it coincides with, and we use it interchangeably with, the Hilbert–Schmidt norm $\|\cdot\|_{\mathrm{HS}}$ on finite-dimensional spaces. We record the squared version and an isometric embedding into Euclidean space.

**Definition 8.10** (Frobenius norm squared)**.** For a (possibly rectangular) matrix $X \in M_{m\times n}(\mathbb{C})$, the *Frobenius norm squared* is

$$\|X\|_F^2 := \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |X_{ij}|^2.$$

**Lemma 8.11** (Trace formula for Frobenius norm)**.** $\|X\|_F^2 = \operatorname{Re}\operatorname{tr}(X^\dagger X)$.

*Proof.* Expand the matrix product and trace entry by entry. ☐

**Definition 8.12** (Euclidean-space embedding)**.** The map $\mathrm{vec} : M_{m\times n}(\mathbb{C}) \to \mathbb{C}^{mn}$ that flattens matrix entries is an isometric embedding with respect to the Frobenius norm: $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

**Lemma 8.13** (Norm of embedded matrix)**.** $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

*Proof.* Expand both sides as sums over entries and use $|z|^2 = \bar{z}z$. ☐

## 8.4 Eigenvalue bound and spectral gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization. For the rigidity theorem, one gauges $A$ and $B$ separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument to a block embedding of a mixed-transfer eigenvector.

**Theorem 8.14** (Eigenvalue bound)**.** *Let $A, B$ be normalized MPS tensors. Every eigenvalue $\mu$ of $F_{AB}$ satisfies $|\mu| \le 1$.*

*Proof.* Expand $F_{AB}^n(X)$ as a sum over words and use Cauchy–Schwarz, together with $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$, to obtain a uniform Frobenius-norm bound $\|F_{AB}^n(X)\|_{\mathrm{HS}} \le C_D \|X\|_{\mathrm{HS}}$ for all $n$, where $C_D$ depends only on $D$. If $F_{AB}(X) = \mu X$ and $X \ne 0$, then $|\mu|^n \|X\|_{\mathrm{HS}} = \|F_{AB}^n(X)\|_{\mathrm{HS}} \le C_D \|X\|_{\mathrm{HS}}$ for every $n$. Hence $|\mu| \le 1$.

The argument uses only the trace-preserving normalization and does not appeal to Perron–Frobenius theory. ☐

**Theorem 8.15** (Spectral radius bound)**.** $\rho(F_{AB}) \le 1$ *for normalized tensors.*

*Proof.* The spectral radius is the supremum of $|\mu|$ over all eigenvalues, and each satisfies $|\mu| \le 1$ by Theorem 8.14. ☐

**Theorem 8.16** (Spectral radius $\ge 1$ implies gauge-phase equivalence)**.** *Let $A, B$ be injective normalized MPS tensors. If $\rho(F_{AB}) \ge 1$, then $A$ and $B$ are gauge-phase equivalent.*

*Proof.* Since $\rho(F_{AB}) \le 1$ (Theorem 15.58) and $\rho(F_{AB}) \ge 1$ by hypothesis, there exists an eigenvalue $\mu$ with $|\mu| = 1$ and eigenvector $X \ne 0$: $F_{AB}(X) = \mu X$. The proof proceeds in five steps.

*Step 1 (Separate gauging).* Choose positive definite fixed points $\rho_A, \rho_B$ of the transfer maps $\mathcal{E}_A, \mathcal{E}_B$ (Theorem 7.9) and gauge each tensor separately to a unital Kraus family (Theorem 7.10):

set $A'^i = \rho_A^{1/2} A^i \rho_A^{-1/2}$ and similarly for $B$. Let $X' = \rho_A^{1/2} X \rho_B^{-1/2}$ denote the correspondingly gauged eigenvector.

*Step 2 (Block Kraus family and off-diagonal embedding).* Form the block Kraus operators $C^i = \begin{pmatrix} A'^i & 0 \\ 0 & B'^i \end{pmatrix}$. These constitute a unital Kraus family on $M_{2D}(\mathbb{C})$. The off-diagonal matrix $Y = \begin{pmatrix} 0 & X' \\ 0 & 0 \end{pmatrix}$ satisfies $E_C(Y) = \mu Y$.

*Step 3 (KS equality and Kraus commutation).* Because $|\mu| = 1$ and $E_C$ is unital and trace-preserving, the Kadison–Schwarz equality for peripheral eigenvectors (Theorem 6.13) gives $E_C(Y^\dagger Y) = E_C(Y)^\dagger E_C(Y)$. By the Kraus commutation relation (Theorem 6.12), $Y$ commutes with every block Kraus operator: $C^i Y = Y C^i$ for all $i$.

*Step 4 (Intertwining identity and invertibility).* Expanding the commutation relation block by block gives $A'^i X' = X' B'^i$ for every $i$. Since $\{A'^i\}$ span $M_D(\mathbb{C})$ (injectivity of $A$), $X'$ is a nonzero intertwiner from $M_D(\mathbb{C})$ to $M_D(\mathbb{C})$. The left multiplicative identity (Theorem 6.14) gives $E_C(Y^\dagger Y) = E_C(Y^\dagger) E_C(Y)$, so $X'^\dagger X'$ is a positive semi-definite fixed point of $\mathcal{E}_{B'}$, positive definite by injectivity. Hence $X'$ is invertible.

*Step 5 (Skolem–Noether conclusion).* The map $M \mapsto X' M (X')^{-1}$ is a $\mathbb{C}$-algebra automorphism of $M_D(\mathbb{C})$ sending $B'^i$ to $A'^i$. By Skolem–Noether (Theorem 4.9) this is an inner automorphism. Undoing the separate gauges gives $B^i = \bar{\mu}\, X\, A^i\, X^{-1}$ for all $i$, establishing gauge-phase equivalence (Definition 2.16).

The crucial point is that one gauges the individual transfer maps to unital form; one does not require the mixed transfer map itself to be simultaneously unital and trace-preserving. □

**Theorem 8.17** (Strict spectral gap)**.** *Let $A, B$ be injective normalized MPS tensors that are* not *gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

*Proof.* $\rho(F_{AB}) \le 1$ by Theorem 15.58. If $\rho(F_{AB}) = 1$, Theorem 8.16 gives gauge-phase equivalence, contradicting the hypothesis. □

## 8.5 Spectral gap — rectangular case

**Theorem 8.18** (Rectangular spectral gap)**.** *Let $A$ be an injective normalized tensor of bond dimension $D_1$ and $B$ an injective normalized tensor of bond dimension $D_2$, with $D_1 \ne D_2$. Then $\rho(F_{AB}^{\mathrm{rect}}) < 1$.*

*Proof.* The same word-expansion argument as in Theorem 8.14 gives $|\mu| \le 1$ for every eigenvalue of $F_{AB}^{\mathrm{rect}}$. If $\rho(F_{AB}^{\mathrm{rect}}) = 1$, choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$.

Gauge $A$ and $B$ separately to unital Kraus families and apply the block-embedding Kadison–Schwarz argument as in the square case (Theorem 8.16, Steps 2–3). The Kraus commutation relation gives $A'^i X' = X' B'^i$ for all $i$. Now consider $\ker(X') \subseteq \mathbb{C}^{D_2}$. If $v \in \ker(X')$, then $X' B'^i v = A'^i X' v = 0$, so $B'^i v \in \ker(X')$ for every $i$. Since $B$ is injective, $\{B'^i\}$ spans $M_{D_2}(\mathbb{C})$, so $\ker(X')$ is invariant under every $D_2 \times D_2$ matrix and is therefore either $\{0\}$ or $\mathbb{C}^{D_2}$. Since $X' \ne 0$, we have $\ker(X') = \{0\}$, so $X'$ is injective as a linear map $\mathbb{C}^{D_2} \to \mathbb{C}^{D_1}$ and $D_2 \le D_1$. Applying the same argument to $X'^\dagger$ (using the commutation relation for the adjoint block family) gives $D_1 \le D_2$. Hence $D_1 = D_2$, contradicting the hypothesis. □

**Theorem 8.19** (Rectangular overlap decay)**.** *If $D_1 \ne D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

*Proof.* The spectral gap $\rho(F_{AB}^{\mathrm{rect}}) < 1$ (Theorem 8.18) implies $(F_{AB}^{\mathrm{rect}})^N \to 0$, so the operator trace converges to 0. By Theorem 8.8, $O_{AB}(N) = \mathrm{Tr}((F_{AB}^{\mathrm{rect}})^N) \to 0$. □

## 8.6 MPV overlap decay

**Theorem 8.20** (Transfer powers converge to zero)**.** *Let $A, B$ be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \to 0$ for every $X \in M_D(\mathbb{C})$.*

*Proof.* The spectral gap $\rho(F_{AB}) < 1$ (Theorem 8.17) implies $\|F_{AB}^n\|_{\mathrm{op}} \to 0$ by the Gelfand formula, so $F_{AB}^n(X) \to 0$ for every $X$. □

**Theorem 8.21** (Overlap decay)**.** *Let $A, B$ be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

*Proof.* By Theorem 8.7, $O_{AB}(N) = \mathrm{Tr}(F_{AB}^N)$. Expand the operator trace over matrix units (Lemma 8.6): $\mathrm{Tr}(F_{AB}^N) = \sum_{p,q}(F_{AB}^N(E_{pq}))_{pq}$. By Theorem 8.20, each $F_{AB}^N(E_{pq}) \to 0$ entry-wise. Since the sum is finite, $O_{AB}(N) \to 0$. □

## 8.7 Block separation

These asymptotic facts are the ingredients used later in the canonical-form separation arguments.

*Remark* 8.22 (Cross-correlation decay). If $A$ and $B$ are injective normalized tensors that are not gauge-phase equivalent, then for every $X \in M_D(\mathbb{C})$ one has $\mathrm{tr}(F_{AB}^N(X)) \to 0$ as $N \to \infty$. This is the scalar trace consequence of the operator convergence $F_{AB}^N(X) \to 0$ from Theorem 8.20.

*Remark* 8.23 (Self-correlation persists). If $\rho$ is a fixed point of $\mathcal{E}_A$, then $\mathrm{tr}(\mathcal{E}_A^N(\rho)) = \mathrm{tr}(\rho)$ for all $N \geq 0$. In particular this applies to the distinguished QPF fixed point from Theorem 7.9.

## 8.8 Spectral gap under irreducible-TP hypotheses

The preceding spectral-gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* combined with trace-preserving normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), following Cirac et al. [CPGSV17b]. The argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

**Theorem 8.24** (Gauge-equivalence from irreducible-TP spectral radius)**.** *Let $A, B$ be irreducible trace-preserving normalized MPS tensors of bond dimension $D$. If $\rho(F_{AB}) \geq 1$, then $A$ and $B$ are gauge-phase equivalent.*

*Proof.* Choose a modulus-one eigenvector as in Theorem 8.16. Instead of requiring injectivity to deduce invertibility of the intertwiner, one uses the Perron–Frobenius fixed points (Theorem 7.9), which are positive definite under irreducibility. The Kadison–Schwarz–Cauchy–Schwarz chain then yields a positive-definite intertwiner, and the Skolem–Noether argument concludes as before. □

**Theorem 8.25** (Strict spectral gap (irreducible-TP))**.** *Let $A, B$ be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

*Proof.* Contrapositive of Theorem 8.24, combined with $\rho(F_{AB}) \leq 1$ (Theorem 15.58). □

**Theorem 8.26** (Transfer power decay (irreducible-TP))**.** *Under the hypotheses of Theorem 8.25, $F_{AB}^n(X) \to 0$ for every $X \in M_D(\mathbb{C})$.*

*Proof.* By the Gelfand formula, $\rho(F_{AB}) < 1$ implies $\|F_{AB}^n\|_{\mathrm{op}} \to 0$. □

**Theorem 8.27** (Overlap decay (irreducible-TP)). *Let $A, B$ be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

*Proof.* Combine the spectral gap (Theorem 8.25) with the overlap–trace identity (Theorem 8.7). □

**Theorem 8.28** (Rectangular spectral gap (irreducible-TP)). *Let $A$ (bond dimension $D_1$) and $B$ (bond dimension $D_2$) be irreducible trace-preserving normalized tensors with $D_1 \neq D_2$. Then $\rho(F_{AB}^{\mathrm{rect}}) < 1$.*

*Proof.* Assume for contradiction that $\rho = 1$. Choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C})$. The Cauchy–Schwarz rigidity argument produces an intertwiner $X'$ with $A'^i X' = X' B'^i$ for all $i$. Since $\{B'^i\}$ spans $M_{D_2}(\mathbb{C})$ (irreducibility), the kernel of $X'$ is a $B'$-invariant subspace, forced to be $\{0\}$; similarly for $X'^\dagger$. This gives $D_1 = D_2$, contradicting the hypothesis. □

**Theorem 8.29** (Rectangular overlap decay (irreducible-TP)). *If $D_1 \neq D_2$ and both tensors are irreducible and trace-preserving normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

*Proof.* Combine Theorem 8.28 with Theorem 8.8. □

## 8.9 Conditional expectation from a faithful fixed point (Wolf Thm. 6.15)

**Definition 8.30** (Conditional expectation predicate). Given a $*$-subalgebra $S \subseteq M_D(\mathbb{C})$, a linear map $P : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a conditional expectation onto $S$ if it is idempotent, unital, has range contained in $S$, and satisfies $P(X) = X$ for all $X \in S$.

**Definition 8.31** (Scalar conditional expectation). For $\sigma \geq 0$ with $\mathrm{tr}(\sigma) \neq 0$, define

$$E_\sigma(X) := \frac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)} \mathbb{1}.$$

**Lemma 8.32** (Evaluation formula). $E_\sigma(X) = \dfrac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)} \mathbb{1}$.

**Theorem 8.33** (Unitality). *$E_\sigma(\mathbb{1}) = \mathbb{1}$.*

**Theorem 8.34** (Idempotence). *$E_\sigma(E_\sigma(X)) = E_\sigma(X)$ for all $X$.*

**Lemma 8.35** (Range is scalar). *For every $X$, there exists $c \in \mathbb{C}$ with $E_\sigma(X) = c\,\mathbb{1}$.*

**Lemma 8.36** (Fixes scalar operators). *For $c \in \mathbb{C}$, $E_\sigma(c\,\mathbb{1}) = c\,\mathbb{1}$.*

**Theorem 8.37** (Absorbs the adjoint map). *If $T(\sigma) = \sigma$, then $E_\sigma \circ T^* = E_\sigma$.*

**Theorem 8.38** (Adjoint map absorbs scalar projection). *If $T$ is trace-preserving, then $T^* \circ E_\sigma = E_\sigma$.*

**Lemma 8.39** (Image lies in adjoint fixed points). *For all $X$, $E_\sigma(X)$ is fixed by $T^*$.*

**Theorem 8.40** (Scalar map is a conditional expectation). *Let $K$ be trace-preserving, let $\rho > 0$ be a positive-definite fixed point of the associated channel $T(\rho) = \sum_i K_i \rho K_i^\dagger$ (so $T(\rho) = \rho$), and assume every adjoint fixed point of $T^*$ is a scalar multiple of $\mathbb{1}$. Then $E_\rho$ is a conditional expectation onto the adjoint fixed-point $*$-subalgebra.*

## 8.10 Stationary support (Wolf §6.4)

This section records the channel-side support-projection infrastructure behind stationary states, following [Wol12, §6.4]. The main formalized statement is `Channel.support_proj_fixed`.

**Theorem 8.41** (Support projection invariance (Wolf Lemma 6.4))**.** *Let $E$ be a quantum channel and $\rho \geq 0$ a positive semidefinite fixed point ($E(\rho) = \rho$). Let $P$ denote the support projection of $\rho$. Then*

$$P\,E(PXP)\,P = E(PXP)$$

*for all $X \in M_D(\mathbb{C})$.*

*Proof.* Write $E$ in Kraus form $E(X) = \sum_i K_i X K_i^\dagger$. From $E(\rho) = \rho$ and $\rho \geq 0$ one deduces $(\mathbb{1} - P)K_i P = 0$ for all $i$, i.e. each Kraus operator maps the range of $P$ into itself. The identity then follows by expanding the Kraus sum and using $K_i P = PK_i P$. ☐

**Definition 8.42** (Stationary state)**.** For an irreducible channel $E$ on $M_D(\mathbb{C})$ (with $D \geq 1$), the *stationary state* is the unique density-matrix fixed point $\rho_\infty$ satisfying $E(\rho_\infty) = \rho_\infty$, $\rho_\infty > 0$, and $\mathrm{tr}(\rho_\infty) = 1$. Uniqueness follows from the quantum Perron–Frobenius theorem (Theorem 7.9).

**Definition 8.43** (Stationary support)**.** The *stationary support* of an irreducible channel $E$ is the support projection of the stationary state (Definition 8.42).

**Theorem 8.44** (Prop. 6.9: stationary support is full)**.** *For an irreducible channel $E$, the stationary support equals the identity:* $\mathrm{supp}(\rho_\infty) = \mathbb{1}$.

*Proof.* The support projection $P$ of $\rho_\infty$ is an orthogonal projection satisfying $PE(PXP)P = E(PXP)$ for all $X$ (Theorem 8.41). By irreducibility, $P = 0$ or $P = \mathbb{1}$. Since $\rho_\infty \neq 0$, we have $P \neq 0$, so $P = \mathbb{1}$. ☐

*Remark* 8.45 (Props. 6.10–6.11: non-trivial formulations). The original Lean declarations for "irreducible iff support is full" and "stationary support minimality" were removed in PR #102 because their previous formulations were vacuous (they followed trivially from the irreducibility hypothesis alone). The correct non-trivial statements are:

- **Prop. 6.10** (converse of Prop. 6.9): if the support projection of *every* PSD fixed point is full, then the channel is irreducible.
- **Prop. 6.11**: minimality of the stationary support among invariant projections, without assuming irreducibility.

These remain TODO in the formalization.

## 8.11 Additional Wolf Chapter 6 declaration wrappers

This section records declaration wrappers used in cross-chapter references.

**Theorem 8.46** (Wolf Thm. 6.2(3): exponential positivity condition)**.** *Let $E$ be an irreducible completely positive map, $A \geq 0$ with $A \neq 0$, and $t > 0$. Then* $\exp(tE)(A)$ *is positive definite. Conversely, if* $\exp(tE)(A)$ *is positive definite for every such $t$, $A$, then $E$ is irreducible (full equivalence ).*

**Theorem 8.47** (Wolf Thm. 6.8: Kraus-span equivalence)**.** *Under normalization, paper primitivity is equivalent to eventual full Kraus rank.*

**Theorem 8.48** (Wolf Thm. 6.8: conjunction wrapper). *Under normalization, paper primitivity is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.*

*Remark* 8.49 (Wolf Thm. 6.8 pairwise equivalence declarations). These are the pairwise equivalence declarations referenced by the packaged Wolf Thm. 6.8 wrappers.

**Theorem 8.50** (Wolf Thm. 6.15 (scalar fixed-point algebra case)). *In the scalar fixed-point algebra setting, the weighted map $X \mapsto \frac{\mathrm{tr}(\rho X)}{\mathrm{tr}(\rho)}\mathbb{1}$ is a conditional expectation onto the adjoint fixed-point $*$-subalgebra.*

## 8.12 Peripheral eigenvalue group structure (Wolf Thm. 6.6)

**Theorem 8.51** (Peripheral eigenvalues: cyclic structure). *Let $E$ be an irreducible unital Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point (trace-preservation is* not *required). Then there exist $m \geq 1$ and a primitive $m$-th root of unity $\gamma$ such that the peripheral eigenvalues of $E$ are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

*Proof.* The peripheral eigenvalues form a finite subset of $\mathbb{C}$, contain 1, and are closed under multiplication and inversion, so they form a finite subgroup of $\mathbb{C}^\times$. By Mathlib's `subgroup_units_cyclic`, this subgroup is cyclic. The generator $\gamma$ has order $m = |\text{peripheral spectrum}|$. □

**Theorem 8.52** (Peripheral eigenvalues form a cyclic group). *Let $E$ be an irreducible unital trace-preserving Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. Then there exist $m \geq 1$ and a primitive $m$-th root of unity $\gamma$ such that $m \mid D$ and the peripheral eigenvalues of $E$ are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

*Proof.* Apply Theorem 8.51 for the cyclic structure. Divisibility $m \mid D$ follows from the cyclic decomposition: $m$ mutually orthogonal projections summing to $\mathbb{1}$ each have equal trace (by trace preservation), so $m \cdot \mathrm{tr}(P_0) = D$ with $\mathrm{tr}(P_0) \in \mathbb{N}$. □

**Theorem 8.53** (Period divides bond dimension). *Let $E$ be as in Theorem 8.52, and let $\gamma$ be a primitive $m$-th root of unity generating the peripheral eigenvalues. Then $m \mid D$.*

*Proof.* Follows from the cyclic projection decomposition and trace preservation, as in the final step of Theorem 8.52. □

## 8.13 Primitive overlap convergence (spectral-gap form)

The theorems in this section are stated directly in terms of the complementary map $E - P$. This is the analytic form needed for the later overlap argument. For primitive normalized transfer maps, Chapter 5 supplies this spectral-gap input under explicit hypotheses.

**Theorem 8.54** (Complementary spectral gap implies trace convergence to 1). *Let $E$ be trace-preserving with fixed point $\rho$ ($\mathrm{tr}(\rho) \neq 0$) and let $P$ be the fixed-point projection (Definition 5.20). If $\rho_{\mathrm{spec}}(E - P) < 1$, then $\mathrm{Tr}(E^n) \to 1$ as $n \to \infty$.*

*Proof.* By the power decomposition (Theorem 5.21), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\mathrm{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \to 0$, so $\mathrm{Tr}(E^n) \to \mathrm{Tr}(P)$. The operator trace of $P$ equals 1: since $P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$, we have $\mathrm{Tr}(P) = \frac{\mathrm{tr}(\rho)}{\mathrm{tr}(\rho)} = 1$. □

**Theorem 8.55** (Self-overlap convergence from a complementary spectral gap)**.** *Let $A$ be a normalized MPS tensor with a nonzero PSD fixed point $\rho$ of the transfer map. If $\rho_{\mathrm{spec}}(\mathcal{E}_A - P) < 1$ (where $P$ is the fixed-point projection), then $O_{AA}(N) \to 1$ as $N \to \infty$.*

*Proof.* By Theorem 8.7 (with $A = B$), $O_{AA}(N) = \mathrm{Tr}(\mathcal{E}_A^N)$. By Theorem 8.54, $\mathrm{Tr}(\mathcal{E}_A^N) \to 1$. $\square$

# Chapter 9

# Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension $D$, the cumulative span of matrix products stabilises after at most $D^2$ steps. The results follow [SPGWC10].

## 9.1 Cumulative span

**Definition 9.1** (Word span)**.** The *word span* at length $n$ is the subspace

$$S_n(A) = \operatorname{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, §II].

**Definition 9.2** (Cumulative span)**.** The *cumulative span* at level $n$ is

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A).$$

**Lemma 9.3** (Monotonicity of cumulative span)**.** *$T_n(A) \leq T_{n+1}(A)$ for all $n$.*

*Proof.* $T_{n+1}$ includes all words of length $\leq n+1$, which contains all words of length $\leq n$. ☐

**Lemma 9.4** (Dimension bound)**.** *$\dim T_n(A) \leq D^2$ for all $n$.*

*Proof.* $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$. ☐

**Theorem 9.5** (Cumulative span stabilisation)**.** *If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \geq n$.*

*Proof.* If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of $S_{n+1}$ by $A^i$ gives an element of $S_{n+2}$, so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m > n$, hence $T_m = T_n$. ☐

**Theorem 9.6** (Dimension growth)**.** *If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.*

*Proof.* $T_n \leq T_{n+1}$ (Lemma 9.3) and $T_n \neq T_{n+1}$ imply strict submodule inclusion, hence strict rank increase. ☐

**Lemma 9.7** (Word span characterises block injectivity)**.** *$S_L(A) = M_D(\mathbb{C})$ if and only if $A$ is $L$-block injective.*

*Proof.* Both conditions assert that the span of all products of length $L$ equals $M_D(\mathbb{C})$. ☐

## 9.2 Fitting decomposition

**Definition 9.8** (Fitting decomposition)**.** A *Fitting decomposition* of a linear endomorphism $f : V \to V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. $f$ is nilpotent on the generalized 0-eigenspace,
2. $f$ is invertible on each generalized $\mu$-eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span $V$,
4. the generalized eigenspaces are linearly independent.

**Theorem 9.9** (Fitting decomposition exists)**.** *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

*Proof.* Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. □

**Theorem 9.10** (Nilpotent power bound)**.** *If $f$ is nilpotent on a space of dimension $n$, then $f^n = 0$.*

*Proof.* A nilpotent endomorphism on an $n$-dimensional space has nilpotency index $\leq n$. □

## 9.3 Nonzero trace product

**Theorem 9.11** (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound)**.** *If $A$ is normal, then there exists $n \leq D^2$ with $T_n(A) = M_D(\mathbb{C})$.*

*Proof.* By Theorem 9.6, if $T_n \neq T_{n+1}$ then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ (Lemma 9.4), after at most $D^2$ steps either $T_n = M_D(\mathbb{C})$ or $T_n$ has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some $L_0$, hence $T_{L_0} = M_D(\mathbb{C})$). □

**Theorem 9.12** (Cumulative span reaches $M_D(\mathbb{C})$)**.** *If $A$ is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

*Proof.* Take $n$ from Theorem 9.11; since $n \leq D^2$ and $T$ is monotone, $T_{D^2} = M_D(\mathbb{C})$. □

**Theorem 9.13** (Nonzero trace product)**.** *If $A$ is normal, then there exists a word $w$ of length $\leq D^2$ such that $\operatorname{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1]. For the trace argument below, the nontrivial case is $D \geq 1$; when $D = 0$ the normality hypothesis is vacuous.*

*Proof.* By Theorem 9.12, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length $\leq D^2$. In the nontrivial case $D \geq 1$, at least one has nonzero trace (since $\operatorname{tr}(\mathbb{1}) = D \neq 0$). □

**Theorem 9.14** (Cumulative span reaches $M_D(\mathbb{C})$ — sharp bound)**.** *If $A$ is normal, then $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$, where $d' = \dim S_1(A) = \operatorname{kr}(A)$. This is the sharp form of [SPGWC10, Lemma 1]: the cumulative span already reaches $M_D(\mathbb{C})$ by index $D^2 - d' + 1 \leq D^2$.*

*Proof.* The key observation is that if $T_n(A) \neq T_{n+1}(A)$, the dimension strictly increases. Since $S_1(A) \subseteq T_1(A)$ and $\dim S_1(A) = d'$, we have $\dim T_1(A) \geq d'$. After at most $D^2 - d'$ additional strict-growth steps the dimension reaches $D^2 = \dim M_D(\mathbb{C})$. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality. Hence $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$. □

**Theorem 9.15** (Sharp nonzero trace product)**.** *If $A$ is normal, then there exists a word $w$ of length $\leq D^2 - \mathrm{kr}(A) + 1$ such that $\mathrm{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1] with the sharp length bound.*

*Proof.* By Theorem 9.14, the identity $\mathbb{1}$ lies in $T_{D^2-d'+1}(A)$, hence is a linear combination of word products of length $\leq D^2 - d' + 1$. At least one such product has nonzero trace. □

**Theorem 9.16** (Sharp nonzero trace product — positive length)**.** *If $A$ is normal and $D \geq 2$, then there exists a word $w$ of* positive *length $1 \leq |w| \leq D^2 - \mathrm{kr}(A) + 1$ such that $\mathrm{tr}(A^w) \neq 0$. This strengthens Theorem 9.15 by requiring $|w| \geq 1$, which is needed for the blocking argument in Theorem 9.28.*

*Proof.* Define the *positive-level* cumulative span $V_n = \mathrm{span}\{A^w : 1 \leq |w| \leq n\}$. For $D \geq 2$, the span $S_0(A) = \mathrm{span}\{\mathbb{1}\}$ has dimension 1, so $V_1 \supseteq S_1(A)$ has dimension $\geq d' \geq 1$. The same stabilization-or-strict-growth dichotomy shows $V_{D^2-d'+1} = M_D(\mathbb{C})$: if $V_n$ stabilises and is not $M_D(\mathbb{C})$, then $V_n$ is closed under left-multiplication by each $A^i$, so $S_N(A) \subseteq V_n \neq M_D(\mathbb{C})$ for all $N \geq 1$, contradicting normality. Since $V_{D^2-d'+1} = M_D(\mathbb{C})$, the identity $\mathbf{1}$ lies in $V_{D^2-d'+1}$ and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace (since $\mathrm{tr}(\mathbf{1}) = D \geq 2 \neq 0$). □

## 9.4 Eigenvector extraction

**Theorem 9.17** (Nonzero trace implies eigenvector)**.** *If $M \in M_D(\mathbb{C})$ has $\mathrm{tr}(M) \neq 0$, then $M$ has a nonzero eigenvalue $\mu \neq 0$ and a corresponding eigenvector $\varphi \neq 0$ with $M\varphi = \mu\varphi$.*

*Proof.* A matrix with nonzero trace has a nonzero eigenvalue (since trace = sum of eigenvalues counted with multiplicity). The Fitting decomposition (Theorem 9.9) ensures that the corresponding generalized eigenspace is nontrivial. On that generalized eigenspace, $M - \mu\mathbb{1}$ is nilpotent, so its kernel is nontrivial. Any nonzero vector in $\ker(M - \mu\mathbb{1})$ is therefore a genuine eigenvector with eigenvalue $\mu$. □

**Theorem 9.18** (Eigenvalue extraction)**.** *If $A$ is normal, then there exist a word $w_0$ with $|w_0| \leq D^2$, a nonzero scalar $\mu \neq 0$, and a nonzero vector $\varphi \neq 0$ such that $A^{w_0}\varphi = \mu\varphi$.*

*Proof.* By Theorem 9.13, there is a word $w_0$ with $\mathrm{tr}(A^{w_0}) \neq 0$ and $|w_0| \leq D^2$. By Theorem 9.17, $A^{w_0}$ has a nonzero eigenvector. □

## 9.5 Eigenvector spreading

**Definition 9.19** (Cumulative vector span)**.** Given an MPS tensor $A$ and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \mathrm{span}_{\mathbb{C}}\{A^w\varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

**Theorem 9.20** (Vector span stabilisation)**.** *If $K_n(A,\varphi) = K_{n+1}(A,\varphi)$, then $K_m(A,\varphi) = K_n(A,\varphi)$ for all $m \geq n$.*

*Proof.* Same argument as Theorem 9.5: left-multiplication by $A^i$ cannot escape the stabilised span. □

**Lemma 9.21** (Vector span from matrix span)**.** *If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $K_N(A,\varphi) = \mathbb{C}^D$.*

*Proof.* Every matrix $M \in M_D(\mathbb{C})$ is in $T_N(A)$, so in particular every rank-one matrix sending $\varphi$ to any target vector $v$ is in $T_N$. Applying such matrices to $\varphi$ recovers all of $\mathbb{C}^D$. □

**Theorem 9.22** (Eigenvector spreading)**.** *Let $A$ be a normal MPS tensor and let $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A,\varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].*

*Proof.* The vector span $K_n$ is monotone and $\dim K_n \leq D$. If $K_n = K_{n+1}$ for some $n < D-1$, the span stabilises (Theorem 9.20). But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 9.12, so Lemma 9.21 gives $K_{D^2}(A,\varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by $\geq 1$ at each step, reaching $D$ by step $D-1$. □

## 9.6 Proof of the cumulative Wielandt bound

**Definition 9.23** (Wielandt analysis)**.** A *Wielandt analysis* for a normal MPS tensor $A$ packages: a word $w_0$ of length $\leq D^2$, a nonzero eigenvalue $\mu$, and a nonzero eigenvector $\varphi$ of the matrix $A^{w_0}$, such that $A^{w_0}\varphi = \mu\varphi$.

**Theorem 9.24** (Wielandt analysis exists)**.** *Every normal MPS tensor admits a Wielandt analysis.*

*Proof.* Immediate from Theorem 9.18. □

**Theorem 9.25** (Wielandt chain)**.** *Every normal MPS tensor admits a Wielandt analysis consisting of data $(w_0, \mu, \varphi)$ with $K_{D-1}(A,\varphi) = \mathbb{C}^D$.*

*Proof.* Combine Theorems 9.24 and 9.22. □

**Theorem 9.26** (Wielandt bound)**.** *If $A$ is a normal MPS tensor with bond dimension $D$, then $T_{D^2}(A) = M_D(\mathbb{C})$. In particular, the cumulative span stabilises at index at most $D^2$. This is the cumulative-span part of [SPGWC10, Theorem 1]. The paper's theorem is stronger: via Lemma 2(b) it upgrades this to a fixed-length conclusion $S_N(A) = M_D(\mathbb{C})$, proved below in Theorem 9.40.*

*Proof.* Immediate from Theorem 9.12. □

*Remark* 9.27 (Explicit blocking-length estimates)*.* Theorem 9.26 gives the $D^2$ cumulative-span bound. The full-rank index bound $\iota(A) \leq (D^2 - d + 1) \cdot D^2$ (Theorem 1 case (1) of [SPGWC10]) is proved in Theorem 9.28 below. The sharp blocking-length estimates for injectivity ($D^4$) and bi-canonical form ($3D^5$) from [SPGWC10] are not proved here.

**Theorem 9.28** (Wielandt's inequality — general bound)**.** *Let $A$ be a normalized MPS tensor ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) that is primitive in the paper sense. Then*

$$\iota(A) \;\leq\; (D^2 - \mathrm{kr}(A) + 1) \cdot D^2,$$

*where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$ is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].*

*Proof.*

1. $D = 1$*:* In this case $\iota(A) = 0$, so the bound holds trivially.

2. $D \geq 2$*:* By Theorem 9.16, there exists a *positive-length* word $w$ with $|w| \leq D^2 - d' + 1$ and $\operatorname{tr}(A^w) \neq 0$, where $d' = \operatorname{kr}(A)$. The matrix $A^w$ has nonzero trace, hence a nonzero eigenvalue $\mu$ and eigenvector $\varphi$ (Theorem 9.17). Set $n = |w|$ and let $B = A^{[n]}$ be the blocked tensor. Then $B$ is normal (Theorem 9.35), and the encoding of $w$ as a single block index gives $B^{i_0} = A^w$.

   (a) If $A^w$ is invertible (invertible case), $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

   (b) If $A^w$ is non-invertible (non-invertible case), the eigenvector $\varphi$ spans a proper subspace, and the Wielandt chain argument gives $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

   In the invertible case, the invertible Kraus operator $B^{i_0}$ allows direct dimension growth, giving $S_{D^2 - k_B + 1}(B) = M_D(\mathbb{C})$ where $k_B = \operatorname{kr}(B)$. In the non-invertible case, the bound $S_{D^2}(B) = M_D(\mathbb{C})$ follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; Theorem 9.40 only gives the qualitative conclusion $\exists N$ : $S_N(B) = M_D(\mathbb{C})$ without this explicit bound. In both cases $\iota(A) \leq D^2 \cdot n \leq D^2 \cdot (D^2 - d' + 1)$.

□

## 9.7 Lemma 2(b): fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next stage is to pass to a span at one fixed length, namely to find $n_0$ such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices).

**Definition 9.29** (Fixed-length vector span)**.** The *fixed-length vector span* at length $n$ is

$$H_n(A, \varphi) = \operatorname{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(A)|\varphi\rangle$ in the notation of [SPGWC10].

**Lemma 9.30** (Word span products)**.** $S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$*: the product of word spans at lengths $m$ and $n$ is contained in the word span at length $m + n$.*

*Proof.* Every product $A^{w_1} A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1 w_2}$ with $|w_1 w_2| = m + n$. □

**Lemma 9.31** (Eigenvector padding)**.** *Suppose $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$. Then for all $n$,*

$$K_n(A, \varphi) = H_n(A, \varphi).$$

*In particular, if $K_n(A, \varphi) = \mathbb{C}^D$ then already $H_n(A, \varphi) = \mathbb{C}^D$.*

*Proof.* Any word $w$ with $|w| \leq n$ can be padded to exact length $n$ by appending $n - |w|$ copies of the index $i_0$. The eigenvector relation gives $A^{w \cdot i_0^k}\varphi = \mu^k \cdot A^w \varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of $K_n$ lies in $H_n$. The reverse inclusion $H_n \leq K_n$ is immediate. □

**Theorem 9.32** (Vector spanning to matrix spanning (assembly step))**.** *Assume:*

1. *$H_n(A, \varphi) = \mathbb{C}^D$ (length-$n$ word products applied to $\varphi$ span all of $\mathbb{C}^D$), and*
2. *for each standard basis vector $e_j$, the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.*

*Then $S_{n+m}(A) = M_D(\mathbb{C})$.*

*Proof.* Fix a matrix unit $E_{ij}$. By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i\varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and $E_{ij} \in S_{n+m}(A)$ by Lemma 9.30. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. $\square$

**Lemma 9.33** (Blocking transfer)**.** *If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$.*

*Proof.* Each length-$n$ word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length-$nL$ word in the original alphabet $\{0, \dots, d-1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. $\square$

*Remark* 9.34 (Relation with Lemma 2(b)). In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with an assembly argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 9.32 is the second step. Theorem 9.39 supplies the rank-one input in blocked form, and Theorem 9.40 then gives a blocked, fixed-length word-span saturation result.

## 9.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) infrastructure to the *blocked tensor* $A^{[L]}$ and develops the "rectangular span" used to produce rank-one elements.

**Theorem 9.35** (Blocking preserves normality)**.** *If $A$ is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.*

*Proof.* Choose $N$ with $S_N(A) = M_D(\mathbb{C})$. Then $\mathbb{1} \in S_N(A)$. For any $M \in S_N(A)$, write $\mathbb{1} = \sum_r c_r A^{w_r}$ with each $|w_r| = N$. Then

$$M = M \cdot \mathbb{1} = \sum_r c_r \, M A^{w_r} \in S_N(A) \cdot S_N(A) \subseteq S_{2N}(A)$$

by Lemma 9.30. Iterating this inclusion gives $S_N(A) \subseteq S_{kN}(A)$ for every $k \geq 1$; taking $k = L$ yields $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$. By Lemma 9.36, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. $\square$

**Lemma 9.36** (Chunking: word span embeds into blocked word span)**.** *$S_{nL}(A) \subseteq S_n(A^{[L]})$ for all $n, L$.*

*Proof.* Every length-$nL$ word in the original alphabet can be split into $n$ consecutive chunks of length $L$, each of which is a single blocked Kraus operator. By induction on $n$, using the word span product lemma (Lemma 9.30). $\square$

**Theorem 9.37** (Word eigenvector gives blocked eigenvector)**.** *If $A^{w_0}\varphi = \mu\varphi$ for a word $w_0$ of length $L$, then $(A^{[L]})^{j_0}\varphi = \mu\varphi$, where $j_0$ is the encoded blocked index corresponding to $w_0$.*

*Proof.* By definition, $(A^{[L]})^{j_0} = A^{w_0}$. □

**Theorem 9.38** (Wielandt blocked assembly)**.** *Suppose $A$ is normal and $L > 0$, and we have:*

1. *a word $w_0$ of length $L$ with eigenvector $A^{w_0}\varphi = \mu\varphi$ ($\mu \neq 0$, $\varphi \neq 0$),*
2. *a word $\tau_0$ of length $L$ with transpose eigenvector $(A^{\tau_0})^T\psi = \nu\psi$ ($\nu \neq 0$, $\psi \neq 0$),*
3. *a rank-one element $\varphi\psi^T \in S_m(A^{[L]})$ for some $m$.*

*Then $S_{((D-1)+m+(D-1))\cdot L}(A) = M_D(\mathbb{C})$.*

*Proof.* Apply the conditional Lemma 2(b) assembly (Theorem 9.32) to the blocked tensor $A^{[L]}$, which is normal by Theorem 9.35. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of $A^{[L]}$ by Theorem 9.37. Transfer back to the original tensor via Lemma 9.33. □

**Theorem 9.39** (Rank-one extraction for blocked tensor)**.** *Suppose $A$ is normal and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words $\sigma_0, \tau_0$ of length $N_0$, nonzero vectors $\varphi, \psi$, nonzero scalars $\mu, \nu$, and $m \in \mathbb{N}$ such that:*

1. *$A^{\sigma_0}\varphi = \mu\varphi$,*
2. *$(A^{\tau_0})^T\psi = \nu\psi$,*
3. *$\varphi\psi^T \in S_m(A^{[N_0]})$.*

*Proof.* Set $B = A^{[N_0]}$. Since $S_{N_0}(A) = M_D(\mathbb{C})$, we have $S_1(B) = M_D(\mathbb{C})$ by definition of the blocked tensor. Hence $\mathbb{1} \in S_1(B)$, so in the nontrivial case $D \geq 1$ not all blocked generators can have trace zero. Choose $j_0$ with $\mathrm{tr}(B^{j_0}) \neq 0$. By Theorem 9.17, $B^{j_0}$ has a nonzero eigenvector $\varphi$ with eigenvalue $\mu \neq 0$. Writing $j_0$ as the blocked index corresponding to a word $\sigma_0$ of length $N_0$ gives $A^{\sigma_0}\varphi = \mu\varphi$. Applying the same argument to the transpose blocked tensor gives a word $\tau_0$ of length $N_0$, a nonzero vector $\psi$, and $\nu \neq 0$ with $(A^{\tau_0})^T\psi = \nu\psi$. Since $A$ is normal and $N_0 \geq 1$, the blocked tensor $B$ is normal by Theorem 9.35. Therefore there exists $M$ with $S_M(B) = M_D(\mathbb{C})$. In particular $\varphi\psi^T \in S_M(B)$. Taking $m = M$ proves (3). □

**Theorem 9.40** (Fixed-length word-span saturation via blocking)**.** *If $A$ is a normal MPS tensor with bond dimension $D$, then there exists $N$ such that $S_N(A) = M_D(\mathbb{C})$. This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].*

*Proof.* By normality, some $N_0$ has $S_{N_0}(A) = M_D(\mathbb{C})$. If $N_0 = 0$ we are done. Otherwise, Theorem 9.39 produces eigenvectors and a rank-one element $\varphi\psi^T \in S_m(A^{[N_0]})$. Theorem 9.38 then gives $S_{((D-1)+m+(D-1))\cdot N_0}(A) = M_D(\mathbb{C})$. □

*Remark* 9.41 (Blocking route to fixed-length spanning). Theorem 9.40 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 9.39) and then applying the fixed-length assembly step (Theorem 9.38). Thus the fixed-length span of $A$ is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.

## 9.9 Primitive MPS tensors

**Definition 9.42** (Primitive MPS tensor)**.** Assume $D > 0$. A tensor $A$ together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \qquad \rho \geq 0, \qquad \rho \neq 0, \qquad \mathcal{E}_A(\rho) = \rho,$$

and, if $P$ is the fixed-point projection associated to $\rho$, then $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$. When the choice of $\rho$ is irrelevant, we simply say that $A$ is a primitive MPS tensor. This is weaker than the paper's notion, which requires $\rho > 0$; see Remark 9.48. The stronger positive-definite hypothesis is imposed separately in Theorems 9.49 and 9.60.

**Theorem 9.43** (Primitive overlap convergence)**.** *If $A$ is a primitive MPS tensor, then $O_{AA}(N) \to 1$.*

*Proof.* Immediate from the self-overlap convergence under a complementary spectral gap (Theorem 8.55). ☐

## 9.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 9.42: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius $< 1$.

**Theorem 9.44** (Primitive MPS tensors have nonzero word products)**.** *If $A$ is a primitive MPS tensor, then for every $n \geq 0$ there exists a word $w$ of length* exactly *$n$ with $A^w \neq 0$.*

*Proof.* The PSD fixed point $\rho$ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. ☐

**Theorem 9.45** (Iterated transfer does not vanish)**.** *If $A$ is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

*Proof.* By Theorem 9.44, there exists a word $w$ of length $n$ with $A^w \neq 0$, so $\mathcal{E}_A^n(\rho)$ has a nonzero summand. ☐

### 9.10.1 From primitivity to normality

The primitivity condition of Definition 9.42 combines TP normalization, a nonzero PSD fixed point, and a complementary spectral gap. It is connected to normality through the following results.

**Theorem 9.46** (Fixed-point uniqueness from spectral gap)**.** *If $A$ is a primitive MPS tensor with PSD fixed point $\rho$, then every fixed point $\sigma$ of $\mathcal{E}_A$ is proportional to $\rho$: $\sigma = \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$.*

*Proof.* Set $\sigma' = \sigma - \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$. Then $\mathrm{tr}(\sigma') = 0$ and $(E - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $E - P_\rho$, contradicting the spectral gap $\rho_{\text{spec}}(E - P_\rho) < 1$. ☐

**Theorem 9.47** (Complement powers tend to zero)**.** *If $A$ is a primitive MPS tensor with PSD fixed point $\rho$, then $(\mathcal{E}_A - P_\rho)^n \to 0$ in operator norm.*

*Proof.* The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the spectral-gap hypothesis. Apply the general result that powers of an operator with spectral radius $< 1$ tend to zero. ☐

*Remark* 9.48 (Primitivity: spectral gap vs. positive definiteness). The primitivity condition used in §9.10.1 has three parts: TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, a nonzero PSD fixed point $\rho$, and the spectral gap $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$. The notion of "primitive" in [SPGWC10, Proposition 3] additionally requires $\rho$ to be positive definite (full rank). Without positive definiteness, the spectral-gap condition alone does *not* imply irreducibility or normality. A counterexample is $D = 2$, $\rho = \mathrm{diag}(1, 0)$.

With the additional hypothesis $\rho > 0$, Theorem 9.49 gives irreducibility. Together with Burnside's theorem (Chapter 10, §10.4.1) and Theorem 9.58, one obtains

$$\text{primitive} + \rho \text{ PosDef} \;\Rightarrow\; \text{irreducible} \;\Rightarrow\; \mathrm{alg}(A) = M_D(\mathbb{C}) \;\Rightarrow\; \text{normal}.$$

The last step uses the aperiodicity condition $\mathbb{1} \in S_1(A)$ from Remark 9.53. After restoring the positive-definite hypothesis, [SPGWC10, Proposition 3] identifies the paper's primitive/strongly irreducible condition with eventual full Kraus rank, i.e. with our notion of normality.

### 9.10.2 Primitive implies irreducible

**Theorem 9.49** (Primitive with PosDef fixed point implies irreducible tensor)**.** *If $A$ is a primitive MPS tensor with PSD fixed point $\rho$ and $\rho$ is positive definite, then $A$ is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV17a, §2.3]).*

*Proof.* By contrapositive. Assume $A$ has an invariant projection $P$ ($P \neq 0$, $P \neq \mathbb{1}$, $(\mathbb{1}-P)A^iP = 0$ for all $i$). Set $\sigma = P\rho P$.

1. *Invariance of $\sigma$ under $\mathcal{E}_A^n$:* the projection relation $(\mathbb{1}-P)A^iP = 0$ implies $(\mathbb{1}-P)\cdot\mathcal{E}_A^n(\sigma) = 0$ for all $n$ (by induction on $n$).
2. *Convergence:* By Theorem 9.47, $\mathcal{E}_A^n(\sigma) \to \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$. Hence $(\mathbb{1} - P) \cdot \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho = 0$.
3. *Nonzero scalar:* Since $P \neq 0$ and $\rho$ is positive definite, $\mathrm{tr}(P\rho P) > 0$, so $\frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)} \neq 0$. Therefore $(\mathbb{1} - P)\rho = 0$.
4. *Contradiction:* Since $\rho$ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(\mathbb{1} - P)\rho = 0$ we get $\mathbb{1} - P = 0$, i.e. $P = \mathbb{1}$, contradicting $P \neq \mathbb{1}$.

□

### 9.10.3 From primitivity to strong irreducibility and normality

The results in this subsection remove the aperiodicity assumption from the normality conclusions. They connect the spectral-gap primitivity of Definition 9.42 directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

**Theorem 9.50** (Primitive MPS with PosDef fixed point has irreducible transfer map)**.** *If $A$ is a primitive MPS tensor with positive-definite fixed point $\rho$, then the transfer map $\mathcal{E}_A$ is irreducible.*

*The proof uses fixed-point uniqueness (Theorem 9.46): every PSD fixed point is proportional to $\rho$, so Wolf's criterion for irreducibility (positive-definite fixed point implies irreducibility) applies directly.*

*Proof.* By Theorem 9.46, if $\sigma \geq 0$ and $\mathcal{E}_A(\sigma) = \sigma$, then $\sigma = \frac{\mathrm{tr}\,\sigma}{\mathrm{tr}\,\rho}\rho$. Since $\rho > 0$, Wolf's uniqueness criterion for irreducibility applies. □

**Theorem 9.51** (Primitive MPS with PosDef fixed point is strongly irreducible)**.** *If $A$ is a primitive MPS tensor with positive-definite fixed point $\rho$, then $A$ is strongly irreducible in the sense of [SPGWC10, Proposition 3(c)]: $\mathcal{E}_A$ is irreducible, the fixed point $\rho$ is positive definite, and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$.*

*Proof.* Combine Theorem 9.50 (irreducibility) with the peripheral-spectrum primitivity from the spectral-gap hypothesis (Theorem 9.43). □

**Theorem 9.52** (Primitive MPS with PosDef fixed point is normal)**.** *If $A$ is a primitive MPS tensor with positive-definite fixed point $\rho$, then $A$ is normal (has eventually full Kraus rank). No aperiodicity assumption is needed.*

*This is the key connection between the spectral-gap primitivity of Definition 9.42 and normality, filling the gap that the aperiodicity-based route in §9.11 leaves open.*

*Proof.* By Theorem 9.51, $A$ is strongly irreducible. The implication from strong irreducibility to normality proceeds as follows: strong irreducibility gives peripheral spectrum $\{1\}$, hence the aperiodicity condition $\mathbb{1} \in S_1(A)$. Combined with irreducibility, Theorem 9.59 then yields normality. □

## 9.11 Cumulative span to word span

Cumulative spanning ($T_N(A) = M_D(\mathbb{C})$, the union of word spans of all lengths $\leq N$) is weaker than normality ($S_L(A) = M_D(\mathbb{C})$ for a single length $L$). The gap is closed by an aperiodicity hypothesis.

*Remark* 9.53 (Aperiodicity condition). We say $A$ satisfies the *aperiodicity condition* if $\mathbb{1} \in S_1(A)$, i.e. the identity matrix lies in the span of the Kraus operators $\{A^i\}$. This is an additional hypothesis. The one-sided TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ does *not* imply $\mathbb{1} \in S_1(A)$; it only gives a quadratic identity among the Kraus operators. For the paper-style primitive condition (peripheral-spectrum primitivity together with a positive-definite fixed point), one expects aperiodicity to follow from spectral analysis of the transfer map. That implication is not used in the present Wielandt argument, so the condition is kept explicit.

*Remark* 9.54 (Counterexample: cumulative spanning does not imply normality). In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ (the off-diagonal matrix units). Then $\mathrm{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd $n$ and only diagonal matrices for even $n$. Hence $S_n(A) \neq M_2(\mathbb{C})$ for every $n$, and $A$ is not normal. The aperiodicity condition $\mathbb{1} \in S_1(A)$ fails: $S_1(A) = \mathrm{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

**Theorem 9.55** (Word span monotonicity from aperiodicity)**.** *If $\mathbb{1} \in S_1(A)$, then $S_n(A) \leq S_{n+1}(A)$ for all $n$.*

*Proof.* For any $M \in S_n(A)$, $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$ by Lemma 9.30. □

**Theorem 9.56** (Cumulative span equals word span under aperiodicity)**.** *If $\mathbb{1} \in S_1(A)$, then $T_n(A) = S_n(A)$ for all $n$.*

*Proof.* By Theorem 9.55, the word spans are monotone: $S_0 \leq S_1 \leq \cdots$. The supremum $T_n = S_0 + \cdots + S_n$ therefore equals the largest term $S_n$. □

**Theorem 9.57** (Cumulative spanning plus aperiodicity implies normality)**.** *If $T_N(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then $A$ is normal.*

*Proof.* By Theorem 9.56, $S_N(A) = T_N(A) = M_D(\mathbb{C})$. $\square$

**Theorem 9.58** (Algebra span plus aperiodicity implies normality)**.** *If* $\mathrm{alg}(A) = M_D(\mathbb{C})$ *and* $\mathbb{1} \in S_1(A)$*, then* $A$ *is normal.*

*Proof.* Since $\mathrm{alg}(A) = M_D(\mathbb{C})$, there exists $N$ with $T_N(A) = M_D(\mathbb{C})$ (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 9.57. $\square$

**Theorem 9.59** (Irreducible tensor plus aperiodicity implies normality)**.** *If* $A$ *is an irreducible tensor,* $D > 0$*, and* $\mathbb{1} \in S_1(A)$*, then* $A$ *is normal. This assembles the full chain:*

$$\textit{irreducible tensor} \xrightarrow{10.20} \textit{irreducible action} \xrightarrow{\textit{Burnside}} \mathrm{alg}(A) = M_D(\mathbb{C}) \xrightarrow{9.58} \textit{normal.}$$

*This corresponds to the irreducibility-to-normality direction of [SPGWC10, Proposition 3]. For the Burnside step, we appeal forward to Theorems 10.20 and 10.21 in Chapter 10; this is a presentation-level forward dependency, but the mathematical dependency is acyclic.*

*Proof.* By Theorem 10.20, $A$ acts irreducibly on $\mathbb{C}^D$. By Theorem 10.21, $\mathrm{alg}(A) = M_D(\mathbb{C})$. By Theorem 9.58 with the aperiodicity hypothesis, $A$ is normal. $\square$

**Theorem 9.60** (Primitive + positive definite fixed point imply normality)**.** *If* $A$ *is a primitive MPS tensor with fixed point* $\rho$ *and* $\rho$ *is positive definite, then* $A$ *is normal. No aperiodicity assumption is needed.*

*Proof.* By Theorem 9.51, $A$ is strongly irreducible. In particular, $A$ is irreducible and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$, so $A$ is aperiodic. Theorem 9.59 then gives normality, hence eventually full Kraus rank. $\square$

*Remark* 9.61 (The normal-canonical route bypasses the aperiodicity argument). The aperiodicity results in §9.11 are needed only for the block-injective route through Definition 2.22, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical route used later in Chapters 10–12 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, packages them as normal canonical form in the sense of [CPGSV17a]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

# Chapter 10

# Canonical Form Reduction

This chapter reduces a general MPS tensor to block-diagonal data via iterated invariant-subspace decomposition. Two reduction schemes are relevant. The block-injective route aims at injective TP-normalized blocks and the canonical form predicate of Chapter 11. The normal-canonical route packages a weighted family of irreducible TP-normalized blocks with primitive transfer maps and pairwise distinct nonzero weight moduli, in the sense of [CPGSV17a]. The Perron–Frobenius eigenvector existence and TP-gauge construction are in Chapter 7; this chapter uses them to produce CFII data and the normal canonical form packaging. The reduction draws on [PGVWC07], [CPGSV21], and [Wol12, Ch. 6].

## 10.1 Block-diagonal construction

**Definition 10.1** (Block-diagonal tensor from blocks)**.** Recall the block-diagonal tensor of Definition 2.26. Given $r$ blocks with bond dimensions $(D_k)_{k=1}^r$, per-block MPS tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, we write $\bigoplus_{k=1}^r \mu_k A_k$ for the tensor of total bond dimension $D = \sum_k D_k$ whose Kraus operators are $A^i = \bigoplus_{k=1}^r \mu_k A_k^i$.

**Theorem 10.2** (Per-block single-block FT)**.** *If all block tensors $A_k$ are injective and the per-block same-MPV condition holds ($A_k$ and $B_k$ generate the same MPV family for each $k$), then each pair $(A_k, B_k)$ is gauge equivalent.*

*Proof.* Apply the single-block fundamental theorem (Theorem 4.11) to each block independently. □

**Theorem 10.3** (Per-block same MPV implies global same MPV)**.** *If $A_k$ and $B_k$ generate the same MPV family for each block $k$, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ also generate the same MPV family.*

*Proof.* By the MPV decomposition (Theorem 2.27), each MPV is $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. Per-block equality of MPV coefficients gives global equality. □

**Theorem 10.4** (Global multi-block fundamental theorem)**.** *If each block $A_k$ is injective and each pair $(A_k, B_k)$ has* $\mathrm{SameMPV}(A_k, B_k)$*, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

*Proof.* Apply the per-block single-block fundamental theorem (Theorem 10.2) to obtain blockwise gauge equivalences from injectivity plus SameMPV, then assemble them into a global gauge for the block-diagonal tensors. □

## 10.2 Transfer map normalization

**Theorem 10.5** (Scaling preserves injectivity). *If $A$ is injective and $\zeta \neq 0$, then $\zeta A$ is injective.*

*Proof.* Scaling by a nonzero scalar preserves linear independence and hence spanning. ☐

**Theorem 10.6** (Transfer map under scaling). $\mathcal{E}_{\zeta A}(X) = |\zeta|^2 \mathcal{E}_A(X)$.

*Proof.* Each summand: $(\zeta A^i)X(\zeta A^i)^\dagger = |\zeta|^2 A^i X (A^i)^\dagger$. ☐

**Theorem 10.7** (Phase-scaling preserves normalization). *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $|\zeta| = 1$, then $\sum_i (\zeta A^i)^\dagger (\zeta A^i) = \mathbb{1}$.*

*Proof.* $(\zeta A^i)^\dagger (\zeta A^i) = \bar{\zeta}\zeta (A^i)^\dagger A^i = (A^i)^\dagger A^i$, since $\bar{\zeta}\zeta = |\zeta|^2 = 1$. ☐

**Theorem 10.8** (MPV under block normalization). *For any nonzero scaling factors $\mu_k$, set $\eta_k := \mu_k / |\mu_k|$ and $B_k := |\mu_k| A_k$. Then $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \eta_k B_k$ generate the same MPV family. Equivalently, the $k$-th block contribution factors as*

$$\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma,$$

*so the modulus contributes only through $|\mu_k|^N$, while the remaining factor $\eta_k$ is a unit-modulus phase.*

*Proof.* By the MPV decomposition (Theorem 2.27), the $k$-th contribution of $\bigoplus_k \mu_k A_k$ is $\mu_k^N V^{(N)}(A_k)_\sigma$. For $B_k = |\mu_k| A_k$, Lemma 2.15 gives $V^{(N)}(B_k)_\sigma = |\mu_k|^N V^{(N)}(A_k)_\sigma$. Hence $\eta_k^N V^{(N)}(B_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma = \mu_k^N V^{(N)}(A_k)_\sigma$ for every block, so the total MPV families agree. ☐

## 10.3 Invariant subspace decomposition

**Theorem 10.9** (Unitary conjugation preserves the MPV family). *If $U$ is a unitary matrix, then $A$ and $U^\dagger A U$ generate the same MPV family.*

*Proof.* By cyclicity of the trace: $\operatorname{tr}((U^\dagger A^{i_1} U) \cdots (U^\dagger A^{i_N} U)) = \operatorname{tr}(A^{i_1} \cdots A^{i_N})$. ☐

**Definition 10.10** (Support projection). For a positive semi-definite matrix $\rho \geq 0$, the *support projection* $P$ is the orthogonal projection onto the range of $\rho$. It is constructed via the spectral decomposition: if $\rho = U \operatorname{diag}(\lambda_1, \ldots, \lambda_D) U^\dagger$, then $P = U \operatorname{diag}(\mathbf{1}_{\lambda_1 > 0}, \ldots, \mathbf{1}_{\lambda_D > 0}) U^\dagger$, where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

**Definition 10.11** (Invariant projection predicate). An MPS tensor $A$ *has an invariant projection* if there exists an orthogonal projection $P$ with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P) A^i P = 0$ for all $i$. The negation of this predicate defines an *irreducible tensor* (Definition 10.14).

**Theorem 10.12** (PSD fixed point gives invariant projection). *Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let $P$ be the support projection of $\rho$. Then for all $i$, $(\mathbb{1} - P) A^i P = 0$: the range of $P$ is invariant under all $A^i$.*

*Proof.* From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i \rho (A^i)^\dagger$, $(\mathbb{1} - P) A^i \rho (A^i)^\dagger (\mathbb{1} - P) = 0$ for each $i$. Since $\rho$ is supported on the range of $P$, this forces $(\mathbb{1} - P) A^i P = 0$.

This is the finite-dimensional version of [Wol12, Prop. 6.10] ("Stationary subspaces": the support projection of any stationary state is sub-harmonic, i.e. invariant under the channel). The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Thm. 6.2]; see also [CPGSV21, §IV.A.1]. ☐

**Theorem 10.13** (Two-block decomposition)**.** *If the transfer map has a PSD fixed point $\rho$ that is* not *positive definite (i.e. $\rho$ has a nontrivial kernel), then $A$ is unitarily equivalent to a two-block upper-triangular tensor. Moreover, the block-diagonal tensor obtained by discarding the off-diagonal blocks generates the same MPV family as $A$.*

*Proof.* The support projection $P$ of $\rho$ is neither 0 nor $\mathbb{1}$ (since $\rho \neq 0$ and $\rho$ is not PD). Theorem 10.12 gives $(\mathbb{1} - P)A^i P = 0$ for all $i$. A unitary change of basis that block-diagonalizes $P$ puts $A^i$ in upper-triangular block form (Theorem 10.9 ensures the MPV is preserved by the conjugation). The trace of the upper-triangular product equals the sum of the diagonal-block traces, so the block-diagonal tensor obtained by discarding the off-diagonal blocks has the same MPV coefficients as the conjugated tensor, and hence as $A$. □

## 10.4 Iterated reduction

**Definition 10.14** (Irreducible tensor)**.** An MPS tensor $A$ is *irreducible* if it has no nontrivial invariant projection, i.e. $\neg\,\mathrm{HasInvariantProj}(A)$.

**Theorem 10.15** (Irreducible block decomposition)**.** *Every MPS tensor $A$ can be decomposed (up to unitary equivalence) into a block-diagonal tensor $\bigoplus_{k=1}^{r} A_k$ where each $A_k$ is irreducible. The block-diagonal tensor generates the same MPV as $A$.*

*Proof.* By strong induction on the bond dimension $D$. If $A$ is already irreducible, take $r = 1$. Otherwise, by Definition 10.14, $A$ has a nontrivial invariant projection $P$. Diagonalizing $P$ yields the same two-block upper-triangular splitting used in Theorem 10.13; the block-diagonal tensor obtained by discarding the off-diagonal part has the same MPV family as $A$, and the two diagonal blocks have strictly smaller bond dimensions. Apply the induction hypothesis to each block and concatenate. (If one prefers a positive-eigenvector route without assuming TP normalization, the correct replacement for the channel-specific Cesàro theorem is Theorem 7.15; the induction itself needs only the invariant projection.)

Both [PGVWC07] (Theorem 4, "TI canonical form") and [CPGSV21] (§IV.A.1) obtain the block decomposition by the same invariant-projection splitting used here. An alternative route, developed in [Wol12, Thm. 6.14], characterises the fixed-point set of a quantum channel as a $*$-algebra and then applies the Wedderburn–Artin decomposition ([Wol12, Eq. (1.39)]) to read off the block structure directly. The block decomposition is obtained by iterating the invariant-projection splitting and using strong induction on $D$ to ensure termination. □

**Theorem 10.16** (Unitary diagonal positive fixed point in TP gauge)**.** *Let $A$ be an irreducible MPS tensor satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume $D > 0$. Then there exist a unitary $U$ and a diagonal positive definite matrix $\Lambda$ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \qquad \textit{and} \qquad \mathcal{E}_B(\Lambda) = \Lambda.$$

*Proof.* The TP hypothesis makes $\mathcal{E}_A$ a channel, so it has a nonzero PSD fixed point. The irreducible tensor condition implies that $\mathcal{E}_A$ is irreducible as a CP map, and Theorem 7.3 upgrades the fixed point to a positive definite one. The spectral theorem diagonalizes this matrix by a unitary conjugation, and unitary conjugation preserves the TP normalization. □

### 10.4.1 From irreducibility to full spanning via Burnside's theorem

The irreducible-tensor predicate (Definition 10.14) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under the Kraus operators.

**Definition 10.17** (Algebra span)**.** For an MPS tensor $A$, let $\mathrm{alg}(A)$ be the unital $\mathbb{C}$-subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}$:

$$\mathrm{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d{-}1\rangle \subseteq M_D(\mathbb{C}).$$

**Definition 10.18** (Invariant submodule)**.** A subspace $W \leq \mathbb{C}^D$ is *$A$-invariant* if $A^i W \subseteq W$ for every $i$.

**Definition 10.19** (Irreducible action)**.** The Kraus operators of $A$ act *irreducibly* on $\mathbb{C}^D$ if every $A$-invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

**Theorem 10.20** (Irreducible tensor implies irreducible action)**.** *If $A$ is irreducible as a tensor (Definition 10.14), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 10.19).*

*Proof.* Let $W \leq \mathbb{C}^D$ be an $A$-invariant subspace. In finite dimension the orthogonal projection $P$ onto $W$ exists. Invariance of $W$ implies $(\mathbb{1} - P)A^i P = 0$ for all $i$. Thus $P$ is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. ☐

**Theorem 10.21** (Burnside's theorem for Kraus algebras)**.** *Assume $D > 0$. If $A$ acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra:*

$$\mathrm{alg}(A) = M_D(\mathbb{C}).$$

*This is the complex finite-dimensional case of Burnside's theorem (equivalently, Jacobson's density theorem); see [Jac09].*

*Proof.* Transport $\mathrm{alg}(A)$ along the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur's lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. ☐

*Remark* 10.22. This Burnside/Jacobson-density step is a substantial external algebraic dependency for the formalization. In the current project it is formalized directly as the theorem tagged above, rather than imported from Mathlib; if Mathlib later acquires a suitable finite-dimensional Burnside theorem, this chapter could be retargeted to that library result.

**Theorem 10.23** (Irreducible action implies eventual cumulative spanning)**.** *Assume $D > 0$. If $A$ acts irreducibly on $\mathbb{C}^D$, then there exists $N$ such that $T_N(A) = M_D(\mathbb{C})$.*

*Proof.* By Theorem 10.21, $\mathrm{alg}(A) = M_D(\mathbb{C})$. Every element of the algebra span is a linear combination of word products, hence belongs to $T_n(A)$ for some $n$. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \leq T_1(A) \leq \cdots$ stabilizes, so one can choose a uniform $N$ with $T_N(A) = M_D(\mathbb{C})$. ☐

## 10.5 Proportional single-block theorem

**Theorem 10.24** (Proportional MPV implies gauge-phase equivalence)**.** *Let $A$ and $B$ be injective MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If $A$ and $B$ generate proportional MPV families ($V^{(N)}(A)_\sigma = c_N V^{(N)}(B)_\sigma$ for some $c_N \in \mathbb{C}$) and $O_{AA}(N) \to 1$, $O_{BB}(N) \to 1$, then $A$ and $B$ are gauge-phase equivalent.*

*Proof.* By contradiction. If $A$ and $B$ are not gauge-phase equivalent, then $O_{AB}(N) \to 0$ (Theorem 8.21). From the proportionality hypothesis, $O_{AB}(N) = c_N O_{BB}(N)$. Since $O_{BB}(N) \to 1$, it follows that $c_N \to 0$. But $O_{AA}(N) = |c_N|^2 O_{BB}(N) \to 0$, contradicting $O_{AA}(N) \to 1$.

In [PGVWC07], the proportional case is handled as part of the canonical form argument. Thus the proportional case is obtained directly from the spectral gap in Chapter 8. □

**Theorem 10.25** (Proportional MPV + primitive implies gauge-phase equivalence)**.** *Let $A$ and $B$ be MPS tensors with primitive transfer maps $E_A, E_B$ and positive definite fixed points $\rho_A, \rho_B$. If $A$ and $B$ generate proportional MPV families, then $A$ and $B$ are gauge-phase equivalent. This is a strengthened convenience form of Theorem 10.24: primitivity supplies left-canonicality, irreducibility, and overlap convergence, reducing the hypothesis count from seven to four.*

*Proof.* From IsPrimitiveMPS, extract left-canonical normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), overlap convergence ($O_{AA}(N) \to 1$), and irreducibility (Theorem 9.49). Apply Theorem 10.24. □

## 10.6 Canonical form from peripheral primitivity

The canonical form predicate (Definition 11.9) includes an explicit hypothesis that the self-overlap $O_{A_k A_k}(N) \to 1$ for each block. The following theorem derives this from per-block peripheral-spectrum primitivity, removing it as an independent assumption.

**Theorem 10.26** (Canonical form from peripheral primitivity)**.** *Let $(\mu_k, A_k)_{k=1}^r$ be a collection of block tensors and scaling factors satisfying:*

1. *each $A_k$ is injective,*
2. *each $A_k$ satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,*
3. *$|\mu_1| > \cdots > |\mu_r|$ (strictly decreasing),*
4. *all $\mu_k \neq 0$,*
5. *each transfer map $\mathcal{E}_{A_k}$ is primitive, i.e. has peripheral spectrum $\{1\}$.*

*Then $(\mu_k, A_k)$ satisfies the canonical form predicate (Definition 11.9).*

*Proof.* Fix a block $k$ and write $E := \mathcal{E}_{A_k}$. By injectivity and TP normalization, Theorem 7.9 provides a nonzero positive definite fixed point for $E$. To apply the spectral-gap criterion (Theorem 5.39), we check its three hypotheses:

1. $E$ is primitive, since the peripheral spectrum is $\{1\}$ by assumption;
2. every eigenvalue of $E$ has modulus at most 1, because $E = F_{A_k A_k}$ is a normalized mixed transfer operator and Theorem 8.14 applies;
3. any trace-zero fixed point of $E$ is zero; this is the trace-zero uniqueness consequence extracted in Chapter 7 from the uniqueness of the PSD fixed point (Theorem 7.5).

Hence $E$ has a spectral gap on the complement of its fixed-point projection. Theorem 8.55 then gives $O_{A_k A_k}(N) \to 1$. Since this holds for every block, the overlap clause required in Definition 11.9 follows from the other hypotheses. □

*Remark* 10.27. Theorem 10.26 shows that the overlap convergence hypothesis in the canonical form predicate is redundant once peripheral-spectrum primitivity is known. The argument factors through the spectral-gap formulation of Chapter 8: peripheral-spectrum primitivity is first converted to that spectral-gap condition, and the overlap limit is then deduced from it. In the paper [CPGSV21], primitivity is obtained from irreducibility plus periodicity removal by blocking to period one. Theorem 10.28 is precisely this periodicity-removal step under irreducibility and TP normalization.

**Theorem 10.28** (Blocking yields a peripheral-spectrum primitive transfer map)**.** *Let $A$ be an irreducible MPS tensor with $D > 0$. Assume that $A$ satisfies the TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that the transfer map $\mathcal{E}_{A^{[p]}}$ of the blocked tensor $A^{[p]}$ is primitive, i.e. its peripheral spectrum is $\{1\}$.*

*Proof.* Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$. The TP normalization of $A$ makes $K$ unital, and irreducibility of $A$ gives irreducibility of the associated Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint map of $K$, so Theorem 5.34 shows that every peripheral eigenvalue is a root of unity. Since peripheral($\mathcal{E}_A$) is finite (Theorem 5.26), Lemma 5.30 gives a common exponent $p$ such that $\mu^p = 1$ for all $\mu \in$ peripheral($\mathcal{E}_A$). Lemma 5.31 then implies peripheral($\mathcal{E}_A^p$) $= \{1\}$. Finally, blocking corresponds to taking a power of the transfer map, so $\mathcal{E}_{A^{[p]}}$ is primitive. □

*Remark* 10.29. Theorem 10.28 is the "periodicity removal by blocking" step in [CPGSV17a, Appendix A] for an irreducible block already placed in TP gauge. The later existence theorems simply reuse this step.

## 10.7 Normal canonical form

**Definition 10.30** (Normal canonical form predicate)**.** A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *normal canonical form predicate* if:

1. each $A_k$ is irreducible,
2. each $A_k$ satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),
4. the moduli are strictly decreasing: $|\mu_1| > |\mu_2| > \cdots > |\mu_r|$,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Here "normal" is used in the spectral sense of [CPGSV17a]: irreducible blocks whose TP-normalized transfer maps are primitive. As noted in the remark after Definition 2.22, this is equivalent to the eventual block-injectivity formulation of Definition 2.22; see [SPGWC10, Proposition 3].

**Theorem 10.31** (Self-overlap is derived in normal canonical form)**.** *If $(\mu_k, A_k)$ satisfies the normal canonical form predicate, then $O_{A_k A_k}(N) \to 1$ for every block $k$.*

*Proof.* Peripheral-spectrum primitivity is part of the data. Together with irreducibility and TP normalization, it yields the spectral-gap primitive theorem needed for overlap convergence, so the self-overlap condition follows from the other hypotheses rather than appearing as a separate assumption. ☐

**Theorem 10.32** (Modulus-one eigenvalue rigidity for irreducible TP blocks)**.** *Let $A$ and $B$ be irreducible MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If the mixed transfer map has spectral radius at least 1 (equivalently, it has a peripheral eigenvalue of modulus 1), then $A$ and $B$ are gauge-phase equivalent.*

*Proof.* The argument is the same as in Theorem 8.16, but the fixed-point input comes from irreducibility rather than injectivity. Take a modulus-one eigenvector of the mixed transfer map. Because $A$ and $B$ are TP-normalized, each transfer map has a nonzero PSD fixed point by Theorem 7.8. Irreducibility upgrades these fixed points to positive definite ones by Theorem 7.3. Theorem 7.10 therefore gauges $A$ and $B$ separately to unital Kraus families. In this unital gauge, the embedded modulus-one eigenvector saturates the Kadison–Schwarz inequality, and the same equality / multiplicative-domain argument as in Theorem 8.16 produces an invertible intertwiner between the two Kraus families. That intertwiner is exactly the required gauge-phase equivalence. ☐

## 10.8 Steps toward canonical form existence

This section combines the components of the canonical-form construction following [CPGSV17a, §2.3 and Appendix A]. The approach here requires blocking to remove non-trivial periodicity. An alternative framework that handles periodic blocks without blocking is developed in [DlCCSPG17]; see Chapter 13 for a discussion of that extension.

**Theorem 10.33** (Irreducible block decomposition)**.** *Every MPS tensor $A$ generates the same MPV family as a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ with each block irreducible.*

*Proof.* Apply Theorem 10.15. ☐

**Theorem 10.34** (CFII data for irreducible TP blocks)**.** *Let $A$ be an irreducible MPS tensor already in TP gauge ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) with $D > 0$. Then there exist a unitary $U$ and a diagonal positive definite matrix $\Lambda$ such that:*

1. *$B^i := U^\dagger A^i U$ is TP,*
2. *$\mathcal{E}_B(\Lambda) = \Lambda$,*
3. *$\Lambda$ is diagonal and positive definite,*
4. *$A$ and $B$ generate the same MPV family.*

*This is the "Canonical Form II" (CFII) step of [CPGSV17a, Appendix A]. The unitary conjugation occurs only after TP normalization has already been achieved; it is a second step inside TP gauge, not a unitary normalization of an arbitrary irreducible block.*

*Proof.* Apply Theorem 10.16 to obtain the unitary and the diagonal PD fixed point. Unitary conjugation preserves trace-preservation and the MPV family (Theorem 10.9). ☐

**Theorem 10.35** (CFII continuation after irreducible block decomposition)**.** *For any MPS tensor $A$, there exist irreducible blocks $(A_k)_{k=1}^r$ that generate the same MPV family as $A$. Moreover, for each block $k$, if that block has already been placed in TP gauge and $D_k > 0$, one obtains CFII data (unitary, diagonal PD fixed point, MPV preservation). The conclusion is CFII data; the normal-canonical-form packaging requires additional downstream steps.*

*Proof.* Combine Theorems 10.33 and 10.34. □

**Theorem 10.36** (Normal canonical form from a primitive block decomposition)**.** *Suppose $A$ generates the same MPV family as a weighted block-diagonal tensor $\bigoplus_k \mu_k A_k$, and assume that:*

1. *each $A_k$ is irreducible,*
2. *each $A_k$ satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,*
3. *each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),*
4. *the moduli $|\mu_k|$ are pairwise distinct,*
5. *all $\mu_k \neq 0$,*
6. *all bond dimensions are positive.*

*Then there exist a blocking length $p > 0$, weights $(\nu_k)$, and blocked tensors $(B_k)$ such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and $(\nu_k, B_k)$ satisfies the normal canonical form predicate.*

*Proof.* No new blocking is needed here: the input blocks are already primitive, so one may take $p = 1$. The theorem is stated with a blocking length because earlier reduction steps may require a common blocking period; under the present hypotheses that period can be chosen to be 1. Since the moduli $|\mu_k|$ are pairwise distinct, there is a reordering of the blocks for which they become strictly decreasing. After this reindexing, set $\nu_k := \mu_k$ and $B_k := A_k$ in the new order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are unchanged by the reordering, so the reordered family satisfies Definition 10.30 and generates the same MPV family as $A$. □

## 10.9 Zero-block separation

In the irreducible block decomposition, some blocks may have all-zero Kraus operators. Such blocks contribute to the MPV only at word length $N = 0$ (where their contribution equals their bond dimension). The following results separate the zero blocks from the "live" (nonzero) blocks.

**Definition 10.37** (Zero MPS tensor)**.** The *zero MPS tensor* of physical dimension $d$ and bond dimension $D$, denoted $\mathbf{0}_{d,D}$, is the tensor whose Kraus operators are all zero.

**Theorem 10.38** (MPV of zero tensor)**.** *For any spin configuration $\sigma$ of length $N$,*

$$V^{(N)}(\mathbf{0}_{d,D})_\sigma = \begin{cases} D & \text{if } N = 0, \\ 0 & \text{if } N \geq 1. \end{cases}$$

*Proof.* For $N = 0$, $(\mathbf{0}_{d,D})^\emptyset = \mathbb{1}_D$, so the MPV equals $\mathrm{tr}(\mathbb{1}_D) = D$. For $N \geq 1$, every word product of a zero tensor is zero. □

**Theorem 10.39** (Zero-block separation)**.** *Every MPS tensor $A$ admits a block decomposition into:*

1. *a* zero tail *of bond dimension $D_0$ (accumulating all all-zero irreducible blocks), and*
2. *a finite family of* live blocks $(B_k)_{k=1}^r$*, each irreducible, each with at least one nonzero Kraus operator, and each with positive bond dimension,*

*such that*

$$V^{(N)}(A)_\sigma = V^{(N)}(\mathbf{0}_{d,D_0})_\sigma + V^{(N)}(\textstyle\bigoplus_{k=1}^r B_k)_\sigma$$

*for every spin configuration $\sigma$ of length $N$. At $N = 0$ this reads $D = D_0 + \sum_{k=1}^r D_k$. For $N \geq 1$ the zero-tail term vanishes.*

*Proof.* Apply the irreducible block decomposition (Theorem 10.15) and partition the blocks by whether each has a nonzero Kraus operator. The all-zero blocks satisfy a dimension bound via irreducibility; their total bond dimension is $D_0$. The MPV identity follows from the block structure. □

## 10.10 Arbitrary-input TP-gauge reduction

**Theorem 10.40** (Arbitrary-input TP-gauge reduction with zero-block separation)**.** *For any MPS tensor $A$, there exist:*

1. *a zero-tail bond dimension $D_0$,*
2. *live blocks $(B_k)_{k=1}^r$ with nonzero weights $(\mu_k)_{k=1}^r$, each block irreducible and left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), with positive bond dimension,*

*such that*

$$V^{(N)}(A)_\sigma = V^{(N)}(\mathbf{0}_{d,D_0})_\sigma + V^{(N)}(\textstyle\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

*This is the furthest unconditional arbitrary-input reduction step available before periodicity removal.*

*Proof.* Apply Theorem 10.39 to separate the zero tail and live blocks. Then apply the blockwise TP-gauge theorem to the live blocks. The MPV identity chains through both steps. □

## 10.11 Blocking infrastructure

**Theorem 10.41** (Blocking distributes over MPV equivalence)**.** *If $A$ generates the same MPV family as $\bigoplus_k \mu_k B_k$, then for any $p \geq 0$,*

$$A^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]}.$$

*Proof.* The identity $V^{(N)}(A^{[p]})_\sigma = V^{(Np)}(A)_{\sigma_{\text{flat}}}$ holds for any tensor by definition of blocking. The MPV equivalence of $A$ and $\bigoplus_k \mu_k B_k$ at length $Np$ then gives the equivalence at length $N$ with weights $\mu_k^p$. □

**Theorem 10.42** (Primitive channels remain primitive under powers)**.** *If $\mathcal{E}$ has peripheral spectrum $\{1\}$ (is primitive) and $m \geq 1$, then $\mathcal{E}^m$ is also primitive.*

*Proof.* The peripheral spectrum of $\mathcal{E}^m$ is the set of $m$-th powers of that of $\mathcal{E}$. Since $\{1\}^m = \{1\}$, the result follows. $\square$

**Theorem 10.43** (Blocking-period divisibility preserves primitivity)**.** *If the transfer map of $A^{[p]}$ is primitive, $p \mid q$, and $q \geq 1$, then the transfer map of $A^{[q]}$ is also primitive.*

*Proof.* Write $q = p \cdot m$. Then $\mathcal{E}_{A^{[q]}} = \mathcal{E}^m_{A^{[p]}}$. Apply Theorem 10.42 with $\mathcal{E} = \mathcal{E}_{A^{[p]}}$. $\square$

**Theorem 10.44** (Common blocking period for a block family)**.** *Let $(B_k)_{k=1}^r$ be a finite family of MPS tensors with positive bond dimensions, each admitting a blocking period $p_k$ making its transfer map primitive. Then $p = \operatorname{lcm}(p_1, \dots, p_r)$ is a common period: $\mathcal{E}_{B_k^{[p]}}$ is primitive for every $k$.*

*Proof.* Each $p_k \mid p$, so Theorem 10.43 applies. $\square$

## 10.12 Cyclic sector decomposition

**Theorem 10.45** (Compression to a supported projection sector)**.** *Let $A$ be an MPS tensor satisfying $\sum_i (A^i)^\dagger A^i = P$ for an orthogonal projection $P$, and suppose $PA^iP = A^i$ for every $i$. Then there exists a left-canonical tensor $C$ of bond dimension $n = \operatorname{tr}(P)$ with*

$$V^{(N)}(C)_\sigma = \operatorname{tr}\big(P \cdot A^\sigma\big)$$

*for every $\sigma$.*

*Proof.* Diagonalize $P$ via a unitary $U$ (spectral theorem). The eigenvalues are in $\{0, 1\}$. Set $n = \operatorname{tr} P$; let $C^i$ be the top-left $n \times n$ block of $U^\dagger A^i U$ (restricted to eigenvalue-1 indices). TP follows from the projection normalization; the MPV identity follows from trace cyclicity. $\square$

**Theorem 10.46** (Block decomposition from commuting projections)**.** *Let $A$ be a left-canonical MPS tensor, and let $(P_k)_{k=1}^m$ be orthogonal projections summing to $\mathbb{1}$ that commute with every Kraus operator: $P_k A^i = A^i P_k$ for all $i, k$. Then there exist left-canonical blocks $(C_k)_{k=1}^m$ with unit weights such that $A$ generates the same MPV family as $\bigoplus_{k=1}^m C_k$.*

*Proof.* For each $k$, the tensor $P_k A^i$ is supported on $P_k$ (using commutativity and $P_k^2 = P_k$). The TP condition $\sum_i (P_k A^i)^\dagger (P_k A^i) = P_k$ follows from $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Apply Theorem 10.45 to each sector. The MPV identity uses $\sum_k P_k = \mathbb{1}$ and linearity of trace. $\square$

**Theorem 10.47** (Block decomposition from adjoint-fixed projections)**.** *Let $A$ be a left-canonical MPS tensor, and let $(P_k)_{k=1}^m$ be orthogonal projections summing to $\mathbb{1}$ that are fixed by the adjoint transfer map: $\mathcal{E}_A^\dagger(P_k) = P_k$ for every $k$. Then the same block decomposition conclusion holds as in Theorem 10.46.*

*Proof.* The Kadison–Schwarz equality condition for a TP map (Theorem 6.12) implies that a projection fixed by $\mathcal{E}_A^\dagger$ commutes with every Kraus operator. Apply Theorem 10.46. $\square$

**Theorem 10.48** (Cyclic sector decomposition for irreducible TP tensors)**.** *Let $A$ be a left-canonical MPS tensor with irreducible transfer map. Then there exist $m \geq 1$ and left-canonical blocks $(C_k)_{k=1}^m$ such that the $m$-blocked tensor $A^{[m]}$ decomposes as $\bigoplus_{k=1}^m C_k$ (with unit weights).*

*Proof.* Derive the conjugate-transposed Kraus family $K_i = (A^i)^\dagger$, its unitality and irreducibility, and a positive-definite fixed point $\rho$ of the transfer map. Apply the cyclic structure theorem (peripheral eigenvalues form a cyclic group of order $m$) to obtain $m$ cyclic spectral projections. Feed into Theorem 10.47. $\square$

## 10.13 Reduction to primitive blocks

**Theorem 10.49** (Arbitrary tensor to primitive block decomposition)**.** *For any MPS tensor $A$, there exist a period $p \geq 1$, a zero-tail bond dimension $D_0$, nonzero weights $(\mu_k)_{k=1}^r$, and left-canonical blocks $(B_k)_{k=1}^r$ with primitive transfer maps and positive bond dimensions, such that*

$$V^{(N)}\big(A^{[p]}\big)_\sigma = V^{(N)}\big(\mathbf{0}_{d^{[p]},D_0}\big)_\sigma + V^{(N)}\big(\textstyle\bigoplus_k \mu_k B_k\big)_\sigma.$$

*Proof.* Apply Theorem 10.40 to get the zero tail and left-canonical live blocks. Apply Theorem 10.44 to find a common blocking period $p$ making all transfer maps primitive. Thread through via Theorem 10.41. □

**Theorem 10.50** (TP-primitive irreducible tensor is normal)**.** *Let $A$ be a left-canonical MPS tensor with $D > 0$, irreducible transfer map, and peripheral spectrum $\{1\}$. Then $A$ is normal.*

*Proof.* Peripheral primitivity and irreducibility yield the spectral-gap condition. The PSD fixed point is positive definite by Theorem 7.3. Apply Theorem 9.52. □

**Theorem 10.51** (Blocking preserves normality)**.** *If $A$ is normal and $p \geq 1$, then $A^{[p]}$ is normal.*

*Proof.* Since $A$ is normal, there exists $N$ with $S_N(A) = M_D(\mathbb{C})$. Then $S_{Np}(A) = M_D(\mathbb{C})$ (word products at multiples still span), and $S_N(A^{[p]}) \supseteq S_{Np}(A) = M_D(\mathbb{C})$. □

## 10.14 Open directions

*Remark* 10.52 (Canonical form reduction: current status)*.* The core reduction steps are in place. The arbitrary-input reduction (Theorem 10.40) produces a zero tail plus left-canonical live blocks. After a common blocking step (Theorem 10.49), the blocked blocks have primitive transfer maps.

The remaining steps to full normal canonical form require: establishing tensor irreducibility for each blocked block (blocking does not in general preserve tensor irreducibility), and establishing pairwise distinct weight norms. The cyclic-sector decomposition (Theorems 10.46 and 10.47) provides the infrastructure for the periodic-to-aperiodic splitting.

# Chapter 11

# Block Permutation and Separation

This chapter establishes the algebraic structure of automorphisms of products of matrix algebras and uses it to prove that the same-MPV condition on block-diagonal tensors descends to per-block gauge equivalence. The results correspond to the canonical form uniqueness of [PGVWC07] and the block separation results of [CPGSV21]. The proof technique used here—induction on blocks via the spectral gap—is specific to this formalization.

## 11.1 Block permutation algebra

**Theorem 11.1** (Ring isomorphisms permute block ideals)**.** *Let $\{R_k\}_{k=1}^r$ be simple rings. Every ring isomorphism $T : \prod_k R_k \to \prod_k R_k$ permutes the block ideals: there exists a permutation $\pi$ such that $T$ maps the $k$-th block ideal to the $\pi(k)$-th block ideal.*

*Proof.* Each block ideal $I_k = \{x : x_j = 0 \text{ for } j \neq k\}$ is an atom in the lattice of two-sided ideals (since $R_k$ is simple). A ring isomorphism preserves the ideal lattice and hence permutes the atoms. Injectivity of the induced map on atoms gives the permutation $\pi$. □

*Remark* 11.2 (Ring level versus algebra level)*.* Theorem 11.1 is a ring-level statement: it only uses the ideal structure of $\prod_k R_k$. Theorems 11.3 and 11.4 require an upgrade to $\mathbb{C}$-algebra automorphisms. The reason is threefold: dimension preservation compares the blocks as $\mathbb{C}$-vector spaces; Skolem–Noether applies to the resulting central simple $\mathbb{C}$-algebras; and the per-block maps extracted from $T$ must commute with scalar multiplication. Thus the formalization separates the ring-level permutation step from the later algebra-level decomposition.

**Theorem 11.3** (Dimension preservation)**.** *If $T$ is an algebra automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$, and $\pi$ is the induced permutation, then $D_{\pi(k)} = D_k$ for all $k$.*

*Proof.* The restriction of $T$ to the $k$-th block ideal induces a $\mathbb{C}$-algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\pi(k)}}(\mathbb{C})$, which forces $D_k = D_{\pi(k)}$ (by comparing dimensions as $\mathbb{C}$-vector spaces). □

**Theorem 11.4** (Decomposition of automorphisms of $\prod M_{D_k}(\mathbb{C})$)**.** *Every $\mathbb{C}$-algebra automorphism $T$ of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$ decomposes as a block permutation $\pi$ followed by per-block inner automorphisms: there exist $\pi \in S_r$ with $D_{\pi(k)} = D_k$ and invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such* *that, for every tuple $M = (M_1, \dots, M_r)$, $T(M)_k = X_k M_{\pi(k)} X_k^{-1}$, where $T(M)_k$ denotes the $k$-th block component of $T(M)$.*

*Proof.* Theorem 11.1 gives the permutation $\pi$. Theorem 11.3 gives $D_{\pi(k)} = D_k$. The restriction of $T$ to the $k$-th block ideal yields a $\mathbb{C}$-algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\pi(k)}}(\mathbb{C})$; after Theorem 11.3 identifies the two matrix sizes, Skolem–Noether (Theorem 4.9) makes this restriction inner.

This is a standard result in the theory of semisimple algebras; see [CPGSV21, §IV.A] for the MPS context. □

## 11.2 Product algebra linear extension

**Definition 11.5** (Per-block linear extension)**.** Given injective tensors $A_k, B_k$ of bond dimension $D_k$ with the same MPV for each block $k$, the *per-block linear extension* $T_k$ is the unique linear map $M_{D_k}(\mathbb{C}) \to M_{D_k}(\mathbb{C})$ satisfying $T_k(A_k^i) = B_k^i$ for all $i$. It exists (and is unique) because $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$ by injectivity.

**Definition 11.6** (Product algebra automorphism)**.** The *product algebra automorphism* is the blockwise assembled map $T : \prod_k M_{D_k}(\mathbb{C}) \to \prod_k M_{D_k}(\mathbb{C})$ defined by $T(M)_k := T_k(M_k)$. Theorem 11.7 shows that each $T_k$ is a $\mathbb{C}$-algebra homomorphism and that this assembled map is indeed a $\mathbb{C}$-algebra automorphism.

**Theorem 11.7** (Per-block linear extensions assemble)**.** *Let $A_k$ and $B_k$ be injective tensors of bond dimension $D_k$ for $k = 1, \dots, r$, and suppose the per-block same-MPV condition holds. Then the per-block linear extensions $T_k : A_k^i \mapsto B_k^i$ assemble into an algebra automorphism of $\prod_k M_{D_k}(\mathbb{C})$, which decomposes as a permutation $\pi$ with $D_{\pi(k)} = D_k$ and per-block conjugation matrices $X_k$.*

*Proof.* Each $T_k$ exists uniquely and is multiplicative by Theorems 4.6 and 4.7. Promotion to algebra homomorphisms (Lemma 4.10) and composition into the product gives an algebra automorphism $T$. Theorem 11.4 decomposes $T$ into a permutation plus per-block inner automorphisms. Because $T$ is assembled blockwise, it preserves each block ideal, so in this theorem the permutation is actually $\pi = \mathrm{id}$. A nontrivial block permutation only appears after block separation identifies which blocks match globally. □

**Theorem 11.8** (Per-block same MPV gives global gauge equivalence)**.** *If all block tensors $A_k$ are injective and per-block same-MPV holds, then each pair $(A_k, B_k)$ is gauge equivalent, and the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are globally gauge equivalent.*

*Proof.* Per-block gauge equivalence follows from the single-block FT (Theorem 4.11) applied to each block. Global gauge equivalence follows by assembling the per-block gauge matrices into a block-diagonal matrix (Theorem 10.4). □

## 11.3 Block separation

Recall from Definition 2.34 and Theorem 8.7 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma \in \{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \mathrm{Tr}(F_{AB}^N).$$

We use this notation throughout the block-separation argument.

**Definition 11.9** (Canonical form predicate)**.** A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *canonical form predicate* if:

1. each $A_k$ is injective,
2. each $A_k$ satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. the moduli are strictly decreasing: $|\mu_1| > |\mu_2| > \cdots > |\mu_r|$,
4. all $\mu_k \neq 0$,
5. the self-overlap converges: $O_{A_k A_k}(N) \to 1$ for each $k$.

Only this one-sided normalization is part of the predicate; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Thm. 6.7, item 3]), $T^n(\rho) \to \rho_\infty$ for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Thm. 6.8] for the completely-positive characterisation and Chapter 8 for the spectral-gap proof.

*Remark* 11.10 (Normalization convention)*.* The convention here differs from [PGVWC07, Thm. 4], which uses the unital gauge $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$. This chapter instead works in the dual TP gauge $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. A positive-definite fixed point of the adjoint transfer map provides a gauge transformation between these two normalizations (Theorem 7.11).

*Remark* 11.11*.* In the literature [CPGSV21], condition (5) is derived from primitivity of the transfer map (peripheral spectrum $= \{1\}$). Theorem 10.26 gives this implication: per-block primitivity implies overlap convergence, so condition (5) is redundant when primitivity holds. The canonical form predicate retains condition (5) explicitly because it can be verified independently of primitivity.

*Remark* 11.12 (Normal-canonical companion)*.* Definition 11.9 is the block-injective predicate. The normal-canonical route instead uses Definition 10.30: injectivity and the explicit self-overlap hypothesis are replaced by irreducibility, TP normalization, and primitive transfer maps. Theorem 10.31 then supplies the self-overlap limit automatically.

**Theorem 11.13** (Vandermonde separation)**.** *If $\mu_1, \ldots, \mu_r$ are pairwise distinct complex numbers and $\sum_{k=1}^r c_k \mu_k^N = 0$ for $N = 0, 1, \ldots, r-1$, then $c_k = 0$ for all $k$.*

*Proof.* The coefficient vector $c$ satisfies $Vc = 0$ where $V$ is the Vandermonde matrix of the $\mu_k$. Injectivity of $\mu$ makes $V$ invertible. □

*Remark* 11.14*.* Theorem 11.13 is not used in the current proof chain. It is retained for potential alternative routes; the current path to Theorem 12.5 uses BNT linear independence (Chapter 11) rather than Newton–Girard identities to match the weight multiset.

**Lemma 11.15** (Block separation core)**.** *Let $((\mu_k), (A_k))$ satisfy the canonical form predicate, and let $(B_k)$ be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ for all $N, \sigma$, then each pair $(A_k, B_k)$ generates the same MPV family.*

*Proof.* By induction on the number of blocks $r$. In the inductive step, take overlaps with the leading block $A_0$; after division by $\mu_0^N$, the contributions from $k \geq 1$ decay because $|\mu_k/\mu_0| < 1$ (strict weight ordering) and the bilinear overlaps $O_{A_0 A_k}(N)$ and $O_{A_0 B_k}(N)$ are bounded (Theorems 8.19 and 8.21 supply the relevant bounds), leaving $O_{A_0 A_0}(N) - O_{A_0 B_0}(N) \to 0$. Condition (5) gives $O_{A_0 A_0}(N) \to 1$, hence $O_{A_0 B_0}(N) \to 1$. Since $O_{A_0 B_0}(N) \to 1 \neq 0$, the bond dimensions must match: $D_{A_0} = D_{B_0}$; otherwise Theorem 8.19 would give $O_{A_0B_0}(N) \to 0$, a contradiction. With equal bond dimensions established, the contrapositive of Theorem 8.21 forces $A_0$ and $B_0$ to be gauge-phase equivalent; the limit $O_{A_0B_0}(N) \to 1$ then makes the phase trivial, so $A_0$ and $B_0$ generate the same MPV family. Peeling off the leading block and dividing by $\mu_0^N$, the induction hypothesis applies. □

**Theorem 11.16** (Block separation from canonical form)**.** *Let $((\mu_k),(A_k))$ satisfy the canonical form predicate, and let $(B_k)$ be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair $(A_k, B_k)$ generates the same MPV family.*

*Proof.* Apply Lemma 11.15 to the identity obtained from the equality of the two block-diagonal MPV families. The canonical form predicate supplies injectivity, TP normalization, strict nonzero weights, and self-overlap convergence for $(A_k)$, while the hypotheses on $(B_k)$ supply the comparison injectivity and TP normalization. □

**Theorem 11.17** (Block separation from normal canonical form)**.** *Let $((\mu_k),(A_k))$ satisfy the normal canonical form predicate, and let $(B_k)$ be irreducible tensors of the same bond dimensions, satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair $(A_k, B_k)$ generates the same MPV family.*

*Proof.* The proof follows the same peeling argument as Theorem 11.16, with Theorem 10.32 (modulus-one eigenvalue rigidity for irreducible tensors) replacing the contrapositive of Theorem 8.21 at the identification step, since the blocks are irreducible rather than injective. Theorem 10.31 replaces the explicit self-overlap hypothesis on the leading block. Since both block families already use the same dimension function, the conclusion is direct per-block same MPV. □

**Theorem 11.18** (Canonical form fundamental theorem, same-structure version)**.** *Under the hypotheses of Theorem 11.16, each pair $(A_k, B_k)$ is gauge equivalent. Hence the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

*Proof.* Theorem 11.16 gives per-block same MPV. Theorem 11.8 converts this to per-block gauge equivalence and then to global gauge equivalence of the block-diagonal tensors. □

*Remark* 11.19 (Relation to the literature). The results of this chapter correspond to the canonical-form uniqueness of [PGVWC07] and the block-separation statements of [CPGSV21]. The proof technique used here—an induction on the number of blocks driven by the spectral gap of the mixed transfer operators—is specific to this formalization.

# Chapter 12

# Basis of Normal Tensors

This chapter introduces the *basis of normal tensors* (BNT) notion, the canonical-form variants with BNT separation, the cross-overlap decay theorem, and the permutation-rigidity arguments used in the proof of the Fundamental Theorem. The chapter concludes with auxiliary results on coefficient convergence and Newton–Girard identities. The main references are [PGVWC07] and [CPGSV21].

## 12.1 Basis of normal tensors

**Definition 12.1** (Basis of Normal Tensors (BNT))**.** A *basis of normal tensors* (BNT) for a total tensor $A_{\mathrm{tot}}$ consists of $g$ block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that:

1. each $A_j$ is normal (Definition 2.22),
2. at each system size $N$, the MPV of $A_{\mathrm{tot}}$ lies in the span of the block MPVs $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$,
3. for sufficiently large $N$, the block MPV vectors $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$ are linearly independent.

This is [CPGSV21, Definition 4.2].

**Definition 12.2** (Canonical form with BNT separation)**.** A family is in *canonical form with BNT separation* if it satisfies Definition 11.9 and, in addition, distinct blocks of the same bond dimension are *not* gauge-phase equivalent.

**Definition 12.3** (Normal canonical form with BNT separation)**.** A family is in *normal canonical form with BNT separation* if it satisfies Definition 10.30 and, in addition, distinct blocks of the same bond dimension are not gauge-phase equivalent.

**Theorem 12.4** (Cross-overlap decay for distinct blocks in canonical form with BNT separation)**.** *If $((\mu_k), (A_k))$ is in canonical form with BNT separation and $j \neq k$, then $O_{A_j A_k}(N) \to 0$.*

*Proof.* Definition 12.2 supplies injectivity and TP normalization for each block, which are the hypotheses required by the spectral-gap theorems. If $D_j \neq D_k$, the rectangular overlap-decay theorem (Theorem 8.19) gives $O_{A_j A_k}(N) \to 0$. If $D_j = D_k$, the separation condition says that $A_j$ and $A_k$ are not gauge-phase equivalent, so the same-dimension overlap-decay theorem (Theorem 8.21) again gives decay to 0. $\square$

**Theorem 12.5** (Canonical form with BNT separation yields a basis of normal tensors)**.** *If* $((\mu_k), (A_k))$ *is in canonical form with BNT separation, then the block tensors form a basis of normal tensors for the block-diagonal tensor* $\bigoplus_k \mu_k A_k$*.*

*Proof.* Each block is injective, so it is normal by taking blocking length 1. The MPV decomposition (Theorem 2.27) gives

$$|V^{(N)}(A_{\mathrm{tot}})\rangle = \sum_k \mu_k^N |V^{(N)}(A_k)\rangle,$$

so the MPV of $A_{\mathrm{tot}}$ lies in the span of the block MPVs at each system size. For eventual linear independence, consider the Gram matrix

$$G_{jk}(N) = \langle V^{(N)}(A_j)|V^{(N)}(A_k)\rangle.$$

By Lemma 2.35, $G_{jk}(N) = \overline{O_{A_jA_k}(N)}$. The self-overlap $O_{A_jA_j}(N) \to 1$ is real-valued under TP normalization, hence $G_{jj}(N) \to \overline{1} = 1$. For $j \neq k$, Theorem 12.4 gives $O_{A_jA_k}(N) \to 0$, so $G_{jk}(N) \to \overline{0} = 0$. Thus $G(N) \to I_g$, hence the Gram matrix is eventually invertible. Therefore the vectors $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$ are linearly independent for all sufficiently large $N$. ☐

*Remark* 12.6 (The normality gap for normal canonical form with BNT separation). Definition 12.3 references the normal canonical form predicate (Definition 10.30), which encodes normality via the spectral characterization (primitive transfer map). Definition 12.1 requires normality via the algebraic characterization (eventual block injectivity, Definition 2.22). These are equivalent by Remark 2.20, but the equivalence must be invoked explicitly: the Wielandt theory of Chapter 9 provides the implication from primitivity to eventual block injectivity needed in the formalization.

## 12.2 Permutation rigidity for bases of normal tensors

**Theorem 12.7** (Permutation rigidity for proportional MPV decompositions)**.** *Let* $(A_j)_{j=1}^{g_A}$ *and* $(B_k)_{k=1}^{g_B}$ *be injective families satisfying the TP normalization* $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ *and* $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$*, whose self-overlaps converge to* 1 *and whose cross-overlaps within each family converge to* 0*. Assume moreover that there exist MPS tensors* $A_{\mathrm{tot}}$ *and* $B_{\mathrm{tot}}$*, coefficient arrays* $a_{N,j}$ *and* $b_{N,k}$*, and a sequence* $c_N$ *with nonzero limits* $a_j^\infty$*,* $b_k^\infty$*,* $c^\infty$ *such that for every* $N$*,*

$$|V^{(N)}(A_{\mathrm{tot}})\rangle = \sum_{j=1}^{g_A} a_{N,j}|V^{(N)}(A_j)\rangle, \qquad |V^{(N)}(B_{\mathrm{tot}})\rangle = \sum_{k=1}^{g_B} b_{N,k}|V^{(N)}(B_k)\rangle,$$

*with* $a_{N,j} \to a_j^\infty \neq 0$ *and* $b_{N,k} \to b_k^\infty \neq 0$*, and also*

$$|V^{(N)}(A_{\mathrm{tot}})\rangle = c_N|V^{(N)}(B_{\mathrm{tot}})\rangle$$

*with* $c_N \to c^\infty \neq 0$*. Then* $g_A = g_B$*, and there is a permutation* $\pi$ *of the common block index set such that* $A_{\pi(k)}$ *and* $B_k$ *have the same bond dimension and are gauge-phase equivalent for each* $k$*. The canonical-form application of these hypotheses is described in Remark 12.8.*

*Proof.* Substitute the two decomposition formulas into the proportionality relation and take overlaps with the block MPVs. Passing to the limit using the convergence of $a_{N,j}$, $b_{N,k}$, and $c_N$ produces a limiting mixed-overlap matrix. The asymptotically orthonormal overlap hypotheses within each family force every row and column of that matrix to contain at most one nonzero entry, hence determine a permutation $\pi$. If one of the matched mixed overlaps came from blocks of different bond dimensions, Theorem 8.19 would force it to decay to 0; if the bond dimensions agree but the tensors are not gauge-phase equivalent, Theorem 8.21 does the same. Therefore every nonzero matched entry comes from blocks of the same bond dimension that are gauge-phase equivalent.

This is the overlap-and-decomposition form of the permutation step corresponding to [CPGSV21, Theorem 4.1]. $\square$

*Remark* 12.8. In the canonical-form application, the abstract data above come from block-diagonal assemblies

$$A_{\mathrm{tot}} = \bigoplus_{j=1}^{g_A} \mu_j A_j, \qquad B_{\mathrm{tot}} = \bigoplus_{k=1}^{g_B} \nu_k B_k,$$

and Theorem 2.27 gives the coefficient arrays $a_{N,j} = \mu_j^N$ and $b_{N,k} = \nu_k^N$. To fit the convergence hypotheses of Theorem 12.7, one first normalizes by the dominant weight on each side: the relevant arrays are $(\mu_j/\mu_0)^N$ and $(\nu_k/\nu_0)^N$, with $(\mu_j/\mu_0)^N \to 0$ for $j \geq 1$ and $(\nu_k/\nu_0)^N \to 0$ for $k \geq 1$ by Theorem 12.10, while the overall factors $\mu_0^N$ and $\nu_0^N$ are absorbed into the proportionality constant $c_N$. Definition 12.2 supplies injectivity, TP normalization, and the within-family overlap limits, so the theorem isolates the abstract permutation argument from this canonical-form packaging.

**Theorem 12.9** (Permutation rigidity with asymptotically orthonormal overlaps)**.** *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If for every system size $N$,*

$$\mathrm{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g_A\} = \mathrm{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \leq k \leq g_B\},$$

*then $g_A = g_B$, and there is a permutation $\pi$ of the common block index set such that $A_j$ and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent for each $j$.*

*Proof.* For each family, the Gram matrix converges to the identity by Lemma 2.35 together with the self-overlap and off-diagonal decay hypotheses. Hence for all sufficiently large $N$ both families of MPV vectors are linearly independent. At such an $N$, the equality of spans in the statement forces the two common subspaces to have the same dimension, so $g_A = g_B$. The span equality then gives change-of-basis matrices between the two families, and the same mixed-overlap matching argument shows that the mixed-overlap matrix has exactly one nonzero entry in each row and column. This produces the permutation $\pi$, while Theorems 8.19 and 8.21 upgrade each nonzero match to equality of bond dimensions and gauge-phase equivalence. $\square$

## 12.3 Coefficient convergence

**Theorem 12.10** (Coefficient ratio decay)**.** *Under the canonical form predicate of Definition 11.9, indexed so that $\mu_0$ is the dominant weight, one has $(\mu_k/\mu_0)^N \to 0$ for every $k \geq 1$ as $N \to \infty$.*

*Proof.* Since $|\mu_k| < |\mu_0|$ (strict ordering of moduli), $|\mu_k/\mu_0| < 1$, so the geometric sequence converges to 0. $\square$

*Remark* 12.11 (Strict-dominance regime). Theorem 12.10 applies when one modulus is strictly dominant: $|\mu_0| > |\mu_k|$ for $k \geq 1$. In the more general periodic setting of [CPGSV21, Eq. (72a)], the coefficient attached to block $j$ can take the form $\sum_q \mu_{j,q}^N$, where the $\mu_{j,q}$ are the individual weights contributed by that block. The oscillatory case with $|\mu_{j,q}| = 1$ is treated separately in the full paper.

## 12.4 Newton–Girard identities

**Theorem 12.12** (Newton–Girard trace recursion)**.** *If $A, B \in M_n(\mathbb{C})$ satisfy $\operatorname{tr}(A^k) = \operatorname{tr}(B^k)$ for all $k \geq 1$, then $A$ and $B$ have the same characteristic polynomial.*

*Proof.* Let $e_m$ denote the $m$th elementary symmetric polynomial in the eigenvalues, with $e_0 = 1$. The Newton–Girard recursion

$$me_m = \sum_{i=1}^{m} (-1)^{i-1} e_{m-i} \operatorname{tr}(A^i)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces $\operatorname{tr}(A^i)$. Equal power sums for $A$ and $B$ therefore give equal coefficients, hence equal characteristic polynomials. □

**Theorem 12.13** (Equal power sums imply equal multisets)**.** *Let $\alpha, \beta : \{1, \dots, n\} \to \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \geq 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.*

*Proof.* Construct diagonal matrices $\operatorname{diag}(\alpha)$ and $\operatorname{diag}(\beta)$. The power-sum hypothesis gives $\operatorname{tr}(\operatorname{diag}(\alpha)^k) = \operatorname{tr}(\operatorname{diag}(\beta)^k)$. By Theorem 12.12, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. □

*Remark* 12.14. Theorems 12.12 and 12.13 remain relevant for the unformalized literature-level equal-MPV Theorem 13.5, where one expects a power-sum step to recover the multiplicity-weight data. The currently formalized explicit-coefficient special case, Theorem 13.6, instead uses linear independence of the block MPV family supplied by a basis of normal tensors.

# Chapter 13

# Proof of the Fundamental Theorem

This chapter proves the Fundamental Theorem of Matrix Product States. We state the Fundamental Theorem in two versions: one for block-injective tensors in canonical form with basis of normal tensors (BNT) separation (CF-BNT), and one for the normal-canonical variant. We then treat the equal-MPV case, including the common-block-structure specialization that deduces gauge equivalence. The results combine the canonical form reduction (Chapter 10), block separation (Chapter 11), spectral gap (Chapter 8), and the BNT permutation arguments of Chapter 12.

The theorems in Sections 13.1–13.3 follow the approach of [CPGSV17a]: they first block the tensor to remove non-trivial periodicity, then apply the aperiodic fundamental theorem. Section 13.4 outlines the stronger "irreducible form" framework of [DlCCSPG17], which gives a fundamental theorem valid for periodic tensors directly, with an extra $Z$-gauge degree of freedom.

## 13.1 Weak Fundamental Theorem

The following theorem combines the full upstream reduction (zero-block separation, TP gauge, blocking, primitivity) with the downstream Fundamental Theorem for normal canonical form data. It is conditional on the extra hypotheses (tensor irreducibility and distinct weight norms) that the automatic reduction does not yet produce.

**Theorem 13.1** (Weak Fundamental Theorem (conditional))**.** *Let $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$ be two families in normal canonical form, each satisfying the non-degeneracy condition that no two distinct blocks in the same family are gauge-phase equivalent. Let $A_{\text{tot}}$ and $B_{\text{tot}}$ be tensors with MPV decompositions*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_j a_{N,j} V^{(N)}(A_j)_\sigma, \quad V^{(N)}(B_{\text{tot}})_\sigma = \sum_k b_{N,k} V^{(N)}(B_k)_\sigma,$$

*with $a_{N,j} \to a_j \neq 0$ and $b_{N,k} \to b_k \neq 0$. Suppose moreover that $V^{(N)}(A_{\text{tot}})_\sigma = c_N V^{(N)}(B_{\text{tot}})_\sigma$ with $c_N \to c \neq 0$. Then $g_A = g_B$ and there is a permutation $\pi$ such that $D_j^A = D_{\pi(j)}^B$ and $A_j$ is gauge-phase equivalent to $B_{\pi(j)}$ for every $j$.*

*Proof.* This packages the normal-canonical-form Fundamental Theorem (Theorem 13.4). The normal canonical form data provide the BNT separation and overlap convergence hypotheses; the non-degeneracy condition ensures the block matching is well-defined. □

## 13.2 Proportional Fundamental Theorem

**Theorem 13.2** (Fundamental Theorem — proportional MPV case). *Let $A_{\mathrm{tot}}$ and $B_{\mathrm{tot}}$ be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits $a_j^\infty$, $b_k^\infty$, $c^\infty$ such that*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\mathrm{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

*with $a_{N,j} \to a_j^\infty \neq 0$ and $b_{N,k} \to b_k^\infty \neq 0$, and suppose also that*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = c_N V^{(N)}(B_{\mathrm{tot}})_\sigma$$

*for a sequence $c_N \to c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation $\pi$ such that $D_j^A = D_{\pi(j)}^B$ and $A_j$ is gauge-phase equivalent to $B_{\pi(j)}$ for each $j$.*

*Proof.* Apply Theorem 12.7 to the two CF-BNT decompositions. The CF-BNT hypotheses provide injectivity, TP normalization, diagonal overlap limits, and off-diagonal decay via Theorem 12.4. With the supplied decomposition identities and convergence hypotheses, Theorem 12.7 yields the equality of block counts, the permutation, and the per-block gauge-phase equivalence. □

*Remark* 13.3 (Coefficient arrays in the CF-BNT setting). In applications the coefficient arrays in Theorem 13.2 come from the block-diagonal decomposition itself. If

$$A_{\mathrm{tot}} = \bigoplus_{j=1}^{g_A} \mu_j^A A_j, \qquad B_{\mathrm{tot}} = \bigoplus_{k=1}^{g_B} \mu_k^B B_k,$$

then Theorem 2.27 gives

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{g_A} (\mu_j^A)^N V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\mathrm{tot}})_\sigma = \sum_{k=1}^{g_B} (\mu_k^B)^N V^{(N)}(B_k)_\sigma.$$

Thus one takes $a_{N,j} = (\mu_j^A)^N$ and $b_{N,k} = (\mu_k^B)^N$, while the sequence $c_N$ is the global proportionality factor from the hypothesis $V^{(N)}(A_{\mathrm{tot}}) = c_N V^{(N)}(B_{\mathrm{tot}})$. Theorem 13.2 keeps these arrays explicit because Theorem 12.7 is formulated in that abstract form.

**Theorem 13.4** (Fundamental Theorem — proportional MPV case, NT route). *Let $A_{\mathrm{tot}}$ and $B_{\mathrm{tot}}$ be MPS tensors in normal canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits $a_j^\infty$, $b_k^\infty$, $c^\infty$ such that*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\mathrm{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

*with $a_{N,j} \to a_j^\infty \neq 0$ and $b_{N,k} \to b_k^\infty \neq 0$, and suppose also that*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = c_N V^{(N)}(B_{\mathrm{tot}})_\sigma$$

*for a sequence $c_N \to c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation $\pi$ such that $D_j^A = D_{\pi(j)}^B$ and $A_j$ is gauge-phase equivalent to $B_{\pi(j)}$ for each $j$.*

*Proof.* Normal canonical form with BNT separation provides irreducibility, TP normalization, and primitive self-overlap convergence via Theorem 10.31. Theorem 10.32 supplies the leading-block identification step: a nondecaying mixed overlap forces the corresponding comparison block to be gauge-phase equivalent. The same peeling argument as in the injective CF-BNT case therefore yields the block permutation and gauge-phase equivalence. Remark 12.6 explains why the argument stops at this separated normal-canonical level. □

## 13.3 Equal-MPV Fundamental Theorem

**Theorem 13.5** (Fundamental Theorem — equal MPV case (target statement))**.** *Let $A_{\mathrm{tot}}$ and $B_{\mathrm{tot}}$ be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. If*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = V^{(N)}(B_{\mathrm{tot}})_\sigma$$

*for every $N$ and $\sigma$, then the block-diagonal tensors $\bigoplus_{j=1}^{g_A} \mu_j^A A_j$ and $\bigoplus_{k=1}^{g_B} \mu_k^B B_k$ are gauge equivalent.*

*Proof.* The intended literature route begins by applying Theorem 13.2 with $c_N = 1$, which yields a permutation of the matched blocks and blockwise gauge-phase equivalences. One then compares the resulting MPV expansions from Theorem 2.27. In the general equal case the weights occur with multiplicities, so recovering the individual weights requires a power-sum argument such as Theorem 12.13. After the weights are identified, the phase factors are absorbed into them and the blockwise conjugacies assemble to a global gauge equivalence. This full route is not yet formalized here. □

**Theorem 13.6** (Fundamental Theorem — equal MPV case with explicit coefficient data)**.** *Let $A_{\mathrm{tot}}$ and $B_{\mathrm{tot}}$ be MPS tensors in canonical form with BNT separation, with associated families $(\mu_j^A, A_j)_{j=1}^{g_A}$ and $(\mu_k^B, B_k)_{k=1}^{g_B}$. Suppose there exist coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits $a_j^\infty$, $b_k^\infty$ such that*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\mathrm{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

*with $a_{N,j} \to a_j^\infty \neq 0$ and $b_{N,k} \to b_k^\infty \neq 0$. If*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = V^{(N)}(B_{\mathrm{tot}})_\sigma$$

*for every $N$ and $\sigma$, then $g_A = g_B$, and there is a permutation $\pi$ such that $D_j^A = D_{\pi(j)}^B$ and, after reindexing the $B$-blocks by $\pi$, the weighted block-diagonal tensors $\bigoplus_{j=1}^{g_A} \mu_j^A A_j$ and $\bigoplus_{j=1}^{g_A} \mu_{\pi(j)}^B B_{\pi(j)}$ are gauge equivalent.*

*Proof.* Apply Theorem 13.2 with $c_N = 1$. This yields $g_A = g_B$, a permutation $\pi$, equal block dimensions, and for each $j$ a gauge-phase equivalence

$$B_{\pi(j)}^i = \zeta_j X_j A_j^i X_j^{-1}.$$

By Theorem 2.27,

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{g_A} (\mu_j^A)^N V^{(N)}(A_j)_\sigma,$$

and similarly for $B_{\mathrm{tot}}$. Reindex the $B$-sum by $\pi$ and use the gauge-phase identity to rewrite

$$V^{(N)}(B_{\pi(j)})_\sigma = \zeta_j^N V^{(N)}(A_j)_\sigma.$$

Then equality of MPVs implies

$$\sum_{j=1}^{g_A} \left((\mu_j^A)^N - (\mu_{\pi(j)}^B \zeta_j)^N\right) V^{(N)}(A_j)_\sigma = 0.$$

By Theorem 12.5 and Definition 12.1, the family $\{V^{(N)}(A_j)\}_{j=1}^{g_A}$ is linearly independent for all sufficiently large $N$, so

$$(\mu_j^A)^N = (\mu_{\pi(j)}^B \zeta_j)^N$$

for every $j$ and all large $N$. Comparing two consecutive values of $N$ and using that the weights are nonzero gives

$$\mu_j^A = \mu_{\pi(j)}^B \zeta_j.$$

Thus the phase factors can be absorbed into the weights, and the blockwise conjugacies assemble to a gauge equivalence of the reindexed weighted block-diagonal tensors. □

**Theorem 13.7** (Fundamental Theorem — equal MPV case, common block structure)**.** *Fix a common block structure: the same number of blocks $g$, the same bond dimensions $(D_k)_{k=1}^g$, and the same weights $(\mu_k)_{k=1}^g$. Let $(\mu_k, A_k)_{k=1}^g$ and $(\mu_k, B_k)_{k=1}^g$ be two families in canonical form with BNT separation on this common data. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair $(A_k, B_k)$ is gauge equivalent, and hence the assembled block-diagonal tensors are gauge equivalent.*

*Proof.* Since both families share the canonical form data, the BNT separation condition is not needed. The canonical-form fundamental theorem (Theorem 11.18) then applies directly to this common $(g, D_k, \mu_k)$ structure, yielding per-block gauge equivalence and hence global gauge equivalence. □

### 13.3.1 Route B: Paper-faithful multiplicity layer

The paper-faithful multiplicity interface keeps the literature's per-block-copy structure separate from the merged canonical-form BNT predicate.

**Definition 13.8** (Paper sector data and decomposition)**.** A *paper sector data* over $g$ basis blocks specifies: multiplicities $r_j > 0$, sector weights $\mu_{j,q} \neq 0$ for $q \in \{1, \dots, r_j\}$, and the *paper coefficient* $\mathrm{coeff}(N, j) := \sum_{q=1}^{r_j} \mu_{j,q}^N$. A *paper decomposition* bundles the basis blocks $(A_j)_{j=1}^g$ with their sector data.

**Theorem 13.9** (Paper decomposition formula)**.** *For a paper decomposition $P$, the MPV of the assembled tensor satisfies*

$$V^{(N)}(P.\mathrm{toTensor})_\sigma = \sum_{j=1}^{g} \mathrm{coeff}(N, j) \cdot V^{(N)}(A_j)_\sigma.$$

**Theorem 13.10** (Eventual coefficient comparison from BNT linear independence)**.** *If two sector data $S, T$ over the same basis produce equal total MPVs and the basis is eventually linearly independent (the BNT property), then $S.\mathrm{coeff}(N, j) = T.\mathrm{coeff}(N, j)$ for all sufficiently large $N$.*

**Theorem 13.11** (Geometric sequence extrapolation)**.** *If two finite families of nonzero complex numbers have equal power sums $\sum_i \alpha_i^k = \sum_j \beta_j^k$ for all $k > N_0$, then equality holds for all $k \geq 0$. The proof uses induction on the number of terms with a telescoping argument: subtract a geometric factor to reduce the number of terms, apply the induction hypothesis, and conclude via a geometric recurrence.*

**Theorem 13.12** (Route B equal-case corollary: weight multiset recovery)**.** *Let $S$ and $T$ be two paper sector data over the same basis with the same multiplicities ($r_j^S = r_j^T$ for all $j$). If the basis is eventually linearly independent and the total MPVs of $S$ and $T$ agree, then for each basis block $j$ the sector weight multisets $\{\mu_{j,q}^S\}$ and $\{\mu_{j,q}^T\}$ are equal.*

*The proof chains:*

1. *BNT LI + equal MPVs $\Rightarrow$ eventual coefficient equality (Theorem 13.10);*
2. *Geometric extrapolation $\Rightarrow$ full coefficient equality for all positive $k$ (Theorem 13.11);*
3. *Newton–Girard $\Rightarrow$ equal weight multisets (Theorem 12.13).*

*Remark* 13.13 (Current formalized endpoint versus the literature)*.* The formalization provides three complementary equal-case results:

1. Theorem 13.6: the explicit-coefficient equal-case theorem (upgrade of Theorem 13.2 with $c_N = 1$), concluding a global gauge equivalence of the reindexed weighted block-diagonal tensors.
2. Theorem 13.7: the common block-structure special case.
3. Theorem 13.12: the Route B equal-case corollary, which from the paper's richer multiplicity data recovers the sector weight multisets without requiring explicit coefficient convergence hypotheses.

The literature-level unconditional Theorem 13.5 ([CPGSV21, Corollary IV.5]) remains `\notready`. To close it, one must compose the Route B weight recovery with: (a) the proportional FT for block matching and gauge-phase equivalences, (b) phase absorption into weights, and (c) global gauge construction via permutation and cast transport (all of which are now formalized individually).

## 13.4 Periodic Fundamental Theorem (de las Cuevas–Cirac–Schuch–Pére García)

The theorems in the preceding sections all require that each block has a *primitive* transfer map, i.e. peripheral spectrum exactly $\{1\}$. This is achieved by blocking the original tensor by a common period. The paper [DlCCSPG17] develops a framework that avoids this blocking step, giving a fundamental theorem that applies directly to periodic tensors.

### 13.4.1 Irreducible form

**Definition 13.14** (Periodic tensor)**.** An irreducible MPS tensor $A$ of bond dimension $D$ is $m$-*periodic* (for some $m \geq 1$) if the peripheral spectrum of its transfer map $\mathcal{E}_A$ is exactly the $m$-th roots of unity $\{e^{2\pi ik/m} : 0 \leq k < m\}$. A 1-periodic tensor is exactly an irreducible tensor with primitive transfer map (Definition 5.37).

**Definition 13.15** (Irreducible form)**.** A tensor $A$ is in *irreducible form* if it is block-diagonal

$$A^i = \bigoplus_{j \in J} \mu_j \, A^i_j,$$

where each block $A_j$ is an $m_j$-periodic tensor, each weight $\mu_j \neq 0$, and the blocks are pairwise non-repeated (no two distinct $j, j'$ have $A_j$ gauge-equivalent to $A_{j'}$). This is the standard form introduced in [DlCCSPG17, §II]. It extends the normal canonical form (Definition 10.30), which requires the stronger condition that every block is 1-periodic.

*Remark* 13.16 (Relation to normal canonical form). Every tensor in normal canonical form (Definition 10.30) is in irreducible form with all periods equal to 1. The irreducible form is strictly more general: it accommodates blocks such as the antiferromagnet, which are 2-periodic.

The current formalization (Chapters 10–13) works exclusively with period-1 blocks, obtaining them by a blocking step. Extending the formalization to the irreducible form of [DlCCSPG17] would require:

1. a definition of $m$-periodic irreducible tensors and their cyclic-sector decomposition (partial infrastructure exists in Section 10.12 of Chapter 10),
2. the off-diagonal structure of a periodic block's transfer matrix (the cyclic permutation of sectors), and
3. the notion of equivalence between periodic blocks that accounts for the $Z$-gauge freedom described below.

### 13.4.2 The $Z$-gauge degree of freedom

**Definition 13.17** (Entrywise periodic $Z$-gauge ratio)**.** Given complex weights $\mu, \nu$, define the entrywise $Z$-gauge ratio by

$$z(\mu, \nu) := \mu/\nu.$$

**Lemma 13.18** (Matched powers imply root-of-unity ratio)**.** *If $\mu^m = \nu^m$ and $\nu \neq 0$, then*

$$z(\mu, \nu)^m = 1.$$

**Lemma 13.19** (Ratio rescales denominator back to numerator)**.** *If $\nu \neq 0$, then*

$$z(\mu, \nu)\, \nu = \mu.$$

**Definition 13.20** (Diagonal periodic $Z$-gauge matrix)**.** For matched weight families $(\mu_i)_i$ and $(\nu_i)_i$ on a finite index type, define the diagonal matrix

$$Z := \operatorname{diag}\big(z(\mu_i, \nu_i)\big)_i.$$

**Lemma 13.21** (Diagonal $Z$-gauge has finite order under matched powers)**.** *If $\mu_i^m = \nu_i^m$ and $\nu_i \neq 0$ for all $i$, then*

$$Z^m = \mathbb{1}.$$

**Lemma 13.22** (Diagonal $Z$-gauge transports diagonal weights)**.** *Under $\nu_i \neq 0$ for all $i$,*

$$Z \ \operatorname{diag}(\nu) = \operatorname{diag}(\mu).$$

**Definition 13.23** ($Z$-gauge equivalence)**.** Let $A$ be an $m$-periodic irreducible tensor of bond dimension $D$. A *$Z$-gauge transformation* of $A$ is a diagonal unitary matrix $Z \in M_D(\mathbb{C})$ satisfying:

1. $Z$ commutes with every Kraus operator: $[A^i, Z] = 0$ for all physical indices $i$, and

2. $Z^m = \mathbb{1}_D$ (all diagonal entries are $m$-th roots of unity).

Replacing $A$ by $ZA$ (i.e. $A^i \mapsto ZA^i$) leaves the MPV family invariant for lengths $N$ that are multiples of $m$, because the factor $\mathrm{tr}(Z^N) = \mathrm{tr}(\mathbb{1}) = D$ for $N \equiv 0 \pmod{m}$. Two tensors $A, B$ are *$Z$-gauge equivalent* if there exists an invertible $Y$ and a $Z$-gauge transformation $Z$ for $A$ such that $ZA^i = YB^iY^{-1}$ for all $i$.

*Remark* 13.24 ($Z$-gauge is trivial for period-1 blocks)*.* When $m = 1$ the condition $Z^1 = \mathbb{1}$ forces $Z = \mathbb{1}$, so $Z$-gauge equivalence reduces to ordinary gauge equivalence (Definition 2.10). The $Z$-gauge is therefore a genuine new degree of freedom that appears only for $m > 1$.

### 13.4.3 Statement of the periodic Fundamental Theorem

The following is Theorem 3.8 ("equal case") of [DlCCSPG17].

**Theorem 13.25** (Periodic Fundamental Theorem)**.** *Let $A$ and $B$ be MPS tensors in irreducible form, with bases of periodic blocks $\{A_j\}_{j\in J}$ and $\{B_k\}_{k\in K}$ respectively. Suppose that $\mathcal{V}(A) = \mathcal{V}(B)$ (equal MPV families for all lengths $N$).*

*Then $|J| = |K|$ and there is a bijection $\pi : J \to K$ such that $A_j$ and $B_{\pi(j)}$ are $m_j$-periodic for the same period $m_j$. Moreover, for each $j \in J$ there exists:*

1. *an invertible matrix $Y_j$ of size $D_j \times D_{\pi(j)}$, and*

2. *a diagonal unitary matrix $Z_j$ of size $D_j \times D_j$ satisfying $Z_j^{m_j} = \mathbb{1}$ and $[A_j^i, Z_j] = 0$ for all $i$,*

*such that*

$$Z_j\, A_j^i = Y_j\, B_{\pi(j)}^i\, Y_j^{-1} \quad \textit{for all physical indices } i.$$

*Equivalently, setting $Y = \bigoplus_j Y_j \otimes \mathbb{1}_{r_j}$ and $Z = \bigoplus_j Z_j \otimes \mathbb{1}_{r_j}$, one has $ZA^i = YB^iY^{-1}$ for all $i$, with $[A^i, Z] = 0$ and $\mathcal{V}(A) = \mathcal{V}(ZA)$.*

*Remark* 13.26 (Comparison with the 1606.00608 approach)*.* Theorem 13.25 and the equal-MPV Fundamental Theorem (Theorem 13.5) both characterize the gauge freedom between two tensors with equal MPV families, but they do so at different levels:

1. **Blocking approach** (Theorem 13.5, following [CPGSV17a]): Block both $A$ and $B$ by a common period $p$ so that all blocks of $A^{[p]}$ and $B^{[p]}$ are primitive (1-periodic). Apply the aperiodic fundamental theorem to the blocked tensors. The conclusion is that $A^{[p]}$ and $B^{[p]}$ are gauge equivalent. The $Z$-gauge freedom does not appear because blocking absorbs the periodicity.

2. **Periodic approach** (Theorem 13.25, following [DlCCSPG17]): Work directly with the periodic blocks of $A$ and $B$. The conclusion includes an extra $Z$-gauge transformation reflecting the non-trivial phase ambiguity that arises for periodic blocks. The $Z$ matrix satisfies $Z^m = \mathbb{1}$ and commutes with all Kraus operators; it is invisible to the MPV family because it only introduces $m$-th roots of unity in the trace.

In the current formalization, only the blocking approach has been implemented. Formalizing the periodic approach would require developing the irreducible form and $Z$-gauge infrastructure outlined in Remark 13.16.

*Remark* 13.27 (Proportional periodic FT). There is also a proportional-case version of the periodic Fundamental Theorem (Theorem 3.4, "proportional case", in [DlCCSPG17]): if $\mathcal{V}(A) = \lambda_N \mathcal{V}(B)$ for scalars $\lambda_N$ with $\lambda_N \to \lambda \neq 0$, then the bases of periodic tensors of $A$ and $B$ are equivalent (up to the same gauge and $Z$ freedom as above). This extends Theorems 13.2 and 13.4 to the periodic setting.

# Chapter 14

# Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Ch. 7].

Section 14.1 establishes definitions and the basic exponential form (Proposition 14.8). Section 14.2 gives the Duhamel formula and perturbation bound. Section 14.3 develops the generator theory and proves the GKSL theorem.

## 14.1 Dynamical semigroups and the exponential form

**Definition 14.1** (Dynamical semigroup)**.** A family of linear maps $T : \mathbb{R} \to (M_D(\mathbb{C}) \to_\ell M_D(\mathbb{C}))$ is a *dynamical semigroup* if

1. (semigroup law) $T(t+s) = T(t) \circ T(s)$ for all $t, s \geq 0$; and
2. (initial condition) $T(0) = \mathrm{id}$.

This is [Wol12, Eq. (7.1)].

**Definition 14.2** (Continuous dynamical semigroup)**.** A dynamical semigroup $T$ is *norm-continuous* if the map $t \mapsto T(t)$ is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

**Definition 14.3** (Exponential semigroup)**.** Given a generator $L \in \mathrm{End}_{\mathbb{C}}(M_D(\mathbb{C}))$, the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}.$$

**Theorem 14.4** (Semigroup law for the exponential)**.** *For all $t, s \in \mathbb{R}$, $e^{(t+s)L} = e^{tL} \cdot e^{sL}$.*

*Proof.* Since $(t \cdot L)$ commutes with $(s \cdot L)$, this follows from the exponential addition formula for commuting elements in a Banach algebra. □

**Theorem 14.5** (Continuity of the exponential semigroup)**.** *The map $t \mapsto e^{tL}$ is continuous in the operator norm topology.*

*Proof.* The exponential is analytic (convergence radius $= \infty$), hence continuous. □

**Theorem 14.6** (Derivative of the exponential semigroup)**.** *For all $t \in \mathbb{R}$,*

$$\frac{d}{dt}e^{tL} = e^{tL} \cdot L.$$

*This is [Wol12, Eq. (7.2)].*

*Proof.* Chain rule applied to the composition $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \mathrm{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \mathrm{End}(M_D(\mathbb{C}))$. □

**Theorem 14.7** (Generator uniqueness)**.** *If $e^{tL} = e^{tL'}$ for all $t \geq 0$, then $L = L'$.*

*Proof.* Both semigroups agree on $[0, \infty)$, so their derivatives within $[0, \infty)$ at $t = 0$ agree. Since $[0, \infty)$ has unique differentials at 0, $L = L'$. □

**Theorem 14.8** (Prop. 7.1: Every continuous semigroup is exponential)**.** *Every norm-continuous dynamical semigroup $T$ on $M_D(\mathbb{C})$ is of the form*

$$T_t = e^{tL}$$

*for some generator $L \in \mathrm{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ and every $t \geq 0$. This is [Wol12, Prop. 7.1].*

*Proof.* Since $T_0 = \mathbb{1}$ and $T$ is continuous, the integral $M_\varepsilon = \int_0^\varepsilon T_s \, ds$ is invertible for small $\varepsilon > 0$. Using the semigroup property, $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$ is differentiable for $t \geq 0$. ODE uniqueness then gives $T_t = e^{tL}$ for all $t \geq 0$. □

## 14.2 Perturbation theory

**Theorem 14.9** (Lem. 7.1: Duhamel formula)**.** *Let $\Delta = L' - L$. For $t \geq 0$,*

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta \, e^{sL'} \, ds.$$

*This is [Wol12, Lem. 7.1].*

*Proof.* Define $f(s) = e^{(t-s)L}e^{sL'}$. Then $f'(s) = e^{(t-s)L}\Delta \, e^{sL'}$ (Theorem 14.6), and $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s) \, ds$. □

**Theorem 14.10** (Cor. 7.1: Perturbation bound)**.** *For $t \geq 0$,*

$$\left\| e^{tL'} - e^{tL} \right\| \leq t \, \|\Delta\| \cdot \sup_{s \in [0,t]} \|e^{sL}\| \cdot \sup_{s \in [0,t]} \|e^{sL'}\|,$$

*where $\Delta = L' - L$. This is [Wol12, Cor. 7.1].*

*Proof.* Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral (Theorem 14.9). □

## 14.3 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 14.39), which shows that such generators have the standard Lindblad form.

### 14.3.1 Generator decomposition and conditional complete positivity

**Definition 14.11** (Generator decomposition)**.** A *generator decomposition* consists of a completely positive map $\phi : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ and a matrix $\kappa \in M_d(\mathbb{C})$, defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger.$$

This is [Wol12, Eq. (7.14)].

**Definition 14.12** (Conditional complete positivity)**.** A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map $\phi$ and a matrix $\kappa$ such that $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. This is condition 1 of [Wol12, Prop. 7.2].

**Definition 14.13** (Trace-annihilating map)**.** A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is *trace-annihilating* if $\operatorname{tr}(L(\rho)) = 0$ for all $\rho \in M_d(\mathbb{C})$. This is the infinitesimal version of trace preservation.

**Definition 14.14** (Trace constraint)**.** A generator decomposition $(\phi, \kappa)$ satisfies the *trace constraint* if $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, where $\phi^*$ is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Eq. (7.20)].

**Theorem 14.15** (Trace constraint implies trace-annihilating)**.** *If $(\phi, \kappa)$ satisfies $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ is trace-annihilating.*

*Proof.* Let $\phi(\rho) = \sum_i K_i\rho K_i^\dagger$. By the cyclic property of trace, $\operatorname{tr}(L(\rho)) = \operatorname{tr}((\sum_i K_i^\dagger K_i)\rho) - \operatorname{tr}(\kappa\rho) - \operatorname{tr}(\kappa^\dagger\rho) = \operatorname{tr}((\kappa + \kappa^\dagger)\rho) - \operatorname{tr}(\kappa\rho) - \operatorname{tr}(\kappa^\dagger\rho) = 0$. □

### 14.3.2 The Lindblad form

**Definition 14.16** (Lindblad form)**.** A *Lindblad form* consists of a Hermitian matrix $H = H^\dagger$ (the Hamiltonian) and a family of matrices $\{L_j\}_{j=0}^{r-1}$ (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \Big( L_j\rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}_+ \Big),$$

where $[A, B] = AB - BA$ and $\{A, B\}_+ = AB + BA$. This is [Wol12, Eq. (7.21)].

**Theorem 14.17** (Lindblad form is trace-annihilating)**.** *Every Lindblad form defines a trace-annihilating linear map.*

*Proof.* The Hamiltonian part satisfies $\operatorname{tr}(i[\rho, H]) = 0$ by the cyclic property. Each dissipator term satisfies $\operatorname{tr}(L_j\rho L_j^\dagger) = \operatorname{tr}(L_j^\dagger L_j\rho) = \operatorname{tr}(\rho L_j^\dagger L_j)$, so the three terms sum to zero. □

**Theorem 14.18** (Lindblad form equals generator decomposition)**.** *A Lindblad form with Hamiltonian $H$ and operators $\{L_j\}$ defines the same linear map as the generator decomposition $(\phi, \kappa)$ with $\phi(\rho) = \sum_j L_j\rho L_j^\dagger$ and $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$. This is [Wol12, Eq. (7.24)].*

*Proof.* Compute $\kappa^\dagger = -iH + \frac{1}{2}\sum_j L_j^\dagger L_j$ using $H^\dagger = H$ and $(A^\dagger A)^\dagger = A^\dagger A$. Setting $S = \sum_j L_j^\dagger L_j$, expand $\phi(\rho) - \kappa\rho - \rho\kappa^\dagger = \sum_j L_j\rho L_j^\dagger - iH\rho - \frac{1}{2}S\rho + i\rho H - \frac{1}{2}\rho S$, which is $i[\rho, H] + \sum_j(L_j\rho L_j^\dagger - \frac{1}{2}L_j^\dagger L_j\rho - \frac{1}{2}\rho L_j^\dagger L_j)$. □

**Theorem 14.19** (Lindblad form is CCP)**.** *Every Lindblad form defines a conditionally completely positive map.*

*Proof.* By Theorem 14.18, the Lindblad form equals a generator decomposition with $\phi$ CP. □

**Definition 14.20** (Commutator form of Lindblad equation)**.** The *commutator form* of the Lindblad equation writes the generator as

$$L(\rho) = i[\rho, H] + \tfrac{1}{2}\sum_j ([L_j,\, \rho L_j^\dagger] + [L_j\rho,\, L_j^\dagger]),$$

where $[A, B] = AB - BA$. This is [Wol12, Eq. (7.22)].

**Theorem 14.21** (Commutator form equals standard Lindblad form)**.** *The commutator form (Definition 14.20) defines the same linear map as the standard Lindblad form (Definition 14.16).*

*Proof.* Expanding the commutator brackets $[L_j, \rho L_j^\dagger]$ and $[L_j\rho, L_j^\dagger]$ yields the standard dissipator terms. □

### 14.3.3 Prop. 7.2: Characterization of conditional complete positivity

**Theorem 14.22** (Prop. 7.2, direction 1→2)**.** *If $L$ is CCP and $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then*

$$P((L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|))P \geq 0.$$

*This is [Wol12, Prop. 7.2].*

*Proof.* Write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. The left- and right-multiplication terms are annihilated by $P$, so the projected Choi matrix reduces to the projected Choi matrix of $\phi$, which is positive. □

**Theorem 14.23** (Prop. 7.2, direction 2→1)**.** *If $L$ is Hermiticity-preserving and $P((L\otimes\mathrm{id})(|\Omega\rangle\langle\Omega|))P \geq 0$ where $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then $L$ is CCP. This is [Wol12, Prop. 7.2].*

*Proof.* Decompose the Hermitian Choi matrix as $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$ with $Q \geq 0$ supported on $|\Omega\rangle^\perp$. The Choi–Jamiolkowski isomorphism applied to $Q$ gives the CP map $\phi$, and $(\kappa\otimes\mathbb{1})|\Omega\rangle = |\psi\rangle$ defines $\kappa$. □

### 14.3.4 Prop. 7.3: CP semigroups and CCP generators

**Theorem 14.24** (Prop. 7.3: CP semigroup implies CCP generator)**.** *If $e^{tL}$ is CP for all $t \geq 0$, then $L$ is CCP. This is [Wol12, Prop. 7.3].*

*Proof.* From $(e^{tL} \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \geq 0$, expand at infinitesimal $t$ and project onto $P$ to get $P(L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|)P \geq 0$. Then Theorem 14.23 gives CCP. □

**Theorem 14.25** (Prop. 7.3: CCP generator implies CP semigroup)**.** *If $L$ is CCP, then $e^{tL}$ is CP for all $t \geq 0$. This is [Wol12, Prop. 7.3].*

*Proof.* Approximate $e^{tL}$ by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^\dagger + h\phi(\rho), \qquad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to $e^{tL}$, and the cone of CP maps is closed. □

**Theorem 14.26** (Prop. 7.3: CP semigroup iff CCP generator)**.** *$e^{tL}$ is CP for all $t \geq 0$ if and only if $L$ is CCP.*

*Proof.* Combine Theorems 14.24 and 14.25. □

### 14.3.5 Prop. 7.4: Freedom in generator representation

**Theorem 14.27** (Traceless Kraus operators exist)**.** *Assume $d > 0$. Given any Kraus operators $\{K_j\}$, there exist $c_j \in \mathbb{C}$ such that $K'_j = K_j + c_j\mathbb{1}$ is traceless. This is part of [Wol12, Prop. 7.4].*

*Proof.* Set $c_j = -\operatorname{tr}(K_j)/d$. □

**Theorem 14.28** (Shift invariance of generators)**.** *If $K'_j = K_j + c_j\mathbb{1}$ and $\kappa'$ is adjusted per [Wol12, Eq. (7.19)], then $(\phi', \kappa')$ and $(\phi, \kappa)$ define the same generator.*

*Proof.* Direct algebraic expansion. □

### 14.3.6 Prop. 7.4 item 2: Uniqueness of the traceless Lindblad form

**Definition 14.29** (Traceless Kraus operators)**.** A Lindblad form has *traceless Kraus operators* if $\operatorname{tr}(L_j) = 0$ for every $j$.

**Theorem 14.30** (Prop. 7.4, item 2: CP-part uniqueness)**.** *If two Lindblad forms $F, F'$ induce the same generator and both have traceless Kraus operators, then their CP parts agree: $\phi = \phi'$. This is part of [Wol12, Prop. 7.4, item 2].*

*Proof.* Compare projected Choi matrices and use Choi injectivity. □

**Theorem 14.31** (Prop. 7.4, item 2: drift uniqueness modulo phase)**.** *If two Lindblad forms $F, F'$ induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar: $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$. This is part of [Wol12, Prop. 7.4, item 2].*

*Proof.* After identifying $\phi = \phi'$ (Theorem 14.30), the generator equality forces $\Delta\kappa\cdot\rho + \rho\cdot(\Delta\kappa)^\dagger = 0$ for all $\rho$. Setting $\rho = \mathbb{1}$ gives $\Delta\kappa + (\Delta\kappa)^\dagger = 0$ (skew-Hermiticity), which converts the equation to $[\Delta\kappa, \rho] = 0$ for all $\rho$. The scalar commutant lemma gives $\Delta\kappa = c \cdot \mathbb{1}$, and skew-Hermiticity forces $c = i\lambda$ for some $\lambda \in \mathbb{R}$. □

**Theorem 14.32** (Prop. 7.4, item 2: combined uniqueness)**.** *If two Lindblad forms $F, F'$ induce the same generator and both have traceless Kraus operators, then $\phi = \phi'$ and $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$.*

*Proof.* Combine Theorems 14.30 and 14.31. □

### 14.3.7 Trace-annihilation and trace preservation

**Theorem 14.33** (Trace-annihilating generators yield trace-preserving semigroups)**.** *If $L$ is trace-annihilating, then $e^{tL}$ is trace-preserving for every $t \in \mathbb{R}$.*

*Proof.* For fixed $\rho$, the function $f(t) = \operatorname{tr}(e^{tL}(\rho))$ has derivative $f'(t) = \operatorname{tr}(L(e^{tL}(\rho))) = 0$, so $f$ is constant and $\operatorname{tr}(e^{tL}(\rho)) = f(t) = f(0) = \operatorname{tr}(\rho)$. □

**Theorem 14.34** (Trace-preserving semigroups have trace-annihilating generators)**.** *If $e^{tL}$ is trace-preserving for every $t \geq 0$, then $L$ is trace-annihilating.*

*Proof.* Differentiate the identity $\operatorname{tr}(e^{tL}(\rho)) = \operatorname{tr}(\rho)$ at $t = 0$ from the right. □

### 14.3.8 Thm. 7.1: The GKSL/Lindblad theorem

**Definition 14.35** (GKSL generator)**.** A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is a *GKSL generator* if $e^{tL}$ is a quantum channel (CPTP) for all $t \geq 0$.

**Theorem 14.36** (Thm. 7.1, Form (i))**.** *If $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with $\phi$ CP and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then $L$ is a GKSL generator. This is [Wol12, Thm. 7.1, Eq. (7.20)].*

*Proof.* CP follows from Theorem 14.25. The trace constraint implies trace-annihilation by Theorem 14.15, so Theorem 14.33 gives trace preservation. □

**Theorem 14.37** (Thm. 7.1, Form (i) conversely)**.** *If $L$ is a GKSL generator, then there exist a CP map $\phi$ and a matrix $\kappa$ such that*

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$$

*and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$.*

*Proof.* Apply Theorem 14.24 to obtain a CCP decomposition. Then apply Theorem 14.34 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$. □

**Theorem 14.38** (Thm. 7.1: GKSL iff CCP + trace-annihilating)**.** *$L$ is a GKSL generator if and only if $L$ is CCP and trace-annihilating.*

*Proof.* Forward: apply Theorem 14.24 to the CP part and Theorem 14.34 to the TP part. Reverse: combine Theorem 14.25 with Theorem 14.33. □

**Theorem 14.39** (Thm. 7.1: GKSL iff Lindblad form)**.** *$L$ is a GKSL generator if and only if*

$$L(\rho) = i[\rho, H] + \sum_j \Big( L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}_+ \Big)$$

*with $H = H^\dagger$. This is [Wol12, Thm. 7.1].*

*Proof.* Forward: apply Theorem 14.37 to write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with trace constraint. Then choose traceless Kraus operators by Theorem 14.27 and rewrite $\kappa$ as $iH + \frac{1}{2}\phi^*(\mathbb{1})$; Theorem 14.18 gives the Lindblad formula. Reverse: Theorems 14.19 and 14.17 show that every Lindblad form satisfies the right-hand side of Theorem 14.38. □

### 14.3.9 Kossakowski matrix form

**Definition 14.40** (Kossakowski form)**.** A *Kossakowski form* consists of $H = H^\dagger$, a finite family of matrices $\{F_k\}_{k=0}^{n-1}$, and a positive semidefinite matrix $C \in M_n(\mathbb{C})$, defining

$$L(\rho) = i[\rho, H] + \tfrac{1}{2} \sum_{k,l} C_{lk} ([F_k, \rho F_l^\dagger] + [F_k \rho, F_l^\dagger]).$$

In Wolf's formula one takes $\{F_k\}$ to be a basis of traceless matrices; the Lean structure keeps only the algebraic data needed for the conversion to Lindblad form.

**Theorem 14.41** (Kossakowski form iff Lindblad form)**.** *A linear map admits a Kossakowski form iff it admits a Lindblad form.*

*Proof.* Forward: diagonalize $C = B^\dagger B$ and set $L_j = \sum_k B_{jk} F_k$. Reverse: keep the same Hamiltonian and family $F_k = L_k$, and take $C = \mathbb{1}$. □

## 14.4 Primitivity and irreducibility of QDS

*Remark* 14.42 (Status note). The Lean declarations for Proposition 7.5 are currently implemented via axioms to keep CI/elaboration time manageable; accordingly, these results are marked `notready` in the blueprint until full proofs are restored. The Proposition 7.6 implication (4)→(2) sorry was closed in PR #85, so the full equivalence chain is now complete.

**Theorem 14.43** (Prop. 7.5: Irreducible semigroup implies primitive). *Let $T_t = e^{tL}$ be a quantum dynamical semigroup. If $T_{t_0}$ is irreducible for some $t_0 > 0$, then $T_t$ is primitive (and irreducible) for every $t > 0$. This is [Wol12, Prop. 7.5].*

*Proof.* The unique faithful density fixed point $\sigma$ of $T_{t_0}$ is also fixed by all $T_u$ (semigroup commutativity), so by uniqueness it is the common fixed point. Irreducibility of each $T_t$ then follows from uniqueness of the positive definite fixed point and the spectral-gap argument. Primitivity is then the one-dimensional peripheral-eigenvalue characterisation, using that $T_t$ irreducible implies all peripheral eigenvalues are roots of unity, and the faithful fixed point forces those eigenvalues to equal 1. ☐

**Theorem 14.44** (Prop. 7.5: Irreducibility iff primitivity for QDS). *For a quantum dynamical semigroup $T_t = e^{tL}$:*

$$(\exists\, t_0 > 0,\ T_{t_0} \text{ irreducible}) \iff (\forall\, t > 0,\ T_t \text{ primitive and irreducible}).$$

*This is [Wol12, Prop. 7.5].*

*Proof.* Forward: apply Theorem 14.43. Reverse: take $t_0 = 1$ (or any positive time); primitivity implies irreducibility by definition. ☐

## 14.5 Kernel of the adjoint Liouvillian

**Theorem 14.45** (Thm. 7.2: Adjoint kernel equals commutant (faithful state)). *Let $L$ be a GKSL generator in Lindblad form with Hamiltonian $H$ and Lindblad operators $\{L_j\}$. Define the adjoint generator $L^*$ by $\mathrm{tr}(\rho\, L^*(A)) = \mathrm{tr}(L(\rho)\, A)$. If $L$ has a faithful stationary state $\rho_0 > 0$ with $L(\rho_0) = 0$, then for any $A$:*

$$L^*(A) = 0 \implies [A, H] = [A, L_j] = [A, L_j^\dagger] = 0\ \forall j.$$

*That is, $\ker(L^*) \subseteq \{H, L_j, L_j^\dagger\}'$. This is [Wol12, Thm. 7.2].*

*Proof.* From $L^*(A) = 0$ and $L^*(A^\dagger) = 0$, compute $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$. Faithfulness of $\rho_0$ and $\mathrm{tr}(\rho_0 L^*(A^\dagger A)) = 0$ force $[A, L_j] = 0$ for all $j$. Similarly $[A, L_j^\dagger] = 0$, and then $L^*(A) = 0$ directly gives $[A, H] = 0$. ☐

**Theorem 14.46** (Thm. 7.2: Adjoint kernel equals commutant (full equivalence)). *Under the same hypotheses as Theorem 14.45,*

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}',$$

*i.e. the adjoint kernel equals the commutant of $\{H, L_j, L_j^\dagger\}$. This is [Wol12, Thm. 7.2].*

*Proof.* The inclusion $\supseteq$ is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse $\subseteq$ is Theorem 14.45. ☐

## 14.6 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Prop. 7.6] are as follows.

**Definition 14.47** (Rank-deficient fixed density)**.** Condition (1): there exists a density matrix $\rho_0$ with nontrivial kernel such that $e^{tL}(\rho_0) = \rho_0$ for all $t \geq 0$.

**Definition 14.48** (Rank-deficient kernel element)**.** Condition (2): there exists a density matrix $\rho_0$ with nontrivial kernel satisfying $L(\rho_0) = 0$.

**Definition 14.49** (Invariant compression)**.** Condition (3): there exists a nontrivial orthogonal projector $P$ such that $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ for all $t \geq 0$.

**Definition 14.50** (Block-upper-triangular Lindblad form)**.** Condition (4): there exists a nontrivial projector $P$ and a Lindblad form $(H, \{L_j\})$ for $L$ such that $(1-P)L_jP = 0$ and $(1-P)\kappa P = 0$ for all $j$, where $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$.

**Theorem 14.51** (Prop. 7.6, (1)↔(2))**.** *A rank-deficient density matrix is a fixed point of $e^{tL}$ for all $t \geq 0$ if and only if it lies in $\ker L$. This is [Wol12, Prop. 7.6, (1)↔(2)].*

*Proof.* The equivalence $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$ for all $t \geq 0$ follows from the generator–semigroup correspondence: differentiate at $t = 0$ for the forward direction, and exponentiate for the reverse. □

**Theorem 14.52** (Prop. 7.6, (1)→(3))**.** *If $L$ is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Prop. 7.6, (1)→(3)].*

*Proof.* Let $\rho_0$ be a rank-deficient fixed density. Let $P$ be the support projection of $\rho_0$. Since $\rho_0$ is rank-deficient, $P \neq \mathbb{1}$; since $\rho_0 \neq 0$, $P \neq 0$. Channel positivity and trace preservation force $e^{tL}$ to preserve $PM_D(\mathbb{C})P$. □

**Theorem 14.53** (Prop. 7.6, (3)↔(4))**.** *For a GKSL generator $L$, the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Prop. 7.6, (3)↔(4)].*

*Proof.* (4)→(3): block-upper-triangular Lindblad operators immediately imply that $L$ preserves the compression $PM_D(\mathbb{C})P$, and hence so does $e^{tL}$. (3)→(4): the semigroup preserving $PM_D(\mathbb{C})P$ implies (by differentiation) that $L$ preserves $PM_D(\mathbb{C})P$; the vanishing of $(1-P)L(PXP)$ for all $X$ forces $(1-P)L_jP = 0$ and $(1-P)\kappa P = 0$ via the sum-of-squares identity. □

**Theorem 14.54** (Prop. 7.6, (4)→(2))**.** *If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector $P$, then there exists a density matrix $\rho_0$ with nontrivial kernel satisfying $L(\rho_0) = 0$. This is [Wol12, Prop. 7.6, (4)→(2)].*

*Proof.* The block-upper-triangular structure implies $L$ preserves the compression $PM_d(\mathbb{C})P$. Therefore the semigroup $e^{tL}$ also preserves $PM_d(\mathbb{C})P$ for all $t \geq 0$. By the Brouwer fixed-point theorem, $e^{(1/m)L}$ has a fixed density matrix $\rho_m$ supported in $PM_d(\mathbb{C})P$ for each $m$. Passing to a subsequential limit gives a density matrix $\rho_0$ with $P\rho_0P = \rho_0$ and $L(\rho_0) = 0$. Since $P \neq \mathbb{1}$, the matrix $\rho_0$ has nontrivial kernel. □

**Theorem 14.55** (Prop. 7.6: Full equivalence)**.** *For a GKSL generator $L : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the four conditions*

1. *rank-deficient fixed density,*
2. *rank-deficient kernel element,*
3. *invariant compression, and*
4. *block-upper-triangular Lindblad form*

*are equivalent. This is [Wol12, Prop. 7.6].*

*Proof.* The cycle (1)$\leftrightarrow$(2), (1)$\rightarrow$(3), (3)$\leftrightarrow$(4), and (4)$\rightarrow$(2) closes the equivalence chain. $\square$

# Chapter 15

# The Algebraic Fundamental Theorem

This chapter formalises the one-dimensional MPS results of [MGRPG+18]: the Fundamental Theorem for injective and normal MPS on a finite periodic chain of fixed length. The argument is purely algebraic, working directly with $M_D(\mathbb{C})$. Two preparatory lemmas—the virtual bond gauge (Lemma 15.12) and the proportionality lemma (Lemma 15.13)—carry the entire argument, yielding the main theorem and its corollaries for translation-invariant chains, non-TI descriptions of TI states, physical symmetries, and normal MPS.

Throughout, $D$ is the bond dimension and $d$ is the physical dimension with $d \geq D^2$. The extensions to injective and normal PEPS on arbitrary graphs and the improved system-size bound ($2L + 1$ in place of $3L$) are left for future work. The chain bookkeeping layer (Definition 4.12), including same chain state, injectivity, and cyclic gauge equivalence, was introduced in Chapter 4.

## 15.1 One-sided inverse and decomposition

If $A$ is injective, the linear combination map $\Phi_A(c) := \sum_i c_i A^i : \mathbb{C}^d \to M_D(\mathbb{C})$ is surjective and admits a right inverse.

**Lemma 15.1** (Existence of a right inverse)**.** *If $A$ is injective, then there exists a $\mathbb{C}$-linear map $\Psi : M_D(\mathbb{C}) \to \mathbb{C}^d$ satisfying $\Phi_A \circ \Psi = \mathrm{id}_{M_D(\mathbb{C})}$.*

*Proof.* Surjectivity of $\Phi_A$ yields a linear section by choice. □

**Definition 15.2** (Decomposition map)**.** Let $A$ be an injective MPS tensor. The *decomposition map* $\Psi_A : M_D(\mathbb{C}) \to \mathbb{C}^d$ is a fixed choice of right inverse of $\Phi_A$.

**Lemma 15.3** (Decomposition property)**.** *If $A$ is injective, then for every $X \in M_D(\mathbb{C})$ there exist coefficients $c \in \mathbb{C}^d$ such that*

$$X = \sum_{i=0}^{d-1} c_i\, A^i.$$

*Proof.* Take $c = \Psi_A(X)$. Then $\Phi_A(c) = \Phi_A(\Psi_A(X)) = X$. □

*Remark* 15.4 (Non-uniqueness of the decomposition). When $d > D^2$, the map $\Phi_A$ has a nontrivial kernel, so the section $\Psi_A$ is not unique. The statements that follow depend only on $\Phi_A \circ \Psi_A = \mathrm{id}$, so the choice does not affect the conclusions.

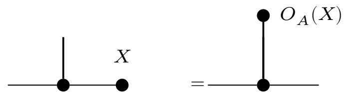
## 15.2 Physical realisation of virtual insertions

For an injective tensor $A$, the physical realisation map encodes the effect of inserting a virtual matrix $X$ on the bond as a $d \times d$ operation on the physical indices.

**Definition 15.5** (Physical realisation map)**.** Let $A$ be an injective tensor. For $X \in M_D(\mathbb{C})$, define the $d \times d$ matrix $O_A(X) \in M_d(\mathbb{C})$ by

$$O_A(X)_{ij} := (\Psi_A(A^i X))_j,$$

so that row $i$ of $O_A(X)$ contains the decomposition coefficients of $A^i X$ in the spanning set $\{A^j\}$. Diagrammatically, the inserted virtual matrix is re-expressed as a physical operation on the same local tensor:




The map $X \mapsto O_A(X)$ is $\mathbb{C}$-linear.

**Lemma 15.6** (Defining property of the physical realisation map)**.** *For every $X \in M_D(\mathbb{C})$ and every physical index $i$,*

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

*Proof.* By definition, $O_A(X)_{ij} = (\Psi_A(A^i X))_j$. Applying $\Phi_A$ gives $\sum_j O_A(X)_{ij} A^j = \Phi_A(\Psi_A(A^i X)) = A^i X$, where the last step uses $\Phi_A \circ \Psi_A = \mathrm{id}$. □

**Theorem 15.7** (Physical realisation is multiplicative – basis case)**.** *Let $A$ be an injective MPS tensor such that $\{A^i\}_{i=0}^{d-1}$ is a* basis *of $M_D(\mathbb{C})$ (i.e. $d = D^2$ and the matrices are linearly independent). For all $X, Y \in M_D(\mathbb{C})$,*

$$O_A(XY) = O_A(X)\, O_A(Y).$$

*Proof.* Applying Lemma 15.6 to $X$ and substituting the resulting formula for $A^j Y$ gives $A^i(XY) = \sum_k (O_A(X)\, O_A(Y))_{ik} A^k$. Applying Lemma 15.6 directly to $XY$ gives $A^i(XY) = \sum_k O_A(XY)_{ik} A^k$. Since $\{A^k\}$ is a basis, the coefficients agree. □

*Remark* 15.8 (General injective case). When $d > D^2$, the decomposition is non-unique, so $O_A(XY) = O_A(X)\, O_A(Y)$ need not hold coefficient-wise even though both sides define the same matrix in $M_D(\mathbb{C})$. The general multiplicativity for arbitrary injective $A$ is tracked as an open problem (GitHub issue #74); the proofs below avoid it by working with trace pairings instead.

## 15.3 Intermediate lemmas

**Lemma 15.9** (Blocking preserves injectivity)**.** *Let $A_0, \dots, A_{m-1}$ be injective MPS tensors with compatible bond dimensions. Then the blocked tensor $(A_0 \otimes \cdots \otimes A_{m-1})^{(i_0, \dots, i_{m-1})} := A_0^{i_0} A_1^{i_1} \cdots A_{m-1}^{i_{m-1}}$ is again injective.*

*Proof.* The span of the blocked matrices contains all products $A_0^{i_0} \cdots A_{m-1}^{i_{m-1}}$. Since each $\{A_k^i\}_i$ spans $M_D(\mathbb{C})$, these products span all of $M_D(\mathbb{C})$. □

*Remark* 15.10 (Lean formalisation note)*.* The Lean proof uses a combined tensor approach () rather than literal tensor blocking: it shows that if any site tensor is injective, the combined tensor (union of all site matrices) spans $M_D(\mathbb{C})$. This serves the same role in the chain fundamental theorem proof.

**Lemma 15.11** (Centre of $M_D(\mathbb{C})$)**.** *If $Z \in M_D(\mathbb{C})$ commutes with every element of $M_D(\mathbb{C})$, then $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}$.*

*Proof.* Standard: $Z$ commutes with every matrix unit $E_{ij}$, which forces all off-diagonal entries to vanish and all diagonal entries to agree. □

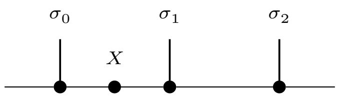
## 15.4 Same state forces a virtual bond gauge

This section proves Lemma 1 of [MGRPG⁺18]: if two injective chains of length $n \geq 3$ generate the same state, then on each bond the virtual insertion algebras are related by an inner automorphism.

**Lemma 15.12** (Virtual bond gauge – Lemma 1 of [MGRPG⁺18])**.** *Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ and $\mathbf{B} = (B_0, \dots, B_{n-1})$ be two injective non-TI periodic chains of length $n \geq 3$ that generate the same chain state. Fix a bond position $k \in \{0, \dots, n-1\}$. Then there exists an invertible matrix $Z_k \in \mathrm{GL}_D(\mathbb{C})$, unique up to a nonzero scalar multiple, such that for every $X \in M_D(\mathbb{C})$,*

$$O_{A,k}(X) = O_{B,k}(Z_k^{-1} X Z_k),$$

*where $O_{A,k}$ (resp. $O_{B,k}$) denotes the physical realisation map obtained by blocking the sites adjacent to bond $k$ in the $\mathbf{A}$-chain (resp. $\mathbf{B}$-chain) to form a three-site chain. Equivalently, inserting $X$ on bond $k$ in the $\mathbf{A}$-chain gives the same physical operation as inserting $Z_k^{-1} X Z_k$ on bond $k$ in the $\mathbf{B}$-chain. After blocking to three sites, the coefficient-level relation formalized in Lean by  is the equality between the blocked $\mathbf{A}$-chain contraction*




*and the corresponding blocked $\mathbf{B}$-chain contraction with $X$ replaced by $Z_k^{-1} X Z_k$.*

*Proof.* Block the sites on either side of bond $k$ into a three-site chain; Lemma 15.9 ensures injectivity, and both $O_{A,k}$ and $O_{B,k}$ are algebra homomorphisms from $M_D(\mathbb{C})$ (established in [MGRPG⁺18] via diagrammatic inversion rather than coefficient uniqueness; see Remark 15.8). Since both chains generate the same state, for every $X \in M_D(\mathbb{C})$ there is a unique $Y \in M_D(\mathbb{C})$ with $O_{A,k}(X) = O_{B,k}(Y)$; set $T(X) = Y$. In the Lean proof, this step is carried out on the blocked three-site coefficient function  and the combined-tensor linear extension, rather than by treating $O_{A,k}$ and $O_{B,k}$ as primitive black boxes. Linearity of $O_{A,k}$ and $O_{B,k}$ and uniqueness of $Y$ make $T$ $\mathbb{C}$-linear, and since both maps are algebra homomorphisms, $T$ is multiplicative. Thus $T$ is a nonzero $\mathbb{C}$-algebra endomorphism of $M_D(\mathbb{C})$, hence bijective by Theorem 4.8. Skolem–Noether (Theorem 4.9) gives $Z_k \in \mathrm{GL}_D(\mathbb{C})$ with $T(X) = Z_k^{-1} X Z_k$, and uniqueness up to a scalar follows because two inner automorphisms $\mathrm{Ad}(Z)$ and $\mathrm{Ad}(Z')$ agree iff $Z' Z^{-1}$ is central (Lemma 15.11). □

## 15.5 Aligned gauges force proportionality

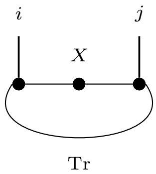
After applying Lemma 15.12 to each bond and absorbing the gauge matrices, one obtains two chains whose virtual-insertion operations agree on every bond. Lemma 2 of [MGRPG+18] shows that this agreement forces the local tensors to be proportional.

**Lemma 15.13** (Aligned gauges force proportionality – Lemma 2 of [MGRPG+18])**.** *Let $A_1, A_2$ and $B_1, B_2$ be injective MPS tensors with the same bond dimension $D$. Suppose that for all $X \in M_D(\mathbb{C})$ (on the internal bond) and all $Y \in M_D(\mathbb{C})$ (on the external bond),*

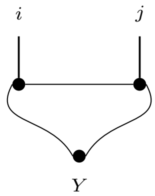
$$\operatorname{tr}(A_1^i\, X\, A_2^j) = \operatorname{tr}(B_1^i\, X\, B_2^j) \qquad \textit{for all } i, j, \tag{15.1}$$

$$\operatorname{tr}(A_2^j\, Y\, A_1^i) = \operatorname{tr}(B_2^j\, Y\, B_1^i) \qquad \textit{for all } i, j. \tag{15.2}$$

*Diagrammatically, (15.1) is the equality obtained by inserting $X$ on the bond between the two sites,*




*while (15.2) is the corresponding equality with the insertion $Y$ on the outer trace bond,*




*so the two hypotheses distinguish the internal and external virtual loops of the two-site periodic chain.*

*Then there exists a nonzero scalar $\lambda \in \mathbb{C}^\times$ such that*

$$A_1^i = \lambda\, B_1^i \qquad \textit{and} \qquad A_2^j = \lambda^{-1}\, B_2^j$$

*for all physical indices $i, j$.*

*Proof.* By cyclicity and non-degeneracy of the trace pairing, conditions (15.1) and (15.2) give

$$A_2^j A_1^i = B_2^j B_1^i \qquad \text{and} \qquad A_1^i A_2^j = B_1^i B_2^j \qquad \text{for all } i, j.$$

In Lean, these two trace-to-product reductions are isolated as the helper lemmas  and . Write $\mathbb{1}_D = \sum_j c_j A_2^j$ using the decomposition map $\Psi_{A_2}$, and define $Z := \sum_j c_j B_2^j \in M_D(\mathbb{C})$. The first product identity gives $A_1^i = Z B_1^i$, and the second gives $A_1^i = B_1^i Z$. Hence $Z$ commutes with every $B_1^i$; since $\{B_1^i\}$ spans $M_D(\mathbb{C})$, $Z$ is central. By Lemma 15.11, $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}^\times$ (nonzero since the tensors are nonzero). Thus $A_1^i = \lambda B_1^i$, and substituting into the first product identity forces $A_2^j = \lambda^{-1} B_2^j$. ∎

## 15.6 The Fundamental Theorem for injective chains

Combining Lemma 15.12 with Lemma 15.13 yields the main theorem of [MGRPG+18] for non-translation-invariant injective MPS.

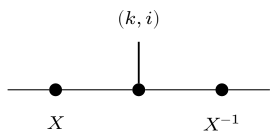
**Theorem 15.14** (Fundamental Theorem for injective chains – Theorem 1 of [MGRPG+18])**.** *Let* $\mathbf{A} = (A_0, \ldots, A_{n-1})$ *and* $\mathbf{B} = (B_0, \ldots, B_{n-1})$ *be two injective non-TI periodic chains of length* $n \geq 3$ *that generate the same chain state. Then* $\mathbf{A}$ *and* $\mathbf{B}$ *are cyclically gauge equivalent: there exist invertible matrices* $Z_0, \ldots, Z_{n-1} \in \mathrm{GL}_D(\mathbb{C})$*, unique up to a common nonzero scalar factor, such that*

$$B_k^i = Z_k^{-1}\, A_k^i\, Z_{k+1}$$

*for every site* $k \in \{0, \ldots, n-1\}$ *and every physical index* $i$*, with* $Z_n = Z_0$*.*

*Proof.* Apply Lemma 15.12 at each bond to obtain $Z_k \in \mathrm{GL}_D(\mathbb{C})$ with $O_{A,k}(X) = O_{B,k}(Z_k^{-1} X Z_k)$. Redefine $\widetilde{B}_k^i := Z_k B_k^i Z_{k+1}^{-1}$; the virtual insertion operations of $\mathbf{A}$ and $\widetilde{\mathbf{B}}$ then agree on every bond. For each adjacent pair $(k, k{+}1)$, block the remaining sites into a two-site chain; the agreement on both bonds gives exactly hypotheses (15.1)–(15.2) of Lemma 15.13, yielding $A_k^i = \lambda_k \widetilde{B}_k^i$ for some $\lambda_k \in \mathbb{C}^\times$. Choosing a configuration $\sigma$ with nonzero trace and substituting $A_k^{\sigma_k} = \lambda_k \widetilde{B}_k^{\sigma_k}$ into the chain-state identity forces $\prod_k \lambda_k = 1$. Set $\mu_k = \prod_{j<k} \lambda_j$ and replace $Z_k$ by $\mu_k Z_k$; then $B_k^i = Z_k^{-1} A_k^i Z_{k+1}$ for all $k$ and $i$, with $Z_n = Z_0$. Uniqueness up to a common scalar follows from the uniqueness clause of Lemma 15.12. □

*Remark* 15.15 (Lean proof route)*.* The formal theorem  does not currently implement the bond-by-bond proof sketched above. Instead, it packages the whole chain into the combined tensor and applies the single-block fundamental theorem to obtain one uniform gauge matrix $X$ acting on every combined-site matrix:




for each site index $k$ and physical index $i$. The resulting uniform gauge is then read as a special case of cyclic gauge equivalence by using the same virtual gauge on every bond.

*Remark* 15.16 (Stronger hypothesis in the Lean formalisation)*.* The Lean theorem  assumes the equal-MPV condition on the combined tensor (trace agreement for mixed-site words of *all* lengths), which is stronger than the paper's equal-state condition (trace agreement at a single chain length $n \geq 3$). The paper closes this gap via a blocking argument requiring $n \geq 3$. This reduction step is now packaged as an abstract interface  that encapsulates the blocking hypothesis. Given this reduction, the theorem  delivers the chain FT endpoint under the weaker equal-state hypothesis; however, the actual proof that the equal-state condition implies equal MPV (i.e. the content of this reduction) has not yet been established in the formalization.

*Remark* 15.17 (Relationship to the translation-invariant all-sizes theorem)*.* Theorem 15.14 handles non-TI chains at fixed length $n \geq 3$, concluding gauge equivalence up to a common scalar. The all-sizes translation-invariant result () instead assumes agreement for every system size; the additional hypothesis eliminates the phase ambiguity, yielding exact gauge equivalence $B^i = X A^i X^{-1}$. The two results are complementary and neither implies the other in full generality.

### 15.6.1 Finite-length MPV agreement

**Definition 15.18** (Finite-length MPV agreement)**.** Two tensors $A$ and $B$ of the same bond dimension satisfy $\mathrm{SameMPVFrom}(N_0)$ if they generate the same MPV family for all system sizes $N \geq N_0$:

$$\mathrm{mpv}(A, \sigma) = \mathrm{mpv}(B, \sigma) \qquad \text{for all } N \geq N_0,\ \sigma \in [d]^N.$$

**Theorem 15.19** (Finite-length agreement implies full agreement)**.** *Let $A$ be an injective tensor and $B$ a tensor of the same bond dimension. If* $\mathrm{SameMPVFrom}(N_0, A, B)$ *holds for some $N_0$, then* $\mathrm{SameMPV}(A, B)$.

*Proof.* Write $1 = \sum_i c_i A^i$ (by injectivity) and set $S = \sum_i c_i B^i$.

1. Show $\mathrm{wordSpan}(B, N_0) = M_D(\mathbb{C})$ via the composition identity $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$ (from trace agreement at length $N_0 + 1$) and a finrank transfer argument.
2. Derive $S = 1$ by trace nondegeneracy: for every $M \in M_D(\mathbb{C})$, $\mathrm{tr}(M(S - 1)) = 0$.
3. Downward induction on $k = N_0 - |w|$: decompose $\mathrm{tr}(w_A)$ and $\mathrm{tr}(w_B)$ using $1 = \sum c_i A^i$ and $S = 1$ respectively, then apply the inductive hypothesis at length $|w| + 1$.

□

**Theorem 15.20** (Strengthened single-block FT (finite-length))**.** *If $A$ is injective and $A$, $B$ agree on all MPV coefficients for system sizes $N \geq N_0$ (for any threshold $N_0$), then $A$ and $B$ are gauge equivalent.*

*Proof.* Apply Theorem 15.19 to obtain $\mathrm{SameMPV}(A, B)$, then invoke the single-block FT (Theorem 4.11). □

## 15.7 Corollaries

### 15.7.1 Translation-invariant chains

**Corollary 15.21** (TI case)**.** *Let $A$ and $B$ be injective MPS tensors of bond dimension $D$. Suppose the two translation-invariant chains $(A, A, \dots, A)$ and $(B, B, \dots, B)$ of length $n \geq 3$ generate the same chain state. Then there exists an invertible $Z \in \mathrm{GL}_D(\mathbb{C})$, unique up to a scalar, and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$, such that*

$$B^i = \lambda\, Z^{-1}\, A^i\, Z$$

*for every physical index $i$.*

*Proof.* The blueprint-level derivation applies Theorem 15.14 to the two constant chains to obtain $Z_0, \dots, Z_{n-1}$ with $B^i = Z_k^{-1} A^i Z_{k+1}$ for all $k$ and $i$. Setting $G_k = Z_k Z_0^{-1}$, the relation $Z_k^{-1} A^i Z_{k+1} = Z_{k+1}^{-1} A^i Z_{k+2}$ shows $A^i = G_k^{-1} A^i G_k$ for each $k$. Since $\{A^i\}$ spans $M_D(\mathbb{C})$, each $G_k$ commutes with all of $M_D(\mathbb{C})$, so $G_k = \mu_k \mathbb{1}$ by Lemma 15.11. Hence $Z_k = \mu_k Z_0$, and the cyclic condition $Z_n = Z_0$ forces $(\mu_1)^n = 1$. Setting $Z = Z_0$ and $\lambda = \mu_1$ yields the result. The current Lean proof route is more direct: the theorem obtains one uniform gauge on the combined tensor immediately, and then packages the result with the scalar parameter $\lambda$, realized in the checked proof by $\lambda = 1$. □

### 15.7.2 Non-TI descriptions of TI states

**Corollary 15.22** (Non-TI realisation of a TI state)**.** *Let* $\mathbf{A} = (A_0, \dots, A_{n-1})$ *be an injective non-TI chain of length* $n \geq 3$*. Suppose the state generated by* $\mathbf{A}$ *is translation-invariant, i.e. it equals the state generated by the cyclically shifted chain* $(A_1, \dots, A_{n-1}, A_0)$*. Then* $\mathbf{A}$ *admits a translation-invariant description with the same bond dimension: there exist an injective tensor* $B$ *such that the TI chain* $(B, \dots, B)$ *generates the same state as* $\mathbf{A}$*. Explicitly,* $B^i = A_0^i Z_0^{-1}$ *(right multiplication,* not *conjugation), where* $Z_0$ *is defined below.*

*Proof.* The cyclic shift $\sigma(\mathbf{A}) = (A_1, \dots, A_{n-1}, A_0)$ generates the same state by hypothesis. Apply Theorem 15.14: there exist invertible $Z_0, \dots, Z_{n-1}$ with $A_{k+1}^i = Z_k^{-1} A_k^i Z_{k+1}$ for all $k$ and $i$ (indices mod $n$). Define $L_k = Z_0 Z_1 \cdots Z_{k-1}$ and $R_k = Z_1 Z_2 \cdots Z_k$ for $k \geq 1$, with $L_0 = R_0 = \mathbb{1}$. Iterating the gauge relation gives $A_k^i = L_k^{-1} A_0^i R_k$. Since $R_k L_{k+1}^{-1} = Z_0^{-1}$ for every $k$, substituting into the trace yields

$$\operatorname{tr}(A_0^{\sigma_0} \cdots A_{n-1}^{\sigma_{n-1}}) = \operatorname{tr}(A_0^{\sigma_0} Z_0^{-1} A_0^{\sigma_1} Z_0^{-1} \cdots A_0^{\sigma_{n-1}} Z_0^{-1}).$$

Hence the state is generated by the TI chain with tensor $B^i := A_0^i Z_0^{-1}$. Since $A_0$ is injective and $Z_0$ is invertible, $B$ is injective. ☐

### 15.7.3 Physical symmetries give virtual gauges

**Corollary 15.23** (Physical symmetry implies virtual gauge)**.** *Let* $A$ *be an injective MPS tensor and let* $U \in \mathrm{GL}_d(\mathbb{C})$ *be a unitary acting on the physical index. Fix* $n \geq 3$ *and suppose the* $n$*-site TI state satisfies*

$$U^{\otimes n} |\psi_n(A)\rangle = |\psi_n(A)\rangle.$$

*Then there exist* $V \in \mathrm{GL}_D(\mathbb{C})$ *and a scalar* $\lambda \in \mathbb{C}^\times$ *with* $\lambda^n = 1$ *such that*

$$\sum_{j=0}^{d-1} U_{ij} A^j = \lambda V^{-1} A^i V$$

*for every physical index* $i$*.*

*Proof.* Define $B^i := \sum_j U_{ij} A^j$. Since $U$ is invertible, $\{B^i\}$ spans $M_D(\mathbb{C})$, so $B$ is injective. The symmetry condition means the TI chains $(A, \dots, A)$ and $(B, \dots, B)$ of length $n$ generate the same state. Corollary 15.21 gives $V \in \mathrm{GL}_D(\mathbb{C})$ and $\lambda$ with $\lambda^n = 1$ and $B^i = \lambda V^{-1} A^i V$ for all $i$. ☐

**Definition 15.24** (Physical-index rotation)**.** Given a matrix $u \in M_d(\mathbb{C})$ and an MPS tensor $A$, the *physical-index rotation* is the tensor $(\text{rotatePhysical } u\, A)^i := \sum_j u_{ij} A^j$.

**Corollary 15.25** (Periodic symmetry corollary — assembly step)**.** *Let* $A$ *be in irreducible form II and let* $u$ *be a matrix such that* $B :=$ rotatePhysical $u\, A$ *is also in irreducible form II and* $\mathcal{V}(A) = \mathcal{V}(B)$*. Then there exists* $m > 0$ *such that* $A$ *and* $B$ *are* $\mathbb{Z}_m$*-gauge equivalent.*

*Proof.* Direct application of the periodic equal-case fundamental theorem (Theorem 13.25) to $A$ and $B$. ☐

### 15.7.4 Virtual representation theorem

With a full group of on-site symmetries (rather than just a single unitary), the individual gauges can be assembled into a projective representation on the bond space.

**Definition 15.26** (Twisted tensor)**.** Given a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$ on the physical index, the *g-twisted tensor* is

$$\widetilde{A}^i_g \;:=\; \sum_{j=0}^{d-1} U(g)_{ij}\, A^j \qquad \text{for each } i \in \{0,\dots,d-1\}.$$

**Definition 15.27** (On-site symmetry)**.** A tensor $A$ is *on-site symmetric* under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$ if $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$ for every $g \in G$.

**Lemma 15.28** (Functoriality of twisting)**.** *The twisted tensor satisfies the composition law*

$$\widetilde{A}_{gh} \;=\; \widetilde{(\widetilde{A}_h)}_g,$$

*i.e. twisting by $gh$ is the same as first twisting by $h$ and then by $g$.*

*Proof.* Expanding both sides using $U(gh) = U(g)\,U(h)$, then swapping the order of summation. ☐

**Lemma 15.29** (Identity twist)**.** *Twisting by the identity group element is trivial: $\widetilde{A}_1 = A$.*

*Proof.* Since $U(1) = I_d$, each component reduces to $\sum_j \delta_{ij}\, A^j = A^i$. ☐

**Lemma 15.30** (Symmetric twists are gauge equivalent)**.** *If $A$ is injective and on-site symmetric under $U$, then for every $g \in G$ the twisted tensor $\widetilde{A}_g$ is gauge equivalent to $A$.*

*Proof.* On-site symmetry gives $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$, and equal MPV for injective tensors implies gauge equivalence. ☐

**Theorem 15.31** (Virtual symmetry equation)**.** *If $A$ is injective and on-site symmetric under $U$, then for every $g \in G$ there exist an invertible matrix $X(g) \in \mathrm{GL}_D(\mathbb{C})$ and a nonzero scalar $\varphi(g) \in \mathbb{C}^\times$ (in fact $\varphi = 1$ in the single-block case) such that $\widetilde{A}^i_g = \varphi(g)\, X(g)\, A^i\, X(g)^{-1}$ for all $i$.*

*Proof.* Apply Lemma 15.30 to obtain $X(g)$, then set $\varphi(g) = 1$. ☐

**Lemma 15.32** (Gauge uniqueness up to scalar)**.** *Let $A$ be an injective tensor. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both satisfy $B^i = X\, A^i\, X^{-1} = Y\, A^i\, Y^{-1}$ for all $i$, then there exists a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

*Proof.* The matrix $Z = Y^{-1}X$ commutes with every $A^i$. Since $A$ is injective, $\{A^i\}$ spans $M_D(\mathbb{C})$, so $Z$ lies in the centre of $M_D(\mathbb{C})$. By Lemma 15.11, $Z$ is a scalar matrix. Since $Z$ is invertible, that scalar is nonzero, and $Y = u \cdot X$. ☐

**Definition 15.33** (Scalar 2-cochain)**.** A *scalar 2-cochain* on a group $G$ is a function $\omega : G \times G \to \mathbb{C}^\times$.

**Definition 15.34** (Projective representation)**.** Given a scalar 2-cochain $\omega$, a *projective representation* of $G$ on $\mathbb{C}^D$ is a map $\rho : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying

$$\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh) \qquad \text{for all } g, h \in G.$$

**Lemma 15.35** (Associativity forces the cocycle condition)**.** *If $\rho$ is a projective representation with factor system $\omega$ and $D > 0$, then $\omega$ satisfies the 2-cocycle identity:*

$$\omega(g,h)\,\omega(gh,k) = \omega(g,hk)\,\omega(h,k).$$

*Proof.* Expanding $\rho(g)\,(\rho(h)\,\rho(k))$ and $(\rho(g)\,\rho(h))\,\rho(k)$ using the projective multiplication law and equating via matrix associativity. The hypothesis $D > 0$ allows cancellation of the invertible matrix factor. □

**Theorem 15.36** (Virtual representation theorem for injective MPS)**.** *Let $A$ be an injective MPS tensor that is on-site symmetric under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$. Then there exist:*

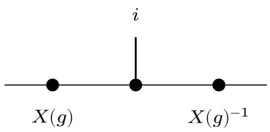
1. *a unit-valued 2-cochain $\omega : G \times G \to \mathbb{C}^\times$, and*
2. *a map $\rho : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying*

$$\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh) \qquad \textit{for all } g, h \in G,$$

*such that, defining the gauge matrices $X(g) := \rho(g^{-1})$, one has*

$$\widetilde{A}^i_g \;=\; X(g)\,A^i\,X(g)^{-1} \qquad \textit{for every } g \in G,\ i \in \{0, \dots, d{-}1\}.$$

*In tensor-network notation, for each fixed $g$ this is the local gauge relation*




*with the black node denoting the tensor named in the surrounding formula and the two side nodes acting on the virtual legs. In particular, $\rho$ is a projective representation of $G$ on the bond space $\mathbb{C}^D$.*

*Proof.* Since $A$ is injective and on-site symmetric, each twisted tensor $\widetilde{A}_g$ is gauge equivalent to $A$ (Lemma 15.30). Choose $X(g) \in \mathrm{GL}_D(\mathbb{C})$ with $\widetilde{A}^i_g = X(g)\,A^i\,X(g)^{-1}$ for all $i$.

By functoriality (Lemma 15.28), the product $X(h)\,X(g)$ also intertwines $A$ with $\widetilde{A}_{gh}$. Since $X(gh)$ does the same, gauge uniqueness (Lemma 15.32) gives a unit scalar $\omega'(h,g)$ with $X(h)\,X(g) = \omega'(h,g)\,X(gh)$.

Defining $\rho(g) := X(g^{-1})$ converts this to the standard law: $\rho(g)\,\rho(h) = \omega(g,h)\,\rho(gh)$ where $\omega(g,h) := \omega'(h^{-1}, g^{-1})$. □

**Corollary 15.37** (2-cocycle identity for the factor system)**.** *If $D > 0$, the factor system $\omega$ from Theorem 15.36 satisfies the 2-cocycle identity:*

$$\omega(g,h)\,\omega(gh,k) = \omega(g,hk)\,\omega(h,k) \qquad \textit{for all } g,h,k \in G.$$

*Proof.* Theorem 15.36 gives $\rho$ and $\omega$. By Lemma 15.35, the cocycle identity follows from associativity of matrix multiplication and invertibility of the representation matrices. □

### 15.7.5 Coboundary equivalence and cohomology classes

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient $H^2(G, \mathrm{U}(1))$.

**Definition 15.38** (Coboundary)**.** A scalar 2-cocycle $\omega$ is a *coboundary* if there exists $\varphi \colon G \to \mathbb{C}^\times$ such that

$$\omega(g,h) = \varphi(g)\,\varphi(h)\,\varphi(gh)^{-1} \qquad \text{for all } g, h \in G.$$

**Definition 15.39** (Cohomologous cocycles)**.** Two cocycles $\omega_1, \omega_2$ are *cohomologous* (written $\omega_1 \sim \omega_2$) if there exists $\varphi\colon G \to \mathbb{C}^\times$ with

$$\omega_1(g,h) = \varphi(g)\,\varphi(h)\,\varphi(gh)^{-1}\,\omega_2(g,h) \qquad \text{for all } g,h \in G.$$

**Theorem 15.40** (Cohomologous-to is an equivalence relation)**.** *The relation "cohomologous to" is reflexive, symmetric, and transitive on the set of scalar 2-cocycles.*

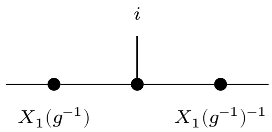
*Proof.* Reflexivity: take $\varphi = 1$. Symmetry: replace $\varphi$ by $\varphi^{-1}$. Transitivity: compose the witnesses $\varphi$ and $\psi$ pointwise. ☐

**Lemma 15.41** (Coboundary iff cohomologous to the trivial cocycle)**.** *A cocycle $\omega$ is a coboundary if and only if $\omega \sim \mathbf{1}$, where $\mathbf{1}(g,h) = 1$ for all $g,h$.*

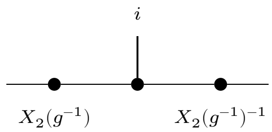
*Proof.* Both directions follow by absorbing or introducing the factor $\mathbf{1}(g,h) = 1$ via $x \cdot 1 = x$. ☐

**Theorem 15.42** (Gauge independence of the cocycle class)**.** *If two projective representations $\rho_1, \rho_2$ (with cocycles $\omega_1, \omega_2$) both arise as virtual representations of the same injective MPS tensor $A$ under an on-site symmetry $U$, then $\omega_1 \sim \omega_2$.*

*Proof.* For each $g$, gauge uniqueness gives a unit scalar $f(g)$ with $X_2(g^{-1}) = f(g)\,X_1(g^{-1})$. Equivalently, for each fixed $g$ the two local gauges




and




describe the same twisted tensor, so they differ by a scalar. Substituting into the projective law and cancelling $X_1(g \cdot h)$ yields $\omega_2(g,h) = f(g)\,f(h)\,f(gh)^{-1}\,\omega_1(g,h)$. ☐

### 15.7.6 Permutation-based physical symmetry

The linear-combination twist $\widetilde{A}^i_g = \sum_j U(g)_{ij}\,A^j$ is the standard form used in the virtual representation theorem. An alternative formulation replaces the matrix action by a permutation of the physical index; this is useful when the symmetry acts by reordering basis states rather than by a general linear map.

**Definition 15.43** (Permutation symmetry data)**.** An *on-site symmetry datum* for a group $G$ on a physical space of dimension $d$ consists of a matrix-valued map $u : G \to M_d(\mathbb{C})$ and a physical-index action $\sigma : G \to \mathrm{End}(\{0,\dots,d{-}1\})$.

**Definition 15.44** (Permutation-twisted tensor)**.** Given on-site symmetry data $(u,\sigma)$, the *permutation-twisted tensor* at group element $g$ is

$$(A^{\sigma_g})^i \;:=\; A^{\sigma(g)(i)}.$$

**Lemma 15.45** (Identity permutation twist)**.** *If $\sigma(1) = \mathrm{id}$, then $(A^{\sigma_1})^i = A^i$ for all $i$.*

*Proof.* Immediate from $\sigma(1)(i) = i$ for all $i$. ☐

**Remark on composition order.** The linear-combination twist (Lemma 15.28) composes as $\widetilde{(\widetilde{A}_h)}_g = \widetilde{A}_{gh}$, applying $h$ first then $g$. The permutation twist below uses the same left-action convention $\sigma(gh) = \sigma(g) \circ \sigma(h)$, so $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$; the inner permutation $\sigma(h)$ acts first on the index.

**Lemma 15.46** (Composition of permutation twists)**.** *If $\sigma(gh) = \sigma(g) \circ \sigma(h)$ for all $g, h \in G$, then*

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}.$$

*Proof.* Expanding both sides: $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$. □

### 15.7.7 String order and local symmetry equivalence

The string order parameter [PGWS+08] detects whether an on-site unitary $u$ acts as a local symmetry on the state generated by an injective MPS tensor.

**Definition 15.47** (Twisted transfer map)**.** Given an MPS tensor $A$ and a physical-index matrix $u$, the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n,n'} \langle n'|u|n\rangle \, A_n \, X \, A_{n'}^\dagger.$$

**Definition 15.48** (String order parameter)**.** The *string order parameter* for twist $u$ and stationary state $\Lambda$ at length $L$ is

$$R_L(u) := \operatorname{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})).$$

**Definition 15.49** (Local symmetry)**.** A state has *local symmetry* under $u$ if $\|R_L(u)\| = \|R_L(\mathbb{1})\|$ for all $L$.

**Definition 15.50** (String order)**.** String order *exists* for $u$ if there is a positive constant $c > 0$ such that $c \leq \|R_L(u)\|$ for all $L$.

**Definition 15.51** (Condition C1 (intertwining))**. Condition C1**: for each physical index $i$, $\sum_j u_{ij} \, A^j = V \, A^i \, V^\dagger$.

**Definition 15.52** (Condition C2 (covariance))**. Condition C2**: the transfer map commutes with virtual conjugation: $\mathcal{E}(VXV^\dagger) = V \, \mathcal{E}(X) \, V^\dagger$.

**Definition 15.53** (Condition C3 (doubled commutation))**. Condition C3**: the doubled transfer matrix commutes with $V \otimes \bar{V}$, expressed as $V \, \mathcal{E}_1(X) \, V^\dagger = \mathcal{E}_1(VXV^\dagger)$.

**Theorem 15.54** (C2 ⇔ C3)**.** *Conditions C2 and C3 are equivalent for any matrix $V$.*

*Proof.* Both sides express the same identity with opposite orientation; the equivalence is `eq_comm`. □

**Lemma 15.55** (Unitary Kraus mixing)**.** *If $u$ is unitary, then replacing Kraus operators $\{A_i\}$ by their unitary mixtures $\{\sum_j u_{ij} A_j\}$ preserves the channel: $\sum_i (\sum_j u_{ij} A_j) \, Y \, (\sum_j u_{ij} A_j)^\dagger = \sum_i A_i \, Y \, A_i^\dagger$.*

*Proof.* Thin adapter over `kraus_same_map_of_unitary_combination` from §5.8. □

**Theorem 15.56** (C1 ⇒ C2)**.** *If $V$ and $u$ are unitary and C1 holds, then C2 holds.*

*Proof.* Insert $V^\dagger V = \mathbb{1}$ to extract conjugated Kraus operators, apply C1, then invoke unitary Kraus mixing. □

**Theorem 15.57** (C2 ⇒ C1 (injective))**.** *For an injective MPS, if $V$ is unitary and C2 holds, then there exists a unitary $u$ satisfying C1.*

*Proof.* Requires the spanning property of $\{A_i\}$ (injectivity) to extract $u$ from the transfer-map covariance relation. Requires CP map spectral theory; see Status section of the Lean module. ☐

**Theorem 15.58** (Spectral radius bound)**.** *For an injective MPS with unitary twist $u$, every eigenvalue $\lambda$ of $\mathcal{E}_u$ satisfies $|\lambda| \leq 1$.*

*Proof.* Requires Cauchy–Schwarz for CP maps and the unique fixed-point property of the transfer map (Perron–Frobenius for CP maps). See Status section of the Lean module. ☐

**Theorem 15.59** (Local symmetry ⇔ spectral radius one)**.** *For a pure FCS, $u$ is a local symmetry iff $\rho(\mathcal{E}_u) = 1$.*

*Proof.* Requires spectral theory of completely positive maps (Perron–Frobenius for CP maps). See Status section of the Lean module. ☐

**Theorem 15.60** (String order ⇔ local symmetry)**.** *For an injective MPS, string order for $u$ exists iff $u$ is a local symmetry.*

*Proof.* Requires spectral theory of completely positive maps (Perron–Frobenius for CP maps). See Status section of the Lean module. ☐

**Theorem 15.61** (Virtual unitary from string order)**.** *If string order exists for $u$ in an injective MPS, then there exists a virtual unitary $V$ and a unit-modulus scalar $\mu$ such that $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$ for all $i$.*

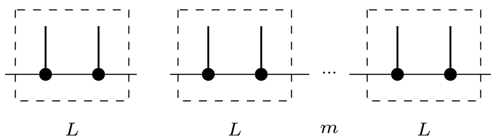
*Proof.* Requires spectral theory of completely positive maps (Perron–Frobenius for CP maps). See Status section of the Lean module. ☐

### 15.7.8 Normal MPS (blocked injectivity)

**Corollary 15.62** (Normal MPS Fundamental Theorem)**.** *Let $\{A_k\}_{k=0}^{n-1}$ and $\{B_k\}_{k=0}^{n-1}$ be two non-TI periodic MPS on $n$ sites. Suppose that blocking any $L$ consecutive sites in either chain yields an injective tensor, and that $n \geq 3L$. If both chains generate the same state, then there exist invertible matrices $Z_0, \ldots, Z_{n-1} \in \mathrm{GL}_D(\mathbb{C})$, unique up to a common scalar, such that for every site $k$ and physical index $i$,*

$$B_k^i = Z_k^{-1} A_k^i Z_{k+1},$$

*with $Z_n = Z_0$. That is, gauge equivalence holds for the* original *(unblocked) tensors. The reduction to the injective case proceeds by grouping the chain into consecutive length-$L$ blocks,*




*and then applying the injective-chain theorem to the blocked local tensors. The checked Lean endpoint currently stops at the blocked chain: proves gauge equivalence for the constant chain of blocked tensors, while the passage back to the original unblocked site tensors is still blueprint-level and is the reason this corollary remains marked .*

*Proof.* Block every $L$ consecutive sites to form a chain of length $m = \lfloor n/L \rfloor \geq 3$ with injective local tensors $A^{[L]}$ and $B^{[L]}$ (injective by hypothesis). The blocked chains generate the same state, so Theorem 15.14 gives invertible gauges $\widehat{Z}_0, \dots, \widehat{Z}_{m-1}$ with $(B^{[L]})_k = \widehat{Z}_k^{-1} (A^{[L]})_k \widehat{Z}_{k+1}$. In the formalization, this blocked endpoint is packaged as , which reduces directly to the injective-chain theorem on the blocked constant chain. Applying Lemma 15.13 within each $L$-block then decomposes the blocked gauge equivalence into site-by-site gauges, yielding $Z_0, \dots, Z_{n-1}$ for the original tensors. □

## 15.8 Product algebra construction and per-block gauge

The chain Fundamental Theorem (Theorem 15.14) handles non-TI periodic chains. A parallel line of argument addresses block-diagonal tensors: given a family of injective block tensors $\{A_k^i\}_{k=0}^{r-1}$ with block dimensions $D_k$ and a second family $\{B_k^i\}$ with per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$, one seeks per-block gauge equivalence $B_k^i = X_k A_k^i X_k^{-1}$ for each $k$ independently, and a global block-permutation plus inner-automorphism decomposition.

### 15.8.1 Per-block linear extension

**Definition 15.63** (Per-block linear extension)**.** Let $A_k, B_k$ be MPS tensors of bond dimension $D_k$ with $A_k$ injective and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block $k \in \{0, \dots, r-1\}$. The *per-block linear extension* $T_k : M_{D_k}(\mathbb{C}) \to M_{D_k}(\mathbb{C})$ is the unique $\mathbb{C}$-linear map satisfying $T_k(A_k^i) = B_k^i$ for every physical index $i$.

**Lemma 15.64** (Per-block multiplicativity)**.** *For each block $k$, the per-block linear extension $T_k$ is multiplicative: $T_k(MN) = T_k(M)\, T_k(N)$ for all $M, N \in M_{D_k}(\mathbb{C})$.*

*Proof.* Since $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$, it suffices to check on spanning elements, where the result follows from the MPV identity. □

**Lemma 15.65** (Per-block bijectivity)**.** *Each $T_k$ is bijective.*

*Proof.* A nonzero multiplicative linear endomorphism of $M_{D_k}(\mathbb{C})$ is automatically bijective: injectivity follows from the simplicity of $M_{D_k}(\mathbb{C})$ (the kernel is a proper ideal, hence zero), and surjectivity from finite dimension. Non-vanishing of $T_k$ is established via the trace pairing: if $T_k = 0$, then all $B_k^i = 0$, contradicting the non-vanishing of the trace pairing for injective tensors. □

### 15.8.2 Product algebra automorphism

**Definition 15.66** (Product algebra automorphism)**.** The *product algebra automorphism* is the $\mathbb{C}$-algebra equivalence

$$T : \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$$

defined by $T(M)_k = T_k(M_k)$, where each $T_k$ is the per-block linear extension. It is an algebra automorphism since each $T_k$ is a multiplicative, unital, bijective linear map. Diagrammatically, this is the blockwise map on the product algebra obtained by applying the corresponding per-block extension to each summand.

**Theorem 15.67** (Block-permutation decomposition)**.** *The product algebra automorphism $T$ decomposes as a block permutation composed with per-block inner automorphisms: there exist a permutation $\sigma \in S_r$ (with $D_{\sigma(k)} = D_k$ for all $k$) and invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that*

$$(T(M))_{\sigma(k)} = X_k \, M \, X_k^{-1}$$

*for all $M \in M_{D_k}(\mathbb{C})$ (viewed as supported in block $k$) and all $k$. The picture is that the product-algebra automorphism first permutes the simple matrix blocks and then acts inside each block by an inner conjugation,*

$$\begin{array}{ccc} \prod_k M_{D_k}(\mathbb{C}) & & \prod_k M_{D_k}(\mathbb{C}) \\ \boxed{A_0} \quad \boxed{A_1} \quad \boxed{A_{r-1}} & \xrightarrow{\sigma} & \boxed{X_0} \quad \boxed{X_1} \quad \boxed{X_{r-1}} \end{array}$$

*where the permutation reindexes the block ideals and the labels $X_k$ denote the resulting inner automorphisms.*

*Proof.* The automorphism $T$ permutes the block ideals of $\prod_k M_{D_k}(\mathbb{C})$ (each block ideal is simple, and an automorphism sends simple ideals to simple ideals of the same dimension). This gives the permutation $\sigma$. On each block, the restricted map is a $\mathbb{C}$-algebra automorphism of $M_{D_k}(\mathbb{C})$; by Skolem–Noether (Theorem 4.9), it is inner. In Lean, this is exactly the call to inside ; the tensor-network picture records the same split into permutation of block ideals followed by inner conjugation on each surviving block. ☐

### 15.8.3 Nondegeneracy of the product trace pairing

**Lemma 15.68** (Nondegeneracy of the product trace pairing)**.** *Let $M = (M_0, \dots, M_{r-1}) \in \prod_k M_{D_k}(\mathbb{C})$. If $\sum_k \mathrm{tr}(M_k N_k) = 0$ for all $N = (N_0, \dots, N_{r-1})$, then $M = 0$.*

*Proof.* Choose $N$ supported on a single block $k$ (all other components zero). The hypothesis reduces to $\mathrm{tr}(M_k N_k) = 0$ for all $N_k$, so $M_k = 0$ by nondegeneracy of the matrix trace pairing on $M_{D_k}(\mathbb{C})$. Since $k$ was arbitrary, $M = 0$. ☐

**Lemma 15.69** (Injectivity of the product Gram map)**.** *Let $\{A_k\}$ be a family of injective block tensors. The product Gram map $M \mapsto (k, i \mapsto \mathrm{tr}(M_k \cdot A_k^i))$ is injective.*

*Proof.* Reduces to per-block injectivity of the Gram map $M_k \mapsto (i \mapsto \mathrm{tr}(M_k \cdot A_k^i))$, which holds by injectivity of $A_k$ (the matrices $\{A_k^i\}$ span $M_{D_k}(\mathbb{C})$, so the trace pairing against them separates points). ☐

## 15.9 The multi-block Fundamental Theorem

**Theorem 15.70** (Multi-block Fundamental Theorem)**.** *Let $\{A_k\}_{k=0}^{r-1}$ and $\{B_k\}_{k=0}^{r-1}$ be two families of MPS tensors with block dimensions $D_k$, each $A_k$ injective, and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all $k$. Then:*

1. *Per-block gauge equivalence: $B_k^i = X_k \, A_k^i \, X_k^{-1}$ for some $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$, for every $k$ and $i$.*
2. *Global gauge equivalence: the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

*The second clause is assembled from the first by putting the per-block gauges on the diagonal,*

$$\overset{\bigoplus_k \mu_k A_k}{\boxed{\mu_0 A_0}\quad\boxed{\mu_1 A_1}\quad\boxed{\mu_{r-1} A_{r-1}}} \xrightarrow{\mathrm{diag}(X_k)} \overset{\bigoplus_k \mu_k B_k}{\boxed{\mu_0 B_0}\quad\boxed{\mu_1 B_1}\quad\boxed{\mu_{r-1} B_{r-1}}}$$

*which is exactly the separation between the two conclusions in .*

*Proof.* Per-block gauge equivalence follows from applying the single-block Fundamental Theorem (existing) to each block, using injectivity of $A_k$ and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. Global gauge equivalence is assembled from the per-block gauges via the block-diagonal structure. The Lean theorem returns these two outputs as an explicit pair: pointwise gauge equivalence of the blocks and gauge equivalence of the weighted direct-sum tensors. □

**Theorem 15.71** (Per-block SameMPV $\Leftrightarrow$ per-block gauge equivalence)**.** *Let $\{A_k\}$ be a family of injective block tensors. Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all $k$ if and only if $A_k$ and $B_k$ are gauge equivalent for all $k$.*

*Proof.* The forward direction is the single-block Fundamental Theorem applied blockwise. The reverse direction holds because gauge equivalence preserves all traces of word evaluations. □

## 15.10 Block separation from canonical form

The multi-block Fundamental Theorem (Theorem 15.70) assumes per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. In practice, one starts from the weaker global hypothesis that the block-diagonal tensors satisfy the same MPV family: $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$. Block separation extracts the per-block identities from the global one under canonical-form normalization. The first formal reduction is the weighted block-sum identity furnished by :

$$\overset{\sum_k \mu_k^N V(A_k)}{\boxed{\mu_0^N V(A_0)}\quad\boxed{\textstyle\sum_{k\geq 1} \mu_k^N V(A_k)}} \quad = \quad \overset{\sum_k \mu_k^N V(B_k)}{\boxed{\mu_0^N V(B_0)}\quad\boxed{\textstyle\sum_{k\geq 1} \mu_k^N V(B_k)}}$$

This is then combined with overlap estimates and the leading-block peel.

**Definition 15.72** (Canonical form)**.** A family of block tensors $\{A_k\}_{k=0}^{r-1}$ with scaling weights $\mu_0, \dots, \mu_{r-1} \in \mathbb{C}^\times$ is in *canonical form* if:

1. each $A_k$ is injective,
2. each $A_k$ is left-canonical: $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}_{D_k}$,
3. the weights are strictly ordered by norm: $|\mu_0| > |\mu_1| > \cdots > |\mu_{r-1}| > 0$,
4. the self-overlaps are normalised: $O_{A_k A_k}(N) \to 1$ as $N \to \infty$.

**Definition 15.73** (Normal canonical form)**.** A family is in *normal canonical form* if it satisfies the canonical-form conditions with each block additionally *irreducible* (no nontrivial invariant projection) and *primitive* (the transfer map has spectral gap).

**Theorem 15.74** (Block separation from canonical form)**.** *Let $\{A_k\}$ be in canonical form with weights $\{\mu_k\}$, and let $\{B_k\}$ be another family with each $B_k$ injective and left-canonical. If*

*$\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$, then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block $k$ individually. The proof move is to isolate the leading block and treat the remaining blocks as an exponentially decaying tail:*

$$V(A_0) = V(B_0)$$

$$\boxed{\mu_0^N V(A_0)} \qquad \boxed{\textstyle\sum_{k\geq 1} \mu_k^N V(A_k)} \qquad = \qquad \boxed{\mu_0^N V(B_0)} \qquad \boxed{\textstyle\sum_{k\geq 1} \mu_k^N V(B_k)}$$

$$O(\rho^N) \qquad\qquad\qquad\qquad O(\rho^N)$$

*with $\rho = |\mu_1|/|\mu_0| < 1$.*

*Proof.* The proof proceeds by strong induction on the number of blocks $r$. For $r = 1$, the weighted sum degenerates and the result is immediate after cancelling $\mu_0^N \neq 0$. In Lean this one-block base case is packaged separately as and then absorbed into the canonical-form wrapper.

For $r \geq 2$, the global identity $\sum_k \mu_k^N \left(V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma\right) = 0$ holds for all $N$ and $\sigma$. Multiplying by the MPV components of $A_0$ and summing over $\sigma$ converts this to an overlap identity. Dividing by $|\mu_0|^{2N}$ and using the strict norm ordering $|\mu_1|/|\mu_0| < 1$, the tail terms ($k \geq 1$) decay geometrically with ratio $\rho = |\mu_1|/|\mu_0|$. The leading term ($k = 0$) therefore converges: $O_{A_0 B_0}(N) \to 1$ as $N \to \infty$. Combined with the normalised self-overlap $O_{A_0 A_0}(N) \to 1$, this gives $\mathcal{V}(A_0) = \mathcal{V}(B_0)$ via the gauge-phase equivalence criterion. Subtracting the leading block and applying the inductive hypothesis yields $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all remaining $k$. In Lean this is not a Vandermonde argument: the core induction is , whose leading-block step is and whose geometric estimate comes from . The weighted identity itself comes from . The diagrams above are meant to track that checked route, not a different proof strategy. □

**Theorem 15.75** (Fundamental Theorem from canonical form)**.** *Under the hypotheses of Theorem 15.74, every block pair $A_k, B_k$ is gauge equivalent, and the block-diagonal tensors are globally gauge equivalent.*

*Proof.* Theorem 15.74 gives per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. Theorem 15.70 converts this to per-block and global gauge equivalence. □

**Theorem 15.76** (Explicit gauge from canonical form)**.** *Under the same hypotheses, there exist invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that $B_k^i = X_k A_k^i X_k^{-1}$ for every block $k$ and physical index $i$.*

*Proof.* Extract the gauge matrices from Theorem 15.75 by choice. □

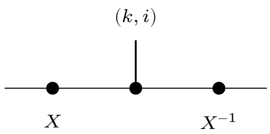
## 15.11 TI reduction corollary

**Theorem 15.77** (TI reduction corollary)**.** *Let $A$ and $B$ be MPS tensors of bond dimension $D$ with $A$ injective. Consider the two constant (translation-invariant) chains $(A, \ldots, A)$ and $(B, \ldots, B)$ of length $n \geq 1$. If the combined tensors of the two chains generate the same MPV family, then:*

1. *$B$ is gauge equivalent to $A$: there exists $X \in \mathrm{GL}_D(\mathbb{C})$ with $B^i = X A^i X^{-1}$ for all $i$.*

2. *$B$ is injective.*

*Proof.* The formal proof does not first derive a family of bond gauges and then collapse them. Instead, it packages the constant chain into the combined tensor, applies the single-block Fundamental Theorem to obtain one uniform gauge




on the combined-site index $(k, i)$, and then reads that back as a site tensor gauge via . Injectivity of $B$ follows from gauge equivalence to the injective tensor $A$. $\square$

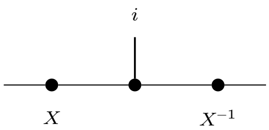
**Theorem 15.78** (TI reduction from same state)**.** *Under the same setup, if $n \geq 3$ and the two constant chains generate the same chain state (rather than the same MPV family on the combined tensor), the same conclusion holds, provided the blocking bridge hypothesis (`SameStateBridgeHyp`) is available to upgrade same-state to same-MPV on the combined tensor.*

*Proof.* The bridge hypothesis converts same-state at chain length $n \geq 3$ to SameMPV on the combined tensor. Then Theorem 15.77 applies. $\square$

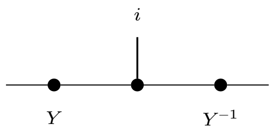
### 15.11.1 Global symmetry via gauge composition

The virtual representation theorem (Theorem 15.36) relies on the fact that composing two gauge transformations yields another gauge up to a scalar factor.

**Lemma 15.79** (Gauge ratio commutes with tensor matrices)**.** *Diagrammatically, the hypothesis is that the two local gauges*




*and*




*produce the same target tensor $B^i$ from the same source tensor $A^i$. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both conjugate an MPS tensor $A$ to the same tensor $B$ (i.e. $B^i = XA^iX^{-1} = YA^iY^{-1}$ for all $i$), then $Y^{-1}X$ commutes with every $A^i$.*

*Proof.* Direct matrix calculation: conjugating both gauge relations and composing gives $(Y^{-1}X)A^i = A^i(Y^{-1}X)$. $\square$

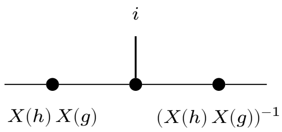
**Lemma 15.80** (Gauge ratio is scalar for injective tensors)**.** *Under the same hypotheses, if $A$ is additionally injective, then $Y^{-1}X = \lambda \mathbb{1}_D$ for some scalar $\lambda \in \mathbb{C}$.*

*Proof.* The same two-gauge picture shows that the ratio $Y^{-1}X$ acts invisibly on the local tensor. By Lemma 15.79, $Y^{-1}X$ commutes with all $A^i$. Since $\{A^i\}$ spans $M_D(\mathbb{C})$, the matrix $Y^{-1}X$ commutes with all of $M_D(\mathbb{C})$. By Lemma 15.11, it is a scalar matrix. $\square$

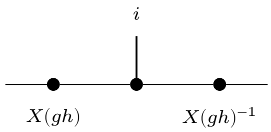
**Theorem 15.81** (Projective multiplication law for gauge matrices)**.** *Let $A$ be an injective MPS tensor with on-site symmetry under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$. For each $g \in G$, let $X(g) \in \mathrm{GL}_D(\mathbb{C})$ satisfy $\widetilde{A}^i_g = X(g)\, A^i\, X(g)^{-1}$ for all $i$. Then for all $g, h \in G$, there exists $c(g, h) \in \mathbb{C}$ such that*

$$X(h)\, X(g) = c(g, h)\, X(g \cdot h).$$

*(In fact $c(g,h) \neq 0$ since both sides are invertible, so $c(g,h) \in \mathbb{C}^\times$; the Lean statement records the existential without the nonzero assertion.) Diagrammatically, both*




*and*




*conjugate $A^i$ to the same twisted tensor $\widetilde{A}^i_{gh}$.*

*Proof.* By the composition law $\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g$ (Lemma 15.28), both $X(h)\,X(g)$ and $X(gh)$ conjugate $A$ to $\widetilde{A}_{gh}$. Lemma 15.80 then gives the scalar factor. The checked Lean theorem in `TNLean/PiAlgebra/GlobalSymmetry.lean` follows this exact route: it first proves that the ratio of the two gauges commutes with every $A^i$, then promotes that commutant statement to a scalar using injectivity. ☐

## 15.12 Closing remarks

*Remark* 15.82. The paper's Appendix [MGRPG+18] strengthens the normal-MPS bound from $n \geq 3L$ to $n \geq 2L+1$ by a more refined blocking argument. The injective PEPS theorem and the normal PEPS theorem on arbitrary graphs both rest on the same two core lemmas (Lemma 15.12 and Lemma 15.13); the graph geometry enters only through the blocking step that reduces to a three-site chain.

# Chapter 16

# Parent Hamiltonians

This chapter introduces the local ground-space construction and the parent interaction for translation-invariant periodic MPS. We follow the standard setup from [CPGSV21, PGVWC07] and align the Lean definitions with the project convention that the $N$-site Hilbert space is represented as coefficient functions

$$\{0, \dots, d-1\}^N \to \mathbb{C},$$

rather than as an explicit tensor product.

## 16.1 Local ground space $G_L(A)$

Fix an MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ with $A^i \in M_D(\mathbb{C})$ and a block length $L \geq 1$. For a word $w = (i_1, \dots, i_L)$, write $A^w := A^{i_1} \cdots A^{i_L}$. The local ground-space map is

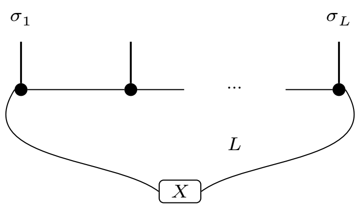
$$\Gamma_L : M_D(\mathbb{C}) \to (\{0, \dots, d-1\}^L \to \mathbb{C}), \qquad \Gamma_L(X)(\sigma) := \mathrm{tr}(A^\sigma X),$$

where $A^\sigma$ abbreviates $A^{\sigma(0)} \cdots A^{\sigma(L-1)}$.

**Definition 16.1** (Ground-space map)**.** The map $\Gamma_L$ above is implemented as a $\mathbb{C}$-linear map

$$\Gamma_L : M_D(\mathbb{C}) \to (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

In tensor-network notation,




the upper chain denotes the word tensor $A^\sigma$ with open physical indices $\sigma_1, \dots, \sigma_L$, and the lower box denotes the boundary matrix $X$ inserted on the virtual bond before taking the trace.

**Definition 16.2** (Local ground space)**.** The *local ground space* is the image

$$G_L(A) := \mathrm{im}(\Gamma_L) \subseteq (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

**Lemma 16.3** (Dimension bound). *For every $L$,*

$$\dim G_L(A) \le D^2.$$

*Proof.* Since $G_L(A)$ is the range of a linear map from $M_D(\mathbb{C})$,

$$\dim G_L(A) \le \dim M_D(\mathbb{C}) = D^2.$$

□

**Lemma 16.4** (Ambient-space dimension). *The coefficient-function model satisfies*

$$\dim(\{0, \dots, d-1\}^L \to \mathbb{C}) = d^L.$$

**Lemma 16.5** (Nontriviality criterion). *If $d^L > D^2$, then $G_L(A)$ is a proper subspace of the local Hilbert space:*

$$G_L(A) \ne (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

*Proof.* If $G_L(A)$ were the whole space, its dimension would be $d^L$ by Lemma 16.4; Lemma 16.3 gives the contradiction $d^L \le D^2$. □

## 16.2 Parent interaction and chain Hamiltonian

The parent interaction at range $L$ is the orthogonal projector onto $G_L(A)^\perp$, implemented via the star-projection (orthogonal complement projection) in the $L$-site Euclidean space.

**Definition 16.6** (Parent interaction). The parent interaction at range $L$ is the orthogonal projector

$$h_L(A) := \Pi_{G_L(A)^\perp},$$

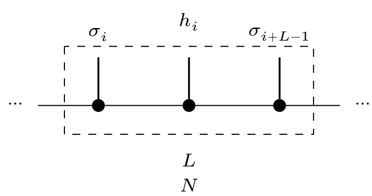
on the local $L$-site space.

**Lemma 16.7** (Parent interaction kills ground-space elements). *For any $v \in G_L(A)$, we have $h_L(A)\, v = 0$.*

*Proof.* Since $h_L(A)$ is the star-projection onto $G_L(A)^\perp$, it annihilates every element of $G_L(A)$. □

For a periodic chain of length $N$, the parent Hamiltonian is the sum of translated copies of the local interaction.

**Definition 16.8** (Translated local term). For each site index $i \in \{0, \dots, N-1\}$, the translated local term $h_i$ embeds the parent interaction into the full $N$-site space at position $i$ (with periodic boundary conditions). Diagrammatically,




the dashed window marks the $L$ consecutive sites on which the local projector acts, while the surrounding chain is left unchanged.

**Definition 16.9** (Parent Hamiltonian)**.** The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i.$$

**Definition 16.10** (Frustration-free condition)**.** A state $\psi$ is frustration-free if each local term annihilates it:

$$\forall\, i \in \{0, \dots, N-1\}, \quad h_i \psi = 0.$$

**Lemma 16.11** (MPV window membership)**.** *For any window of $L$ consecutive sites starting at position $i$ in an $N$-site periodic chain ($L \leq N$), the restriction of the MPV state to that window lies in $G_L(A)$.*

*Proof.* The witness is the product of $A$-matrices on the complement sites. Trace cyclicity rotates the full $N$-site product so that window indices come first, matching the ground-space map definition. ☐

**Lemma 16.12** (Local term annihilates the MPV)**.** *For each site $i$, $h_i\, V^{(N)}(A) = 0$.*

*Proof.* Combines MPV window membership with the fact that the parent interaction kills ground-space elements. ☐

**Lemma 16.13** (Parent Hamiltonian annihilates the MPV)**.** *For every $N$, the MPV state satisfies*

$$H_N(A, L)\, V^{(N)}(A) = 0.$$

*Proof.* Sum of zeros, since each local term annihilates the MPV. ☐

**Lemma 16.14** (Frustration-freeness of the MPV)**.** *The MPV state is frustration-free for the parent model.*

*Proof.* Immediate from `localTerm_annihilates_mpv`. ☐

## 16.3 Intersection property and unique ground state

This section develops the intersection property of local ground spaces for injective MPS and the resulting uniqueness of the ground state on the periodic chain.

### 16.3.1 Restriction maps

Given a state $\psi$ on $L + 1$ sites, we define two restriction maps.

**Definition 16.15** (Left restriction)**.** Fix the last physical index $j$:

$$(\text{restrictLast}\ \psi\ j)(\sigma) := \psi(\sigma_1, \dots, \sigma_L, j).$$

**Definition 16.16** (Right restriction)**.** Fix the first physical index $i$:

$$(\text{restrictFirst}\ \psi\ i)(\sigma) := \psi(i, \sigma_1, \dots, \sigma_L).$$

**Definition 16.17** (Left ground membership)**.** A state $\psi$ on $L + 1$ sites satisfies the left ground condition if restrictLast $\psi$ $j \in G_L(A)$ for every $j$.

**Definition 16.18** (Right ground membership)**.** A state $\psi$ on $L+1$ sites satisfies the right ground condition if restrictFirst $\psi$ $i \in G_L(A)$ for every $i$.

### 16.3.2 Forward direction: ground space restricts

**Lemma 16.19** (Left restriction). *If $\psi \in G_{L+1}(A)$, i.e. $\psi(\sigma) = \mathrm{tr}(A^\sigma X)$ for some $X$, then* restrictLast $\psi\ j \in G_L(A)$ *with witness $A^j X$.*

*Proof.* Direct computation: $\psi(\sigma_1, \dots, \sigma_L, j) = \mathrm{tr}(A^{\sigma_1} \cdots A^{\sigma_L} \cdot A^j \cdot X) = \Gamma_L(A^j X)(\sigma)$. □

**Lemma 16.20** (Right restriction). *If $\psi \in G_{L+1}(A)$ with $\psi(\sigma) = \mathrm{tr}(A^\sigma X)$, then* restrictFirst $\psi\ i \in G_L(A)$ *with witness $XA^i$.*

*Proof.* By trace cyclicity: $\psi(i, \sigma_1, \dots, \sigma_L) = \mathrm{tr}(A^i A^\sigma X) = \mathrm{tr}(A^\sigma \cdot XA^i) = \Gamma_L(XA^i)(\sigma)$. □

### 16.3.3 Injectivity of the ground-space map

**Theorem 16.21** (Ground-space map injectivity). *If $A$ is injective and $L \geq 1$, then $\Gamma_L$ is injective.*

*Proof.* If $\Gamma_L(X) = 0$ then $\mathrm{tr}(A^\sigma X) = 0$ for every length-$L$ word $\sigma$. Since $A$ is injective, the products $\{A^\sigma\}_\sigma$ span $M_D(\mathbb{C})$, so the trace pairing gives $X = 0$. □

**Theorem 16.22** (Ground-space dimension). *If $A$ is injective and $L \geq 1$, then $\dim G_L(A) = D^2$.*

*Proof.* The upper bound $\dim G_L(A) \leq D^2$ is Lemma 16.3. Injectivity of $\Gamma_L$ gives $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$. □

### 16.3.4 The intersection property

**Theorem 16.23** (Intersection property). *If $A$ is injective and $L \geq 2$, then a state $\psi$ on $L+1$ sites satisfying both the left and right ground conditions lies in $G_{L+1}(A)$:*

$$\psi \in G_L^{\mathrm{left}} \cap G_L^{\mathrm{right}} \implies \psi \in G_{L+1}(A).$$

*Proof.* The "invert-and-regrow" argument:

1. From the right ground condition, for each physical index $i$ there exists a unique $Y_i \in M_D(\mathbb{C})$ (by injectivity of $\Gamma_L$) such that $\psi(i, \sigma) = \mathrm{tr}(A^\sigma Y_i)$.
2. Similarly, from the left ground condition, for each $j$ there exists a unique $Z_j$ with $\psi(\sigma, j) = \mathrm{tr}(A^\sigma Z_j)$.
3. Matching on the $(L-1)$-site overlap and using nondegeneracy of the trace pairing: $A^j Y_i = Z_j A^i$ for all $i, j$.
4. Apply the decomposition map (one-sided inverse, Theorem **??**): summing over $i$ with the decomposition coefficients of the identity yields $Y_i = X' A^i$ for a single matrix $X' = \sum_j \Psi(\mathbf{1})_j \cdot Z_j$.
5. By trace cyclicity, $\psi(\sigma) = \mathrm{tr}(A^\sigma X')$, so $\psi \in G_{L+1}(A)$.

□

**Theorem 16.24** (Intersection characterization). *For injective $A$ and $L \geq 2$: $\psi \in G_{L+1}(A) \iff \psi \in G_L^{\mathrm{left}} \cap G_L^{\mathrm{right}}$.*

*Proof.* Immediate from the two forward-direction lemmas and Theorem 16.23. □

### 16.3.5 Unique ground state

**Lemma 16.25** (Iterated contiguous-window intersection). *If an $N$-site state satisfies the $L$-site ground condition on every non-wrapping contiguous window (with $L \geq 2$), then it lies in $G_N(A)$.*

*Proof.* Strong induction on window size, combining adjacent windows via Theorem 16.23 at each step. □

**Definition 16.26** (MPV submodule). The subspace spanned by the MPV state $V^{(N)}(A)(\sigma) = \operatorname{tr}(A^{\sigma_0} \cdots A^{\sigma_{N-1}})$.

**Lemma 16.27** (MPV lies in the ground space). *$V^{(N)}(A) \in G_N(A)$ for every $N$, with boundary matrix $X = I$.*

*Proof.* $V^{(N)}(A)(\sigma) = \operatorname{tr}(A^\sigma) = \operatorname{tr}(A^\sigma \cdot I) = \Gamma_N(I)(\sigma)$. □

**Definition 16.28** (Unique ground state). A submodule $S$ has a unique ground state if $\dim S = 1$.

**Theorem 16.29** (One-dimensionality criterion). *For finite-dimensional $S$, $\dim S = 1$ iff there exists a nonzero $\psi_0 \in S$ such that every $\psi \in S$ is of the form $c\,\psi_0$.*

*Proof.* Apply the Mathlib characterization `finrank_eq_one_iff_of_nonzero'` in both directions. □

**Definition 16.30** (Periodic chain ground space). For $N > 0$ and $L \leq N$, this is the intersection over cyclic window positions and outside configurations of comaps of $G_L(A)$ under the cyclic restriction maps; otherwise it is set to $\top$ by convention.

**Theorem 16.31** (Unique ground state on the periodic chain). *For an injective tensor $A$ on a periodic chain of $N \geq 2$ sites with interaction range $L$, the chain ground space is one-dimensional, spanned by the MPV. The proof iterates the intersection property across the chain and uses the periodic boundary condition to constrain the boundary matrix $X$ to a scalar multiple of $I$.*

**Theorem 16.32** (Unique ground state for block-injective tensors). *If $A$ is $L_0$-block-injective with $L_0 > 0$, the parent Hamiltonian with interaction range $2L_0$ has a unique ground state on the periodic chain.*

**Theorem 16.33** (Optimal unique ground state for normal tensors). *If $A$ is normal (hence $L_0$-block-injective for some $L_0 > 0$) and in normal form, the interaction range can be reduced to $L_0 + 1$.*

## 16.4 Spectral gap

A frustration-free Hamiltonian is *gapped* if the first excited eigenvalue is bounded away from zero uniformly in the chain length $N$. For MPS parent Hamiltonians, the spectral gap follows from a martingale criterion due to Kastoryano and Lucia [KL18].

**Theorem 16.34** (Martingale criterion for spectral gap). *Let $H = \sum_{i=1}^n h_i$ be a frustration-free Hamiltonian with projector interaction terms ($h_i^2 = h_i$). Suppose that for every pair of overlapping terms $h_i, h_j$ there exist constants $c_{ij} \geq 0$ and $\gamma > 0$ satisfying*

$$h_i h_j + h_j h_i \geq -c_{ij}(1-\gamma)(h_i + h_j),$$

*with $\sum_{j:\, j \neq i} c_{ij} \leq 1$ for every $i$. Then the smallest positive eigenvalue of $H$ satisfies $\lambda^+_{\min}(H) \geq \gamma$.*

*Proof.* From $h_i^2 = h_i$,

$$H^2 = \sum_i h_i + \sum_{\substack{i<j \\ \text{overlapping}}} (h_i h_j + h_j h_i) + \sum_{\substack{i<j \\ \text{non-overlapping}}} (h_i h_j + h_j h_i).$$

Non-overlapping projectors commute ($h_i h_j = h_j h_i$), so their product $h_i h_j$ is itself a projector and hence PSD; thus the last sum is $\geq 0$. For the overlapping sum, the martingale condition gives $h_i h_j + h_j h_i \geq -c_{ij}(1-\gamma)(h_i + h_j)$. Since $h_i h_j + h_j h_i$ is symmetric in $i, j$, we may replace $c_{ij}$ and $c_{ji}$ by $\bar{c}_{ij} := \max(c_{ij}, c_{ji})$; the pairwise bounds are preserved. In the MPS application the constants $c_{ij} = 1/(2(L-1))$ depend only on the cyclic distance $d_N(i,j)$ and are therefore already symmetric, so symmetrization does not affect the row-sum bound. (In general, replacing $c$ by the symmetrized $\bar{c}$ may inflate row sums; the abstract statement should either assume symmetric constants or verify that symmetrization preserves the row-sum hypothesis.) Summing over unordered pairs $\{i, j\}$:

$$\sum_{\{i,j\},\,\text{overlap}} (h_i h_j + h_j h_i) \geq -(1-\gamma) \sum_{\{i,j\},\,\text{overlap}} c_{ij}(h_i + h_j).$$

Collecting terms for each $h_i$ (using $c_{ij} = c_{ji}$, so that summing over unordered pairs containing $i$ gives $\sum_{j \neq i} c_{ij}$):

$$= -(1-\gamma) \sum_i \Big( \sum_{j \neq i,\,\text{overlap}} c_{ij} \Big) h_i \geq -(1-\gamma) H,$$

where the last step uses the row-sum hypothesis $\sum_{j \neq i} c_{ij} \leq 1$. Combining: $H^2 \geq H - (1-\gamma)H = \gamma H$. For any eigenvector with $H\psi = \lambda\psi$ and $\lambda > 0$, this gives $\lambda^2 \geq \gamma\lambda$, hence $\lambda \geq \gamma$ [KL18]. □

**Lemma 16.35** (Martingale condition for MPS). *Let $A$ be injective and fix an interaction range $L \geq 2$ as in Definition 16.6. Let $h_L(A) := \Pi_{G_L(A)^\perp}$ be the $L$-site parent interaction, and for each chain length and site $i$ let $h_i$ denote its translate as in Definition 16.8. Then for every pair of overlapping terms $h_i$ and $h_j$ with $0 < d_N(i,j) < L$, where $d_N(i,j) := \min(|i-j|, N - |i-j|)$ is the cyclic distance on the ring (i.e. whose $L$-site supports share at least one site under periodic boundary conditions), the martingale condition of Theorem 16.34 holds with constants $c_{ij}$ and $\gamma > 0$ depending only on $A$ (not on $N$).* Note: *The proof below fully establishes the martingale constants only for $L = 2$; for $L > 2$ the existence of suitable constants is asserted without complete proof (see the remark at the end of the proof). Only the $L = 2$ case is used in this chapter (Theorem 16.36). (When $L = 1$ the interaction terms have disjoint supports, so there are no overlapping pairs and the spectral gap follows directly without the martingale machinery.)*

*Proof.* Two terms $h_i, h_j$ with $0 < d_N(i,j) < L$ have supports overlapping in $L - d_N(i,j)$ sites (where $d_N$ is the cyclic distance on the $N$-site ring). Since $A$ is injective, the intersection property (Theorem 16.23) gives the qualitative identity $\ker(h_i) \cap \ker(h_j) = G_{[i,j+L-1]}(A)$. Because the local Hilbert space is finite-dimensional, the subspaces $\ker(h_i)$ and $\ker(h_j)$ live in a fixed finite-dimensional space. The intersection identity shows that $\ker(h_i) \cap \ker(h_j)$ is exactly $G_{[i,j+L-1]}(A)$; in particular there is no non-trivial vector in one kernel that is orthogonal to the other yet lies in both. Hence the Friedrichs angle $\theta_{d_N(i,j)}$ between $\ker(h_i)$ and $\ker(h_j)$ is strictly positive (for the fixed injective tensor $A$). Because the angle depends only on the overlap pattern $d_N(i,j) \in \{1, \dots, L-1\}$ and not on the absolute position, the number of distinct angles is at most $L - 1$. Let $\alpha := \max_{1 \leq s < L} \cos\theta_s < 1$ be the worst-case cosine of the Friedrichs angles.

**Row-sum bound.** Since $L \geq 2$, each site $i$ overlaps with at most $2(L-1)$ neighbours (those $j$ with $0 < d_N(i,j) < L$). Set $c_{ij} := \frac{1}{2(L-1)}$ for every overlapping pair and $c_{ij} := 0$ otherwise. Then $\sum_{j \neq i} c_{ij} \leq 2(L-1) \cdot \frac{1}{2(L-1)} = 1$, satisfying the row-sum hypothesis of Theorem 16.34.

**Positivity of $\gamma$.** The Friedrichs angle bound gives, for each overlapping pair, $h_i h_j + h_j h_i \geq -\alpha\,(h_i + h_j)$. The martingale inequality holds with $\gamma = 1-2(L-1)\alpha$, which requires $c_{ij}(1-\gamma) = \frac{1}{2(L-1)} \cdot 2(L-1)\alpha = \alpha$ (matching the Friedrichs bound exactly). Since $\alpha < 1$ (the Friedrichs angle is strictly positive), it remains to show $\gamma > 0$, i.e. $\alpha < \frac{1}{2(L-1)}$. In the blocked picture of Theorem 16.36, $L = 2$ and each site overlaps with at most 2 neighbours, so the condition reduces to $\alpha < \frac{1}{2}$. For $L = 2$ with an injective tensor, the two projectors $h_i$ and $h_{i+1}$ act on a finite-dimensional space, and the intersection property (Theorem 16.23) guarantees that $\ker(h_i) \cap \ker(h_{i+1})$ is exactly $G_{[i,i+2]}(A)$; hence $\|P_i P_{i+1}\| < 1$ (the Friedrichs cosine is strictly less than 1). On a finite-dimensional space with two projectors, the Friedrichs cosine satisfies the quantitative bound $\alpha \leq 1 - c_{\min}$ where $c_{\min} > 0$ depends on the minimal principal angle between the two kernels. The bound $\alpha < \frac{1}{2}$ then follows from the Nachtergaele finite-size criterion [Nac96], which establishes that the overlap $\|P_i P_{i+1}\|$ for injective MPS is bounded away from 1 by a constant depending only on the tensor $A$. For general $L > 2$, the uniform constants $c_{ij} = \frac{1}{2(L-1)}$ make the condition $\alpha < \frac{1}{2(L-1)}$ increasingly restrictive as $L$ grows. A more refined choice of separation-dependent constants $c_{ij} = c_{d_N(i,j)}$ that weight small-separation (large-overlap) pairs more heavily can exploit the exponential decay of the Friedrichs cosine $\alpha_s$ in $s$ to satisfy the row-sum constraint. However, this refinement is not needed for the main result: in Theorem 16.36 below, the argument is applied only with $L = 2$ in the blocked picture. In all cases, the constants depend only on $A$ and are independent of $N$ [FNW92, KL18]. □

**Theorem 16.36** (Spectral gap for MPS parent Hamiltonians)**.** *For an $L_0$-block injective MPS tensor $A$, let $L := 2L_0$ be the interaction range (two blocked sites in the $L_0$-blocked picture). There exists $\gamma > 0$ such that for all $N \geq 2L$ with $L_0 \mid N$,*

$$\lambda^+_{\min}(H_N(A, L)) \geq \gamma,$$

*i.e. the parent Hamiltonian is uniformly gapped.*

*Proof.* Since $A$ is $L_0$-block injective, the blocked tensor $A^{[L_0]}$ is injective with physical dimension $d^{L_0}$ and bond dimension $D$. Each blocked site represents $L_0$ original sites, so the range-$L = 2L_0$ interaction on the original chain corresponds exactly to a range-2 interaction on the blocked chain (two blocked sites). *Remark on interaction range:* Theorem 16.32 establishes ground-state uniqueness at range $L_0+1$, whereas here we use range $2L_0$. The ground spaces are nested: increasing the interaction range can only shrink (or preserve) the ground space, so uniqueness at range $L_0+1$ implies uniqueness at range $2L_0$. The larger range is needed here so that, after $L_0$-blocking, each blocked interaction term spans exactly 2 blocked sites, enabling the martingale machinery with $L = 2$. The divisibility condition $L_0 \mid N$ ensures the blocking is compatible with the periodic chain. Apply Lemma 16.35 to the injective tensor $A^{[L_0]}$ with range 2 to obtain the martingale constants $c_{ij}$ and $\gamma > 0$. These constants depend only on $A^{[L_0]}$ (hence only on $A$ and $L_0$), not on $N$. Theorem 16.34 then gives the uniform gap $\lambda^+_{\min}(H_N) \geq \gamma$ [FNW92, Nac96]. □

## 16.5 Ground space of non-injective MPS

For a general (non-injective) MPS tensor in block-diagonal canonical form, the ground space is no longer one-dimensional. Instead, it is spanned by the MPVs of the individual normal blocks — the basis of normal tensors (BNT) from Chapter 12 [FNW92, PGVWC07].

**Theorem 16.37** (Ground space equals span of BNT)**.** *Let $A = \bigoplus_{k=1}^{g} \mu_k A_k$ be in canonical form with BNT separation (Definition 12.3), where $\{A_1, \ldots, A_g\}$ is a basis of normal tensors*

*(Definition 12.1). For $N \geq 2(L_0+1)$ (where $L_0$ is the maximum injectivity length of the blocks), the ground space of the parent Hamiltonian satisfies*

$$\ker(H_N(A, L_0{+}1)) = \operatorname{span}\{V^{(N)}(A_k) : k = 1, \dots, g\}.$$

*Proof.* $\supseteq$: Each $A_k$ is normal, so Lemma 16.13 (applied to the block-diagonal structure) gives $H_N(A, L_0{+}1)\, V^{(N)}(A_k) = 0$. Linearity of the kernel yields the inclusion.

$\subseteq$: Take $\psi \in \ker(H_N)$. The block-diagonal structure of $A$ gives, at the interaction range $L_0{+}1$ used in the Hamiltonian, the containment $G_{L_0+1}(A) \subseteq \sum_k G_{L_0+1}(A_k)$: any local ground state of $A$ on $L_0{+}1$ sites is a sum of local ground states of the individual blocks, because the off-diagonal blocks vanish in the transfer map (the canonical form ensures that distinct blocks have orthogonal images under $\Gamma_{L_0+1}$ for $L_0{+}1 \geq L_0$, so the sum is in fact direct). Each normal block $A_k$ has injectivity length $L_k \leq L_0$, so the $L_k$-blocked tensor $A_k^{[L_k]}$ is injective. Applying Theorem 16.23 to this injective blocked tensor (in unblocked coordinates, as in Theorem 16.32) and iterating across the chain for each block independently, and using Theorem 16.33 for each normal block, the component of $\psi$ associated with block $A_k$ must be proportional to $V^{(N)}(A_k)$. This gives the inclusion [FNW92, PGVWC07]. □

# Chapter 17

# Exponential Decay of Correlations

This chapter records the transfer-matrix expression for thermodynamic-limit correlators of a normal MPS and the resulting exponential decay estimate. The key mechanism is spectral: after removing the leading eigenvalue contribution, the connected part is governed by subleading eigenvalues of the transfer map.

When the MPS tensor generates a parent Hamiltonian (Chapter 16), the correlator quantifies spatial correlations in the ground state. The exponential decay rate is set by the spectral properties of the transfer map, i.e. by the modulus of its subleading eigenvalues.

The spectral analysis of the transfer map and the mixed transfer operator, including the eigenvalue bound $\rho(F_{AB}) \leq 1$, the strict spectral gap $\rho(F_{AB}) < 1$ for non-equivalent blocks, and the resulting overlap decay, is developed in Chapter 8. This chapter focuses on the *correlator* side: one-point and two-point expectations, their spectral decomposition, and quantitative bounds on decay rates.

## 17.1 Connected correlators from the transfer map

**Definition 17.1** (One-point expectation)**.** Fix an MPS tensor $A$ and a right fixed point $\rho^R$ of its transfer map (whose existence, and uniqueness up to scaling under the hypotheses of Theorem 7.9, are guaranteed by the quantum Perron–Frobenius theorem). For a local observable $X \in M_D(\mathbb{C})$ define

$$\langle X_0 \rangle := \operatorname{tr}(X\rho^R).$$

**Definition 17.2** (Two-point expectation at distance $n$)**.** For observables $X, Y \in M_D(\mathbb{C})$ and $n \geq 0$,

$$\langle X_0 Y_n \rangle := \operatorname{tr}\big(Y\, \mathcal{E}_A^n(X\rho^R)\big)\,.$$

Equivalently: insert $X$ at site 0, propagate by $\mathcal{E}_A^n$, then insert $Y$ and take the trace.

**Definition 17.3** (Connected correlator)**.** The connected two-point function is

$$C(X, Y; n) := \langle X_0 Y_n \rangle - \langle X_0 \rangle\langle Y_0 \rangle.$$

## 17.2 Spectral expansion and exponential decay

**Theorem 17.4** (Sum of exponentials)**.** *For a normal MPS, the connected correlator admits a spectral expansion of the form*

$$C(X,Y;n) = \sum_{j=1}^{D^2-1} c_j\, \lambda_j^n,$$

*where $\lambda_j$ are subleading transfer-matrix eigenvalues.*

*Proof.* The leading eigenvalue contribution ($\lambda_1 = 1$) equals $\langle X_0\rangle\langle Y_0\rangle$ and cancels in the connected part. Apply spectral decomposition to $\mathcal{E}_A^n$. □

**Theorem 17.5** (Exponential decay bound)**.** *If $|\lambda_j| \le |\lambda_2| < 1$ for all subleading eigenvalues, then there exists $C_{XY} \ge 0$ such that*

$$|C(X,Y;n)| \le C_{XY}\, |\lambda_2|^n.$$

*Proof.* Combine the sum-of-exponentials expansion with the triangle inequality. □

*Remark* 17.6 (Spectral hypothesis and the spectral gap). The hypothesis "$|\lambda_2| < 1$" in Theorem 17.5 is precisely the *spectral gap* condition for the transfer map $\mathcal{E}_A$. For primitive channels, the complementary spectral gap $\rho(\mathcal{E}_A - P) < 1$ is established in Theorem 5.39 (Chapter 5); for irreducible trace-preserving tensors, the analogous result is Theorem 8.25. Thus the exponential decay of connected correlations follows directly from the block separation theory once the spectral decomposition is in hand.

**Definition 17.7** (Correlation length)**.** The correlation length associated with $\lambda_2$ is

$$\xi := -\frac{1}{\log|\lambda_2|}.$$

**Lemma 17.8** (Correlation length is positive)**.** *If $|\lambda_2| < 1$ and $\lambda_2 \neq 0$, then $\xi > 0$.*

*Proof.* Since $0 < |\lambda_2| < 1$, we have $\log|\lambda_2| < 0$, so $\xi = -1/\log|\lambda_2| > 0$. □

## 17.3 Quantitative spectral gap bounds

The spectral gap theorems in Chapter 5 establish $\rho(\mathcal{E}_A - P) < 1$ for primitive channels qualitatively. This section records the *quantitative* refinements that give explicit constants and rates for the single-tensor transfer map $\mathcal{E}_A$. Two of the three results below are stubs in the formalization (TODO #22); the spectral gap from injectivity is fully proved.

**Theorem 17.9** (Exponential convergence of primitive channels)**.** *Let $A$ be a primitive trace-preserving MPS tensor with positive definite fixed point $\rho$ and fixed-point projection $P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$. Then there exist $C > 0$ and $0 < \delta \le 1$ such that for all $n \ge 0$ and $X \in M_D(\mathbb{C})$,*

$$\|\mathcal{E}_A^n(X) - P(X)\| \;\le\; C\,(1-\delta)^n\,\|X\|.$$

*The spectral gap $\delta$ exists by primitivity (Theorem 5.39, the complementary spectral gap).*

**Theorem 17.10** (Correlation length bound)**.** *For an injective trace-preserving MPS tensor, there exist $C > 0$ and $\xi > 0$ such that for all $n \geq 0$ and all traceless $X \in M_D(\mathbb{C})$ (i.e.* $\operatorname{tr}(X) = 0$*),*

$$\|\mathcal{E}_A^n(X)\| \;\leq\; C\, e^{-n/\xi}\, \|X\|.$$

*Traceless matrices lie in* $\ker P$*, where $P$ is the fixed-point projection. Since $\mathcal{E}_A - P$ has spectral radius strictly less than 1, the Gelfand formula gives exponential decay at rate $\xi = -1/\log\rho(\mathcal{E}_A - P)$.*

**Theorem 17.11** (Spectral gap from injectivity)**.** *For an injective trace-preserving MPS tensor, there exists $\delta > 0$ such that every eigenvalue $\mu \neq 1$ of the transfer map satisfies $|\mu| \leq 1 - \delta$.*

*Proof.* The proof combines: injectivity implies `HasEventuallyFullKrausRank` (at index 1), which implies primitivity (via `IsPrimitivePaper` → `IsPeripherallyPrimitive`), primitivity implies a spectral gap on the complementary map, and the finite number of eigenvalues gives a uniform bound. □

*Remark* 17.12 (Status of quantitative bounds). Two of the three theorems above are formalized as stubs (with `sorry`) in `TNLean/Spectral/QuantitativeGap.lean`, tagged TODO #22; the spectral gap from injectivity is fully proved. The qualitative results they refine — spectral gap existence, overlap decay, and correlation convergence — are fully proved in Chapter 8. Discharging the sorrys requires extracting explicit constants from the Jordan normal form and the Wielandt-bound argument.

## 17.4 Relation to parent Hamiltonians

*Remark* 17.13. When the MPS tensor $A$ generates a parent Hamiltonian $H_N(A, L)$ (Definition 16.9) whose ground state is the matrix product vector (Lemma 16.13), the exponential decay bound (Theorem 17.5) shows that ground-state correlations decay with a finite correlation length $\xi$.

*Remark* 17.14. The identity $\mathcal{E}^2 = \mathcal{E}$ is equivalent to vanishing subleading spectrum; equivalently, $\xi = 0$ and connected correlations vanish at nonzero separation.

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